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Epidemic Modelling Frameworks for Studying Infectious Disease Dynamics

Daniele Baccega, Marco Beccuti, Simone Pernice, Irene Terrone

13TH IEEE International Conference on Healthcare Informatics
University of Calabria, Rende, Italy
18th-21st June, 2025

Outline

- Who are we?
- Overview of computational models in epidemiology ($\approx$ 15 min).
- Macro-level models ($\approx$ 1 hour).
 - Case study: [GreatMod](#).
- Micro-level models ($\approx$ 1 hour).
 - Case study: [Forge4Flame](#).
- From macro to micro ($\approx$ 15 min).
- Conclusion ($\approx$ 5 min).

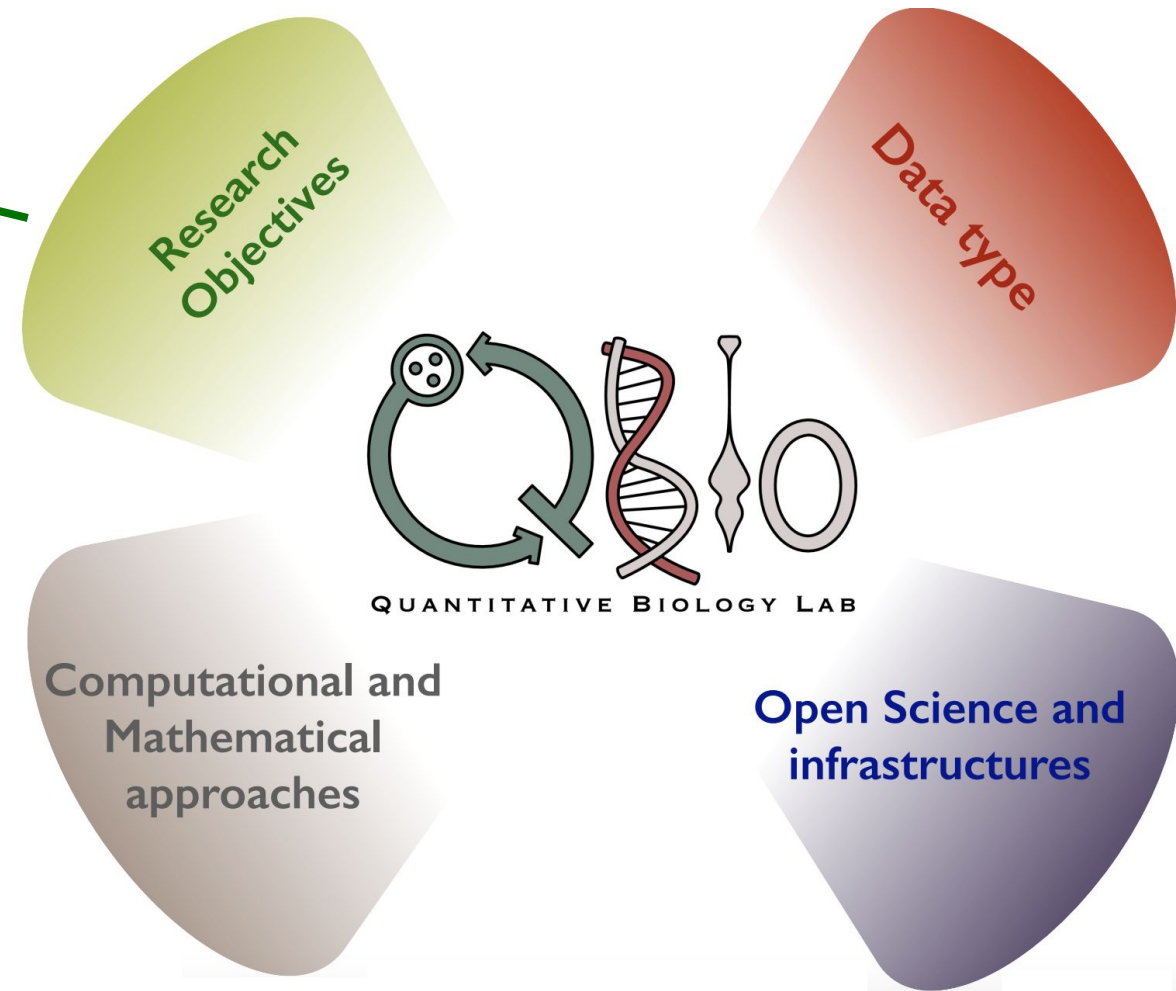


<http://github.com/qBioTurin/IEEEICHI2025>

Who are we?

Research Objectives:

- Patient stratification
- Identification of diagnostic and prognostic biomarkers
- Epidemiological models
- Mathematical models for cancer progression and treatment response
- Evolutionary Genomics and Modelling
- Basic Research: Cancer Genomics and Epigenomics



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Who are we?



Marco Beccuti
Associate Professor

Marco Beccuti is currently associate professor at the Department of Computer Science of the University of Turin, scientific coordinator of InfoLife National Laboratory of CINI, scientific coordinator of the laboratory “HPC for biomed and AI” in ICxT, Chair of the Scientific Committee of the “HPC4AI” laboratory” at the University of Turin.



Simone Pernice
RTDA

Simone Pernice is currently assistant professor at the Department of Computer Science of the University of Turin, coordinator of Young InfoLife group of the Infolife National Laboratory of CINI. His current research activities mainly involve computational epidemiology, modeling of complex biological systems , and longitudinal data analysis.



Daniele Baccega
Post Doc

Daniele Baccega is a Post Doc researcher at the Department of Computer Science of the University of Turin. He holds a National PhD in Artificial Intelligence (Health and Life Sciences). His current research activities mainly involve computational epidemiology, deep learning, and data analysis.



Irene Terrone
Ph.D. Student

Irene Terrone is a PhD student at the Department of Computer Science of the University of Turin. Her research project focuses on the creation and study of biological models to analyze the spread and management of epidemics, integrating it with machine learning and generating models.

Who's presenting what?



Simone Pernice

RTDA

Introduction
Macro-level models



Daniele Baccega

Post Doc

Micro-level models
From macro to micro level models



Irene Terrone

Ph.D. Student

Case studies:

- GreatMod
- Forge4Flame



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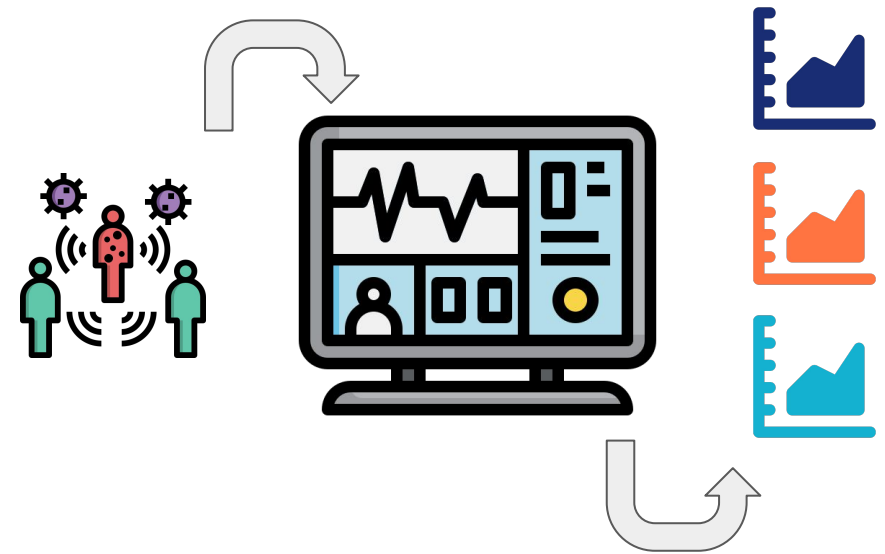
Introduction



Computational epidemiology

Definition

A multidisciplinary field that uses computational methods to simulate, analyze, and predict the spread of diseases within populations.



Key Components:

- **Epidemiological Models:** SIR, SEIR, ...
- **Computation:** Simulation approach, parameter estimation, data assimilation
- **Data Sources:** Case counts, mobility, demographics, contact networks

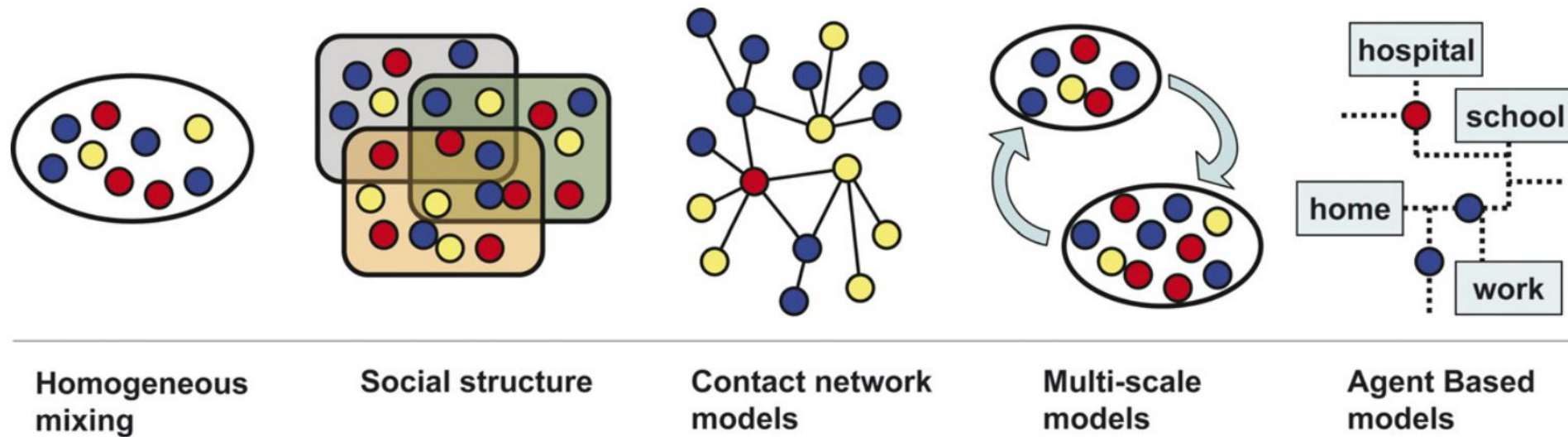


Handles **complex systems** with many interacting parts

Enables **what-if scenarios** and policy testing

Allows for **rapid analysis** using real-time data

Different modelling approaches

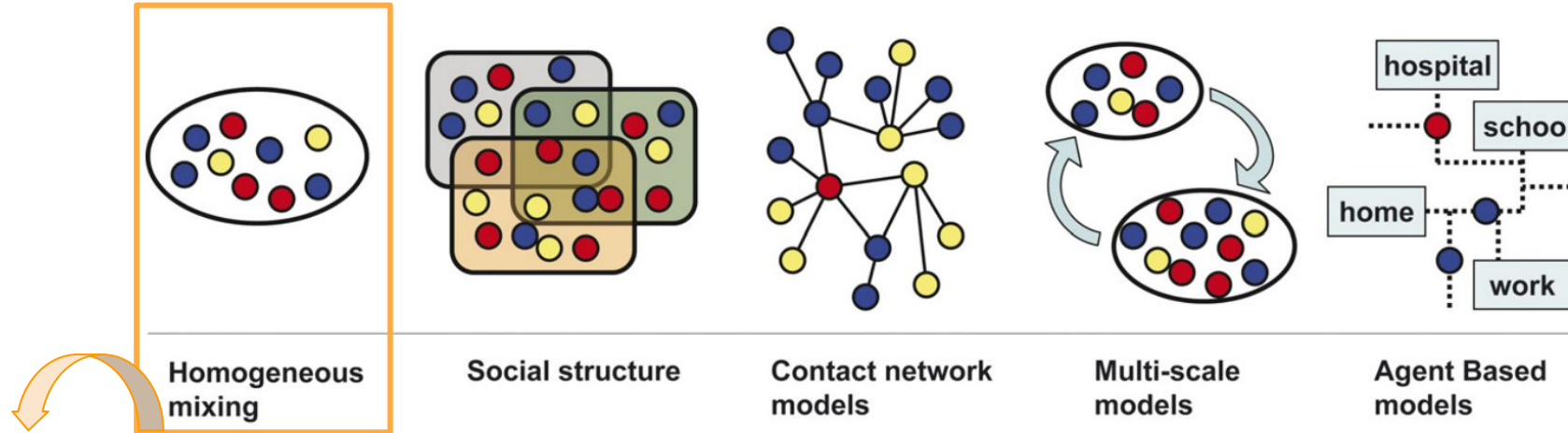


Macro-level

Abstraction

Micro-level

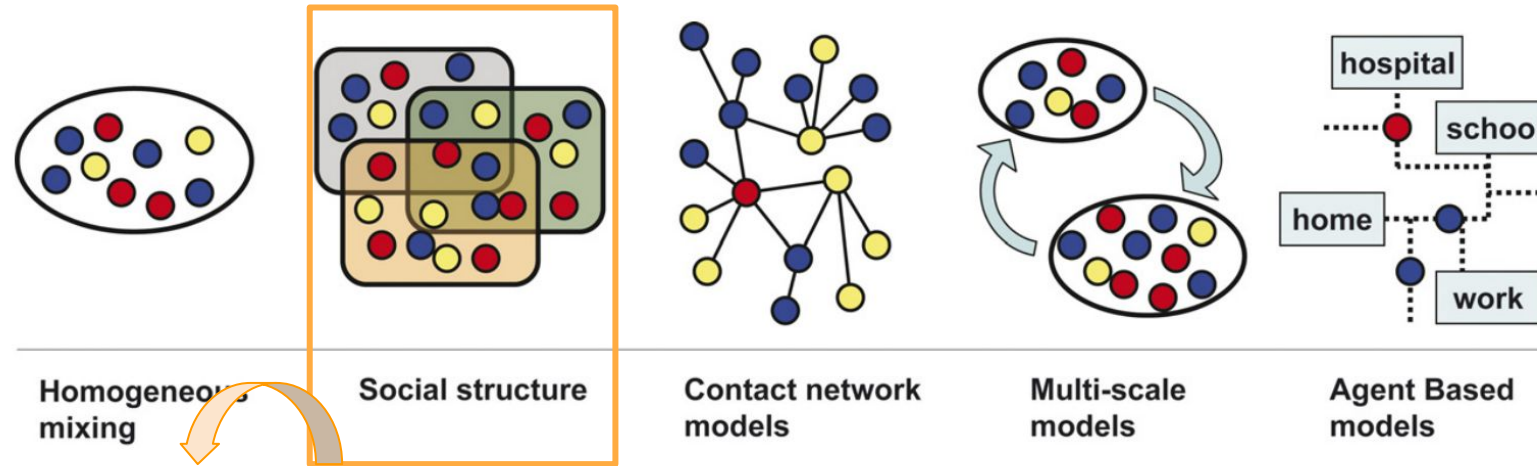
Different modelling approaches



Homogeneous mixing: assumes that every individual in a population is equally likely to interact with every other individual and treats the population as a well-mixed group without any structure, where interactions happen randomly and uniformly.

Model Type: SIR or SEIR compartmental model using **ordinary differential equations (ODEs)**, assuming population is like a "well-stirred soup"

Different modelling approaches



Social structure: accounts for the fact that individuals belong to different social groups or communities, based on factors such as age, economic status, region, and other demographic characteristics.

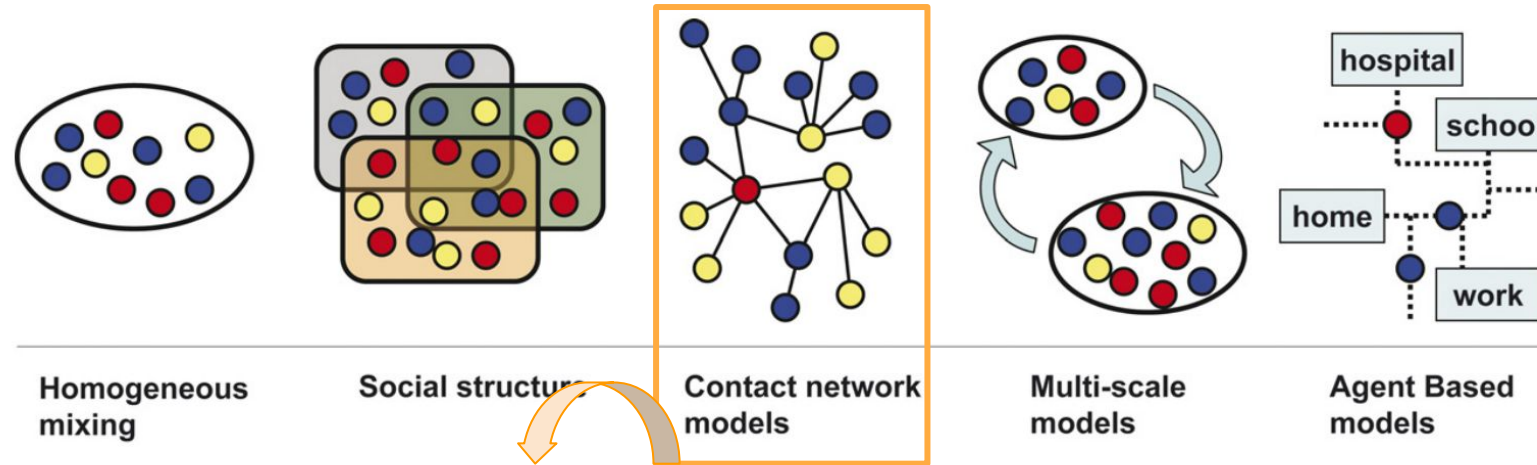
Real-World Use Case: *Evaluating the impact of closing schools vs. workplaces and adding and age structure, to simulate how transmission changes:*

- If schools are closed but workplaces stay open
- If elderly reduce contact but younger groups don't

Model Type:

- Structured compartmental models
- Age-stratified or region-based SIR/SEIR
- Matrix of contact rates between groups

Different modelling approaches



Contact network models: represent the population as a network, where nodes represent individuals, and edges represent the interactions or connections between them.

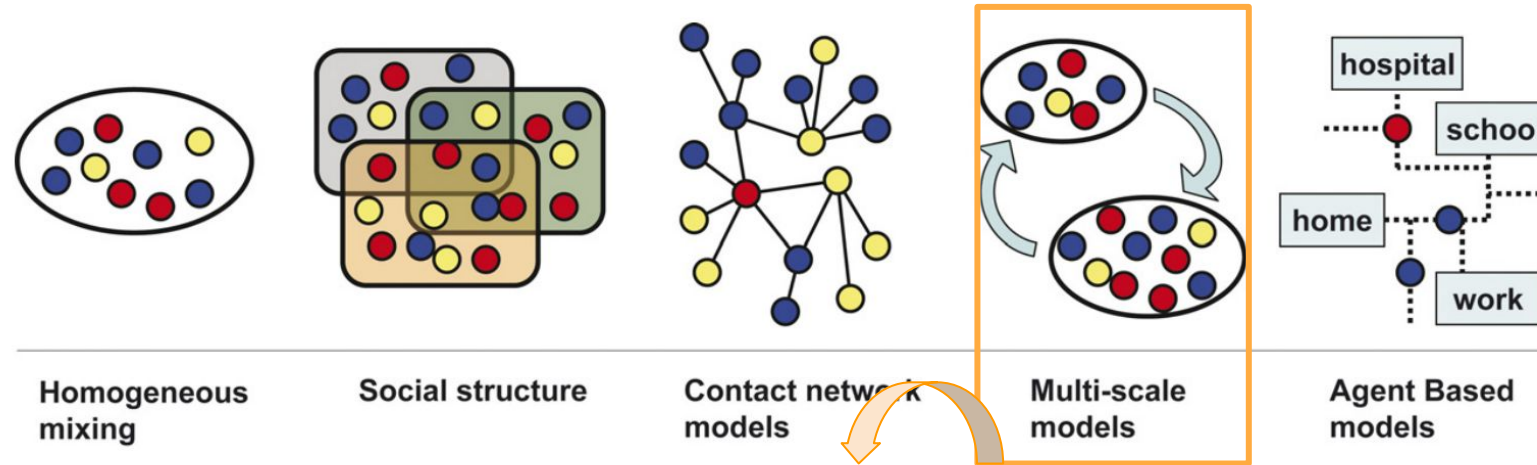
Real-World Use Case: *Identifying and managing super-spreader events.* The model identifies individuals with many connections (hubs) who could become super-spreaders. It helps simulate:

- What happens if these hubs isolate?
- How fast the outbreak could spread through friend networks

Model Type:

- Graph-based models
(nodes = individuals, edges = interactions)
- Network epidemiology

Different modelling approaches



Multi-scale models: integrate multiple levels of interaction, such as individual, group, and population levels, and consider the effects across these different scales.

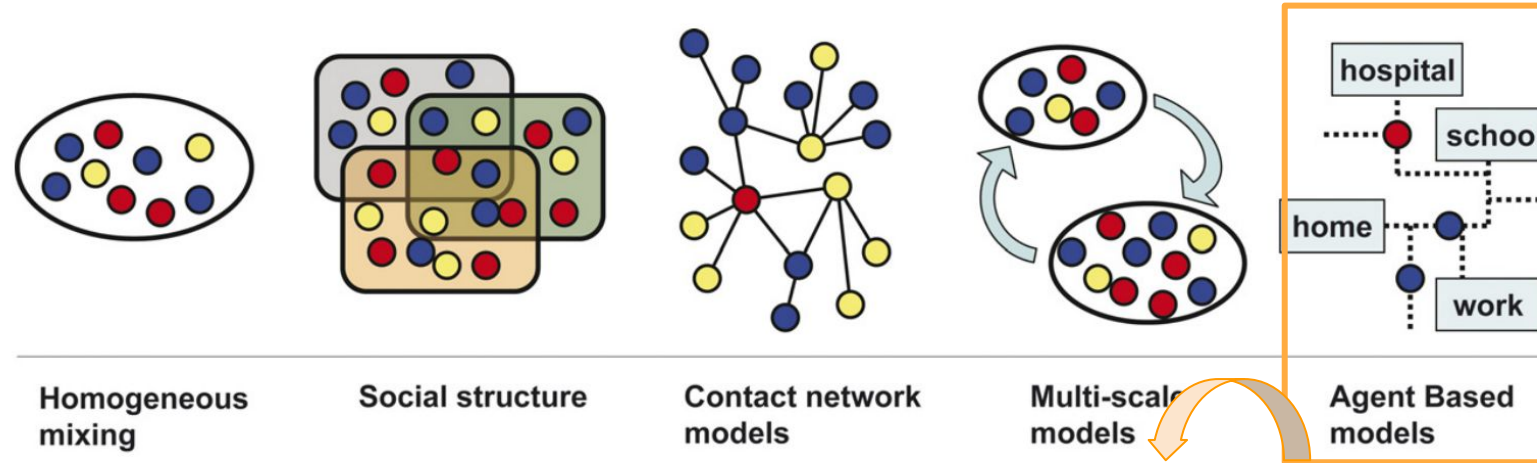
Real-World Use Case: *Understanding national-level spread and regional control strategies.*

- People interact at home, work, school
- Movement happens between cities (commuting, travel)
- Local healthcare capacity varies by region

Model Type:

- Hybrid models combining ODEs, networks, and spatial interactions
- Useful for modeling regional or national dynamics with local resolution

Different modelling approaches



Agent-based models: simulate the actions and interactions of individual agents (such as people or organizations) within specific environments. They consider different locations or settings (e.g., city, neighborhood, hospital, school, work, home) and model how agents move between these locations and interact based on defined rules.

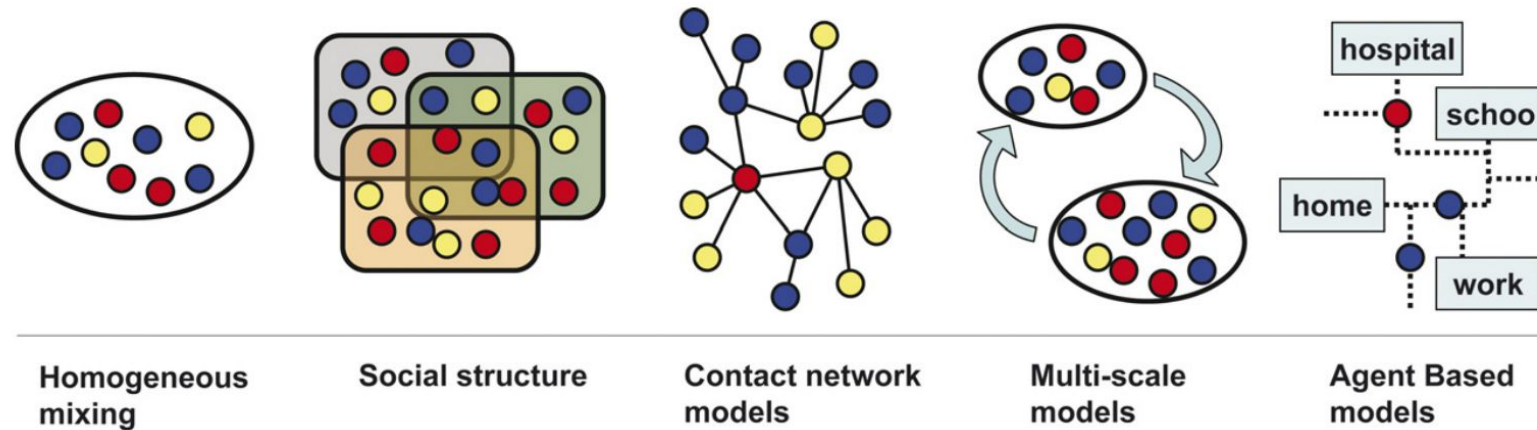
Real-World Use Case: *Testing the effectiveness of behavioral and spatial interventions.* The model explores:

- What if 60% of people wear masks?
- What if gyms close but offices remain open?
- How does work-from-home affect the spread?

Model Type:

- Individual-based simulation
- High computational cost but rich insights
- Allows for testing interventions like mask mandates

Macro- and micro-level models



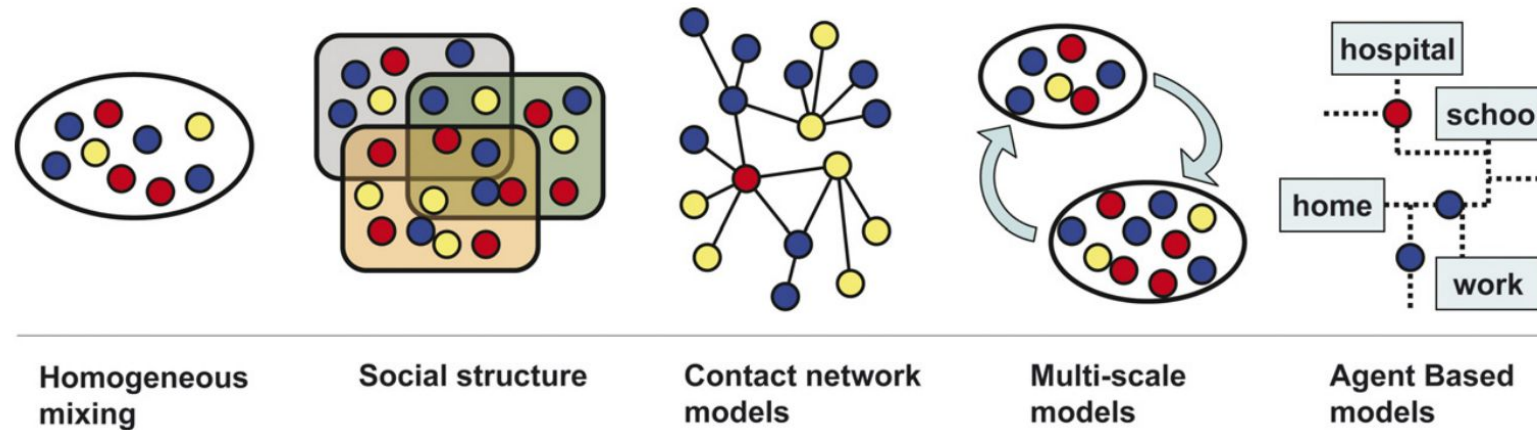
Macro-level Abstraction Micro-level

- Require ***fewer data*** and computational resources.
- Allow ***fast simulations on large populations***.
- Useful for broad forecasts and general policy decisions.
- Capture ***individual behaviors*** and interactions.
- Allow detailed, ***context-specific simulations***.
- Ideal for testing targeted interventions.



Pros

Macro- and micro-level models



Pros

- Require ***fewer data*** and computational resources.
- Allow ***fast simulations on large populations***.
- Useful for broad forecasts and general policy decisions.



Cons

- ***Do not capture individual behavior*** or local dynamics.
 - ***Assume uniform*** or simplified mixing ***patterns***.
 - Less accurate in heterogeneous or small-scale settings.
- Capture ***individual behaviors*** and interactions.
 - Allow detailed, ***context-specific simulations***.
 - Ideal for testing targeted interventions.

What do you expect from this tutorial?

1. Can we estimate the impact of interventions like lockdowns or vaccines?
2. How do individual behaviors affect the epidemic?
3. What's the trade-off between simplicity and detail in a model?
4. When should we use an macro model, and when is a micro model better?

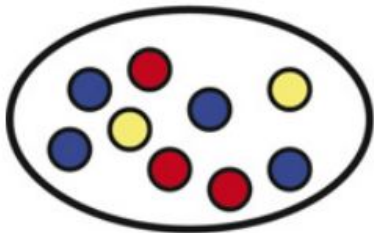


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Macro-level models



Macro-level models



The first major class of modeling approaches: **Macro-Level Models**, often called **compartmental models**.

They divide the entire population into **groups**, for example:

- Susceptible
- Infected
- Recovered

The beauty of these models is in their **simplicity**. They help answer questions like:

- *How fast will the outbreak grow?*
- *What happens if we vaccinate 50% of the population?*



We assume the population is well mixed (like stirring soup) where **everyone has an equal chance of interacting with everyone else.**



So while they don't give detailed answers about individual behavior, they're **powerful for seeing the big picture.**

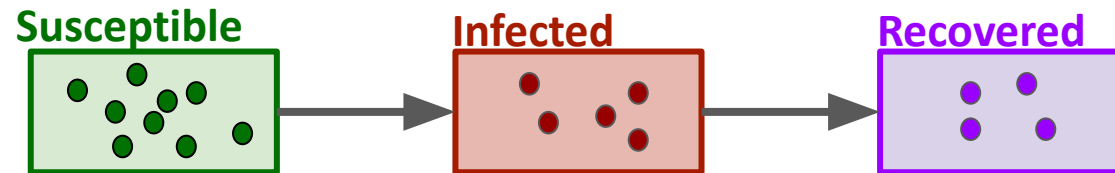


Compartmental model

Almost everyone starts **susceptible** (S).

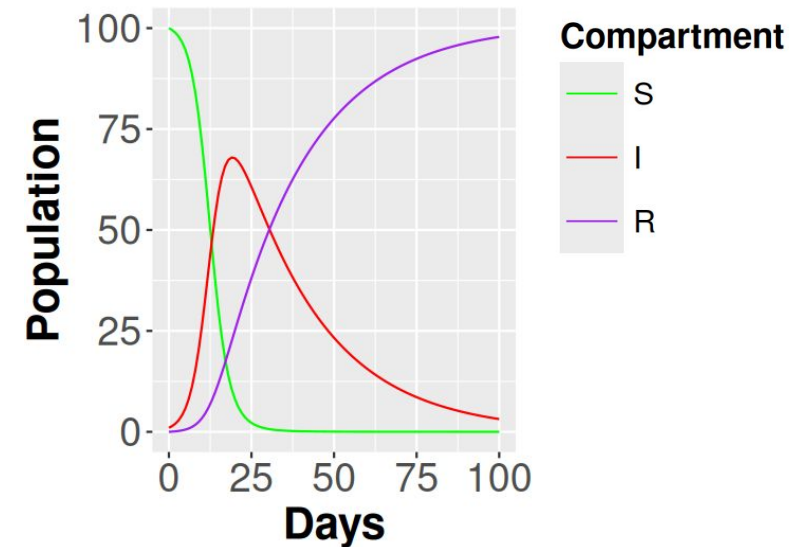
When they come into contact with an infected person, they can become **infected** (I).

After some time, they **recover** (R) and are no longer infectious.

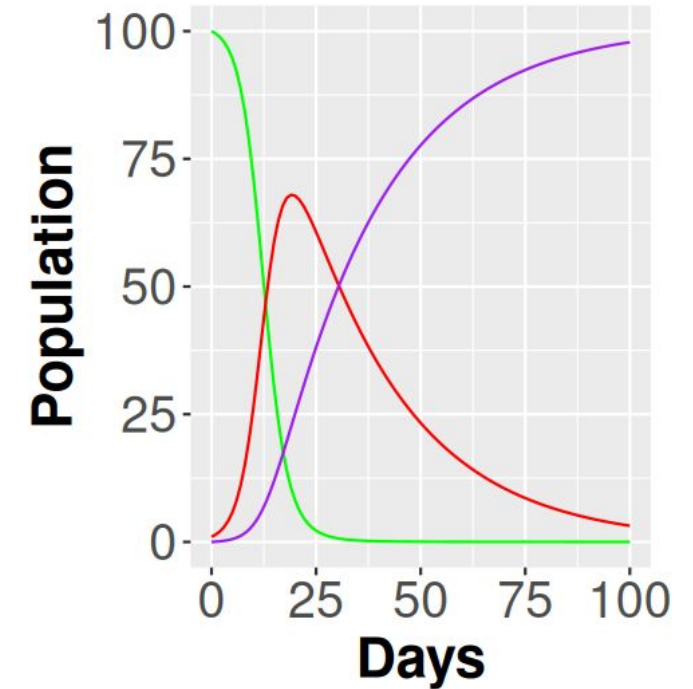
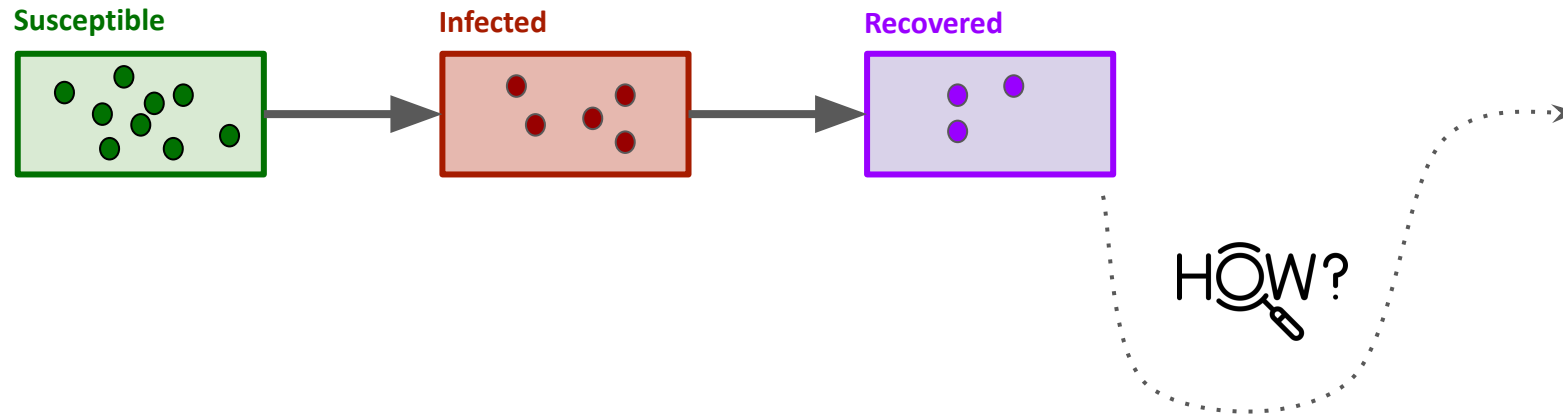


Individuals can move from the S compartment to the I compartment, but not back. They can also move from the I compartment to the R compartment, but not back. The rates at which these movements between compartments occur determine the **dynamics of the model**.

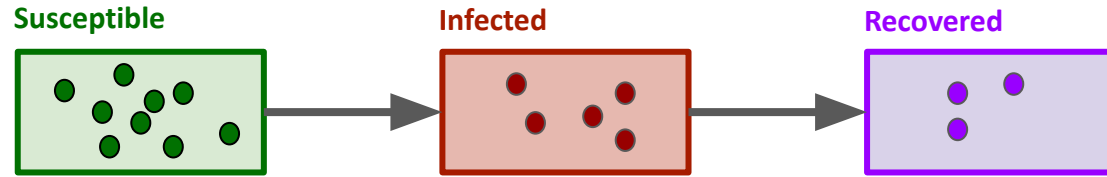
SIR compartmental model as a flow diagram, that helps us visualize **what we're modeling**



The mathematical processes underlying



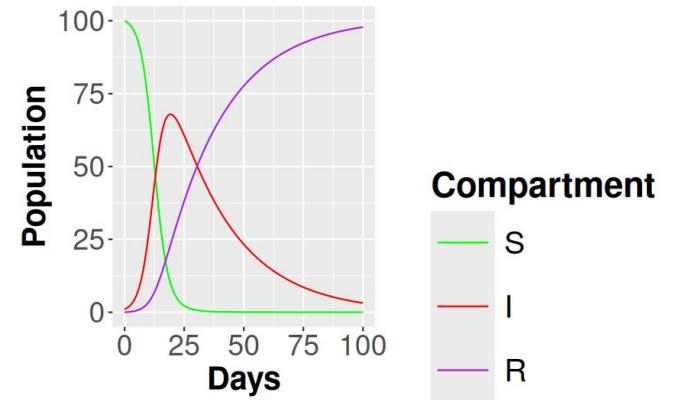
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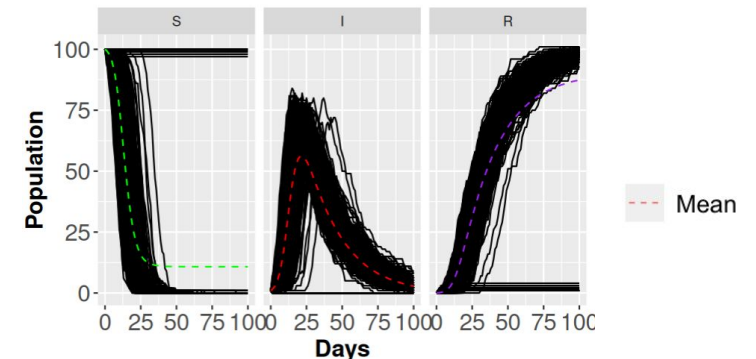
Once we have a flow diagram we can go in **two directions**:



Deterministic approach: we assume the population is large enough that randomness averages out. This leads to **ordinary differential equations (ODEs)**



Stochastic approach: we recognize that especially in small populations or early outbreak stages, random events matter. This leads to **Continuous Time Markov Chain (CTMC)**.



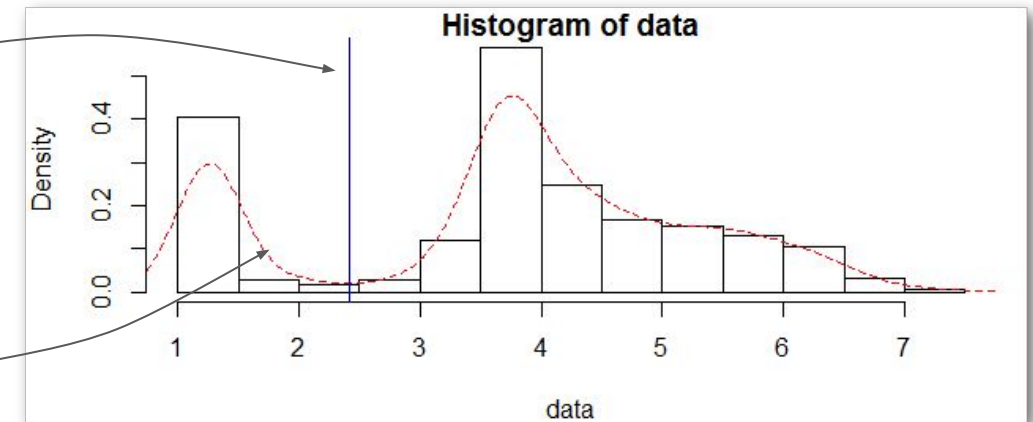
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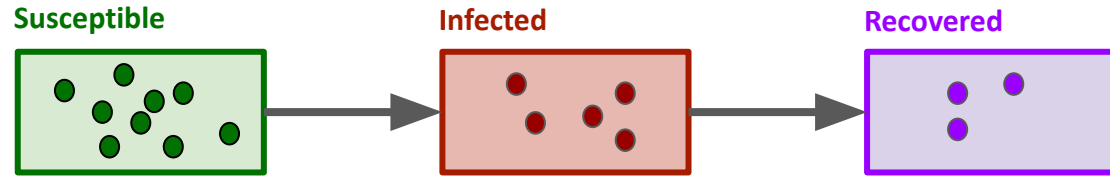
Deterministic	Stochastic
Ordinary differential equations	Continuous-time Markov chains
Future is “predictable” given present knowledge	Includes randomness; every simulation is different
Wide range of techniques available for analysis	Not as many techniques for analysis; often rely on simulations
Good for large number of individuals; qualitative analysis	Better for simulating dynamics with small numbers of individuals
Represents population average	Represents population variability

Deterministic models can be solved easier, but they provide only the average system behavior.

Stochastic models are more computationally demanding, but they can capture the stochastic nature of a biological process.



Stochastic approach



Events:

Infection:

$(S, I, R) \rightarrow (S-1, I+1, R)$
with rate β

Recovery:

$(S, I, R) \rightarrow (S, I-1, R+1)$
with rate γ

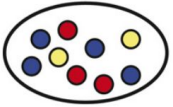
The system is in a discrete state:

$(\#S, \#I, \#R)$

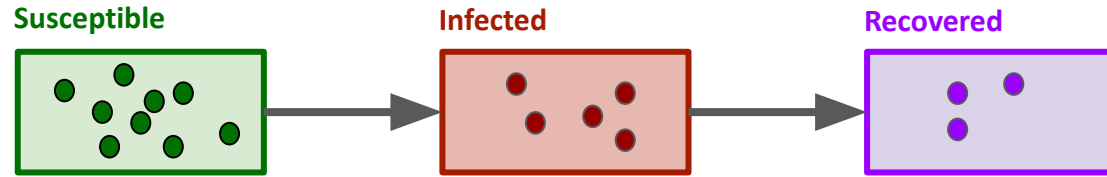
where $\#S$, $\#I$, and $\#R$ are **integer counts** of individuals in each compartment.

In the example, the state is:

$(\#S = 8, \#I = 5, \#R = 4)$



Stochastic approach



Two possible
stochastic events:

Infection:
 $(S, I, R) \rightarrow (S-1, I+1, R)$
with rate β

Recovery:
 $(S, I, R) \rightarrow (S, I-1, R+1)$
with rate γ

The system is in a discrete state:

$(\#S, \#I, \#R)$

where $\#S$, $\#I$, and $\#R$ are **integer counts** of individuals in each compartment.

In the example, the state is:

$(\#S = 8, \#I = 5, \#R = 4)$

These Kolmogorov equations (also called **master equations**) describe how those probabilities to be in a specific state change over time:

$$\frac{dP(S, I, R, t)}{dt} = \beta(S+1)(I-1)P(S+1, I-1, R, t) + \gamma(I+1)P(S, I+1, R-1, t) - [\beta SI + \gamma I]P(S, I, R, t)$$

$P(S, I, R, t)$ is the probability the system is in state (S, I, R) at time t

probability **into** state (S, I, R) from a transition **into** it (someone got infected in the previous state).

probability from someone recovering **subtracts** probability, because once someone gets infected or recovers, we leave the current state.

Stochastic approach



Kolmogorov equations:

$$\frac{d\pi(s_i, \nu)}{d\nu} = \sum_{s_k} \pi(s_k, \nu) q_{k,i}$$



One Kolmogorov equation per
state

State Space Explosion

$\pi(s_i, \nu)$ = probability to be in the state s_i at time ν

$q_{k,i}$ = rate from s_k to s_i

- [1] Gillespie (1977) *The journal of physical chemistry*
- [2] Gillespie (2001) *The journal of physical chemistry*
- [3] T.G. Kurtz (1970) *Journal of Applied Probability*

Stochastic approach



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One Kolmogorov equation per state

State Space Explosion



These equations are hard to solve directly for large systems, but they form the foundation of stochastic simulations like **Gillespie's algorithm**.



Stochastic Simulation Algorithm [1]:
exact algorithm to simulate the events
that might occur in the system

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Stochastic Simulation Algorithm [1]:
exact algorithm to simulate the events that
might occur in the system



Longer simulation with an
increasing number of events!!



τ -leaping algorithm [2] a
Poisson approximation to leap
over many fast events

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Stochastic approach

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One Kolmogorov equation per state

State Space Explosion

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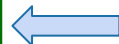
Stochastic Simulation Algorithm [1]:
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Longer simulation with an increasing
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τ -leaping algorithm [2] a Poisson
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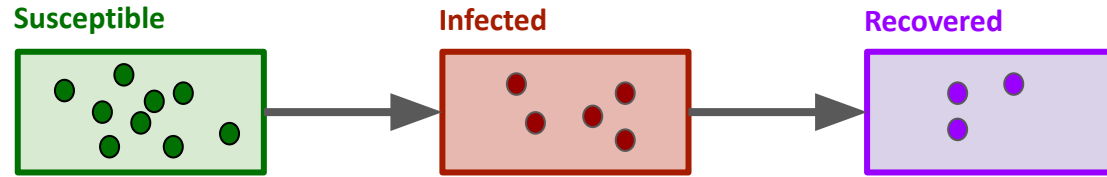


Deterministic Approximation

Mean field analysis (or fluid
approximation) [3]

- [1] Gillespie (1977) The journal of physical chemistry
- [2] Gillespie (2001) The journal of physical chemistry
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Deterministic approximation



Kolmogorov equations:
$$\frac{d\pi(s_i, \nu)}{d\nu} = \sum_{s_k} \pi(s_k, \nu) q_{k,i}$$

$\pi(s_i, \nu)$ = probability to be in the state s_i at time ν
 $q_{k,i}$ = rate from s_k to s_i



Deterministic Approximation

$$\frac{dX_i^N(\nu)}{d\nu} = f(X^N(\nu), \nu)$$

$X_i^N(\nu)$ Number of elements in each variable i -th at time ν

Number of Kolmogorov equations

=

Number of all the possible states of the SIR model that can be reached from a specific initial state

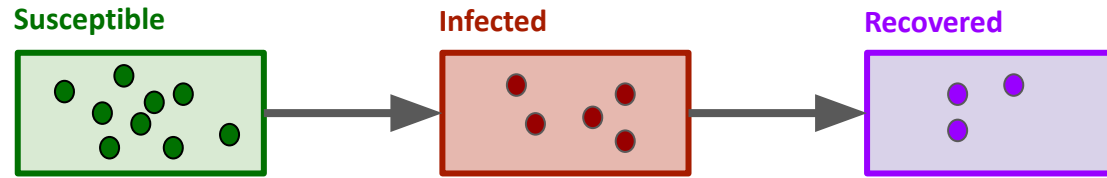
If the initial state is
(#S = 8, #I = 5, #R = 4)
we have more than 60 states



One Ordinary Differential equation (ODE) per variable that characterizes the state!

The state (#S, #I, #R): 3 Variables!

Deterministic approach

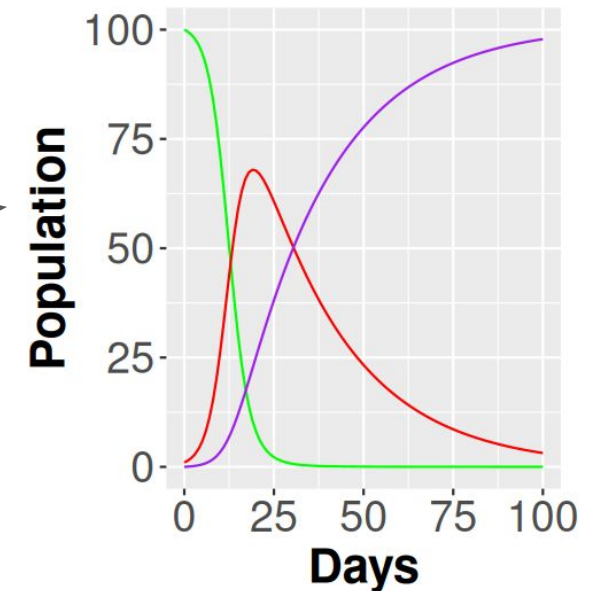


People are leaving the susceptible compartment

People are recovering and moving to the R compartment

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Solving
(integrating)
the equations



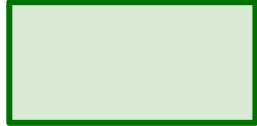
β : the transmission rate — how often a susceptible and infected person interact

γ : the recovery rate — how fast people leave the infected state

Several compartment models



Susceptible



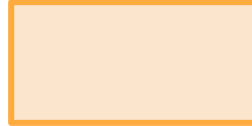
Infected



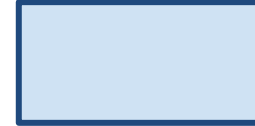
Recovered



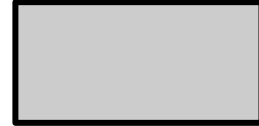
Exposed



Quarantine



Death



.....

SEIQR Model (Susceptible-Exposed-Infected-Quarantined-Recovered)

- **Used for:** Diseases with **quarantine or isolation policies**

SIRV Model (Susceptible-Infected-Recovered-Vaccinated)

- **Used when** vaccination is implemented.

SAIR Model (Susceptible-Asymptomatic-Infected-Recovered)

- **Used for:** Diseases with **asymptomatic carriers**

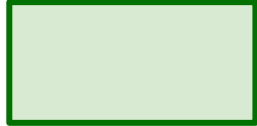
SEIRD Model (Susceptible-Exposed-Infected-Recovered-Deceased)

- Combines latency (**E**) and death (**D**) aspects. Realistic for diseases like **Ebola or COVID-19**

Several compartment models



Susceptible



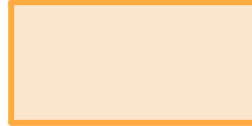
Infected



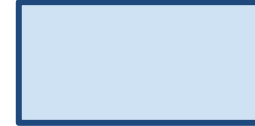
Recovered



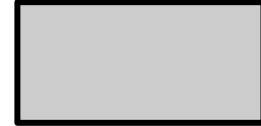
Exposed



Quarantine



Death



.....

SEIQR Model (Susceptible-Exposed-Infected-Quarantined-Recovered)

- Used for: Diseases with **quarantine or isolation policies**

SIRV Model (Susceptible-Infected-Recovered-Vaccinated)

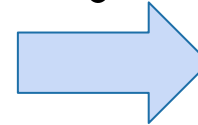
- Used when vaccination is implemented.

SAIR Model (Susceptible-Asymptomatic-Infected-Recovered)

- Used for: Diseases with **asymptomatic carriers**

SEIRD Model (Susceptible-Exposed-Infected-Recovered-Deceased)

- Combines latency (**E**) and death (**D**) aspects. Realistic for diseases like **Ebola or COVID-19**



*It could be useful to have a **general modelling framework** to simulate the models easily!!!!*



Graphical User Interface:

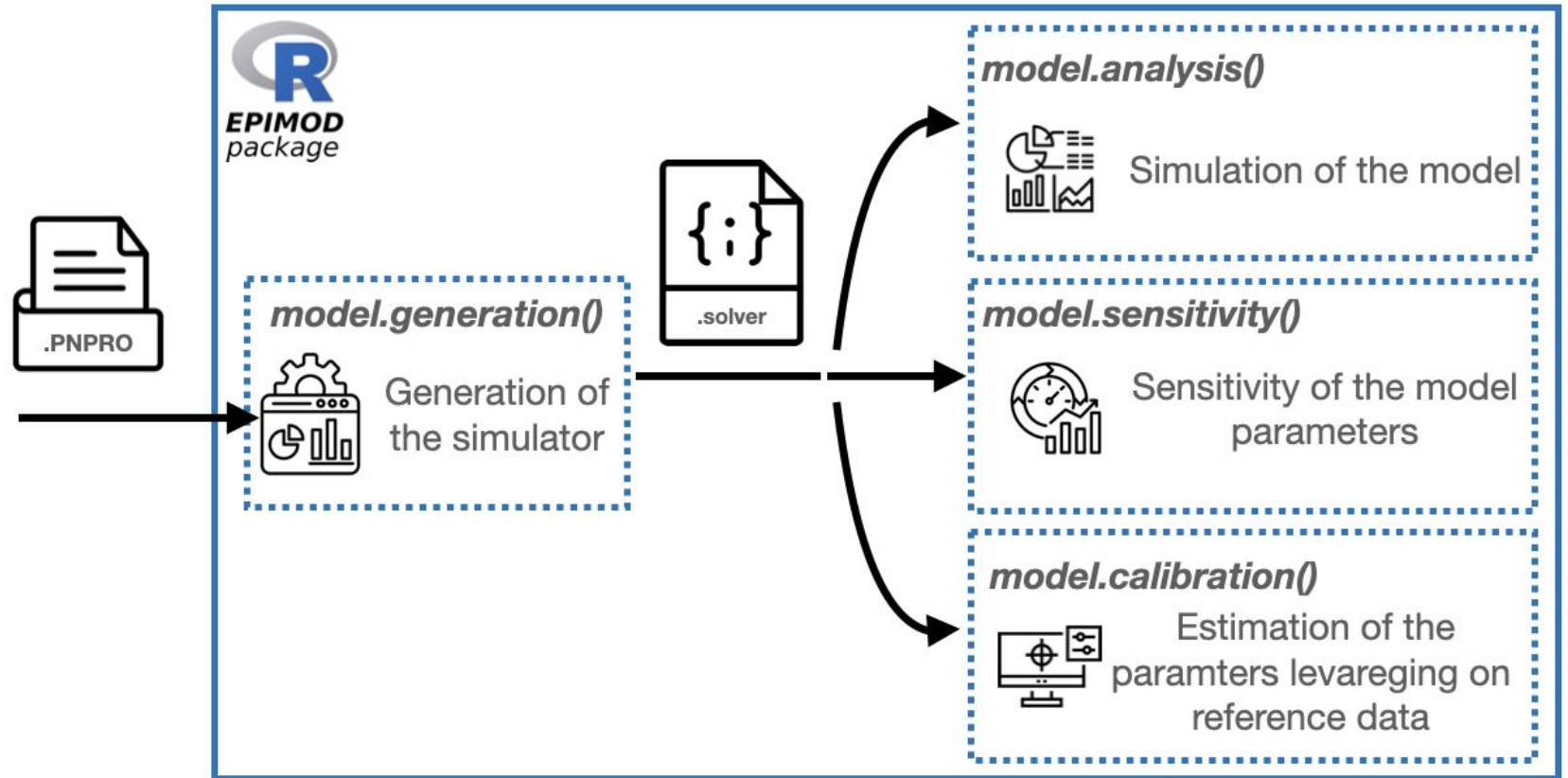
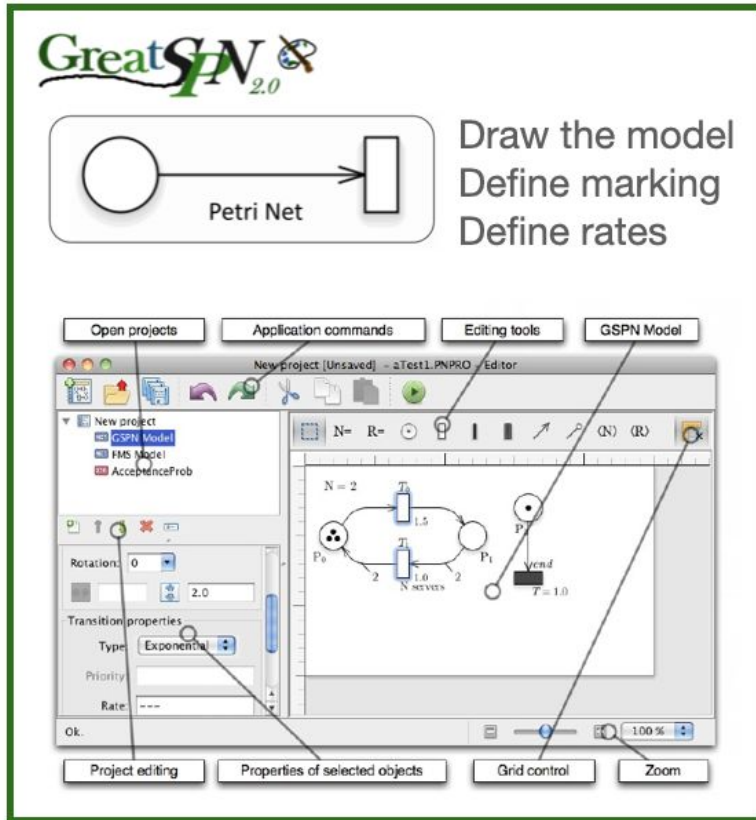
- Interactive modeling via Petri Net formalism
- Construction and visualization of system dynamics
- User-friendly interface for dynamic system modeling

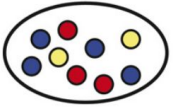
Epimod Library:

- Simulations of complex systems
- 'What-if' scenario analysis
- Detailed result checking (plots, data visualization)

Docker Containers:

- Simulations reproducibility
- Simplifies deployment, updates, and maintenance





Petri Net: a graphical formalism to draw model

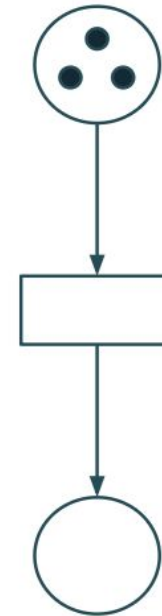
Petri Net (PN) is graphical formalism (bipartite graph), conveniently used for the analysis of complex models. It allows us to derive qualitative and quantitative properties of the system.

Places (circles): Represent *states* or *conditions* (e.g., a molecule present, a person infected).

Transitions (rectangles): Represent *events* or *actions* that change the state (e.g., infection, recovery).

Tokens (dots inside places): Indicate the *current state* (how many resources or individuals are in a place).

Arcs (arrows): Show how places and transitions are connected (direction matters).





Arcs Multiplicity:

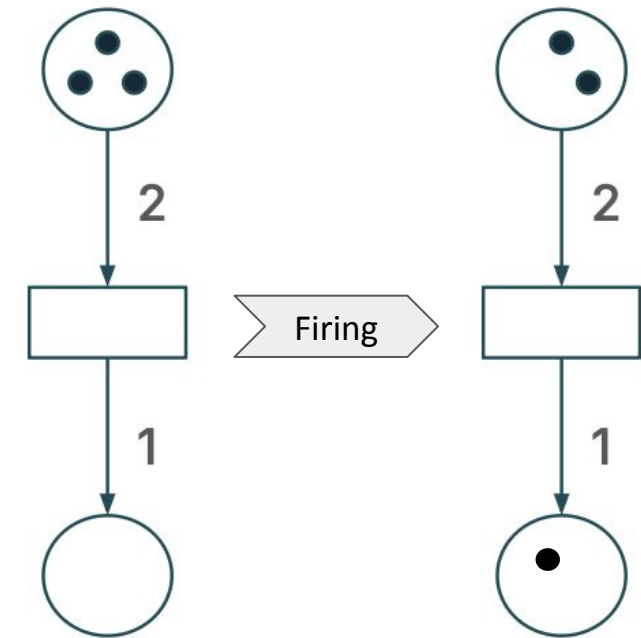
- Arcs can have multiplicities, indicating the number of tokens transferred between places and transitions.

Transitions Firing:

- A transition 'fires' when the input places contain enough tokens based on the arc multiplicities.
- Firing consumes tokens from input places and produces tokens in output place

Firing Rate:

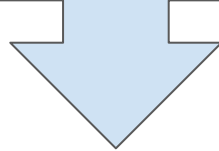
- Determines how quickly a transition converts input tokens into output tokens



Stochastic Petri Net



Exponentially distributed random delays are associated with transition firings (time to wait before the tokens are removed from/ added to places).

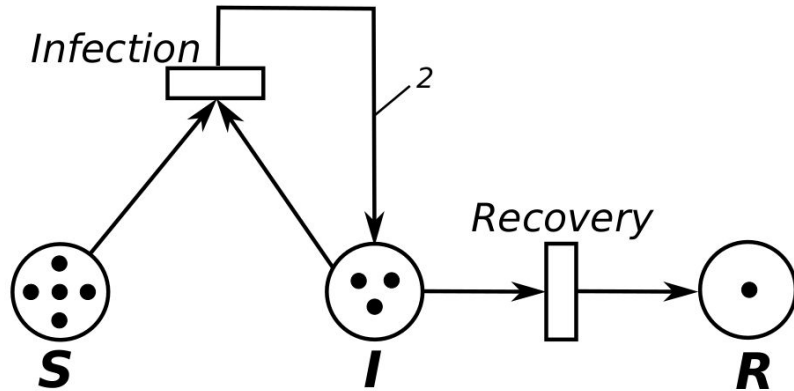


The **introduction of time** allows to model the **temporal dynamics** of the system

The process representing the dynamic of the SPN model is:

Continuous Time Markov Chain

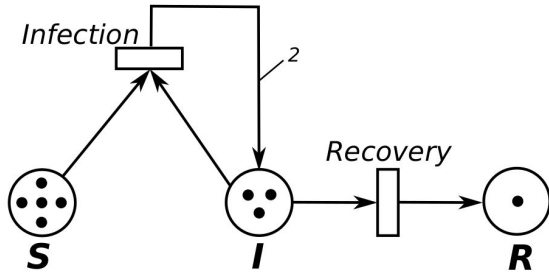
Some examples



SIR model,
one of the simplest
compartmental models

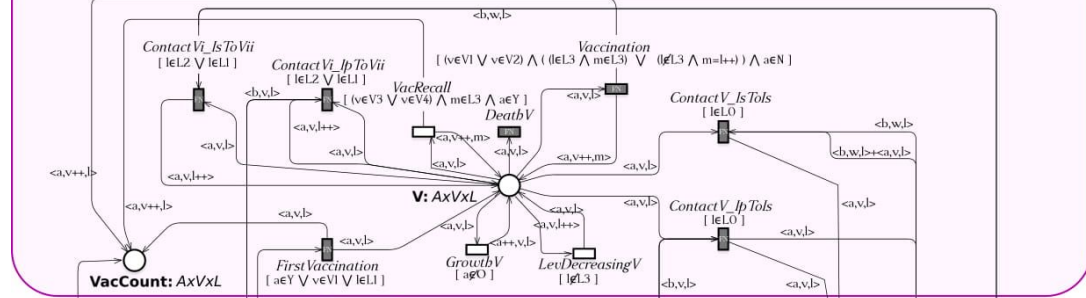


Some examples

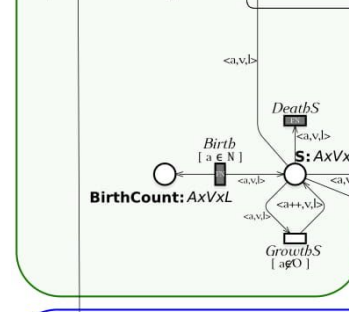


Investigation of the pertussis epidemiology in Italy

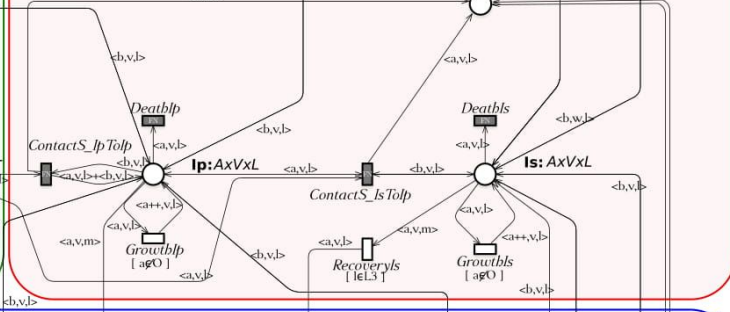
4) Under vaccination



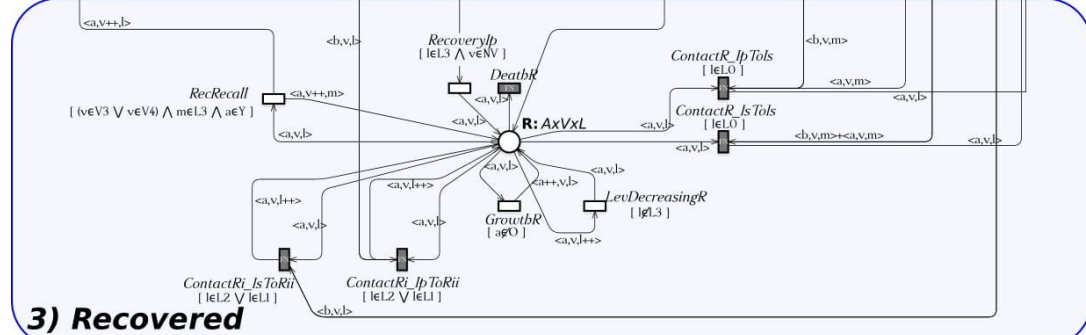
1) Susceptible



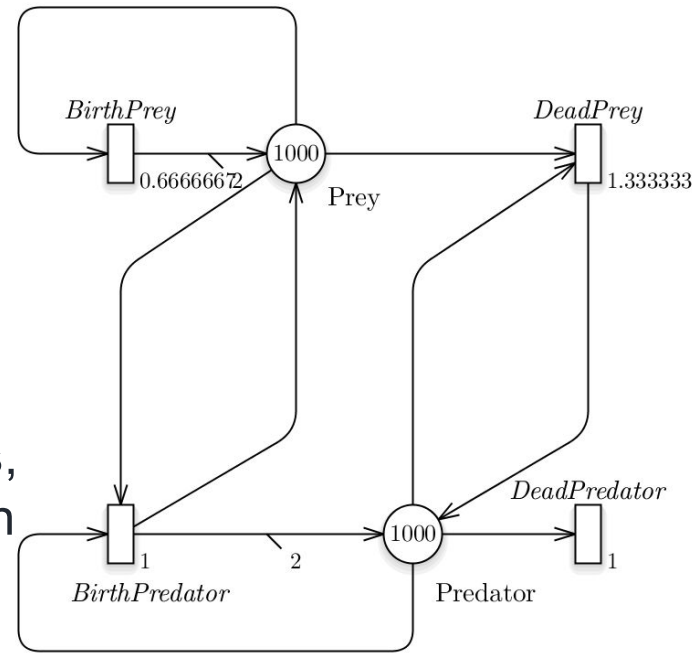
2) Infected



3) Recovered



SIR model,
one of the simplest
compartmental models



Predator-prey equations,
biological systems in which
two species interact



COVID-19 model

domain $A \times S = \text{Age} \times \text{Symptom}$

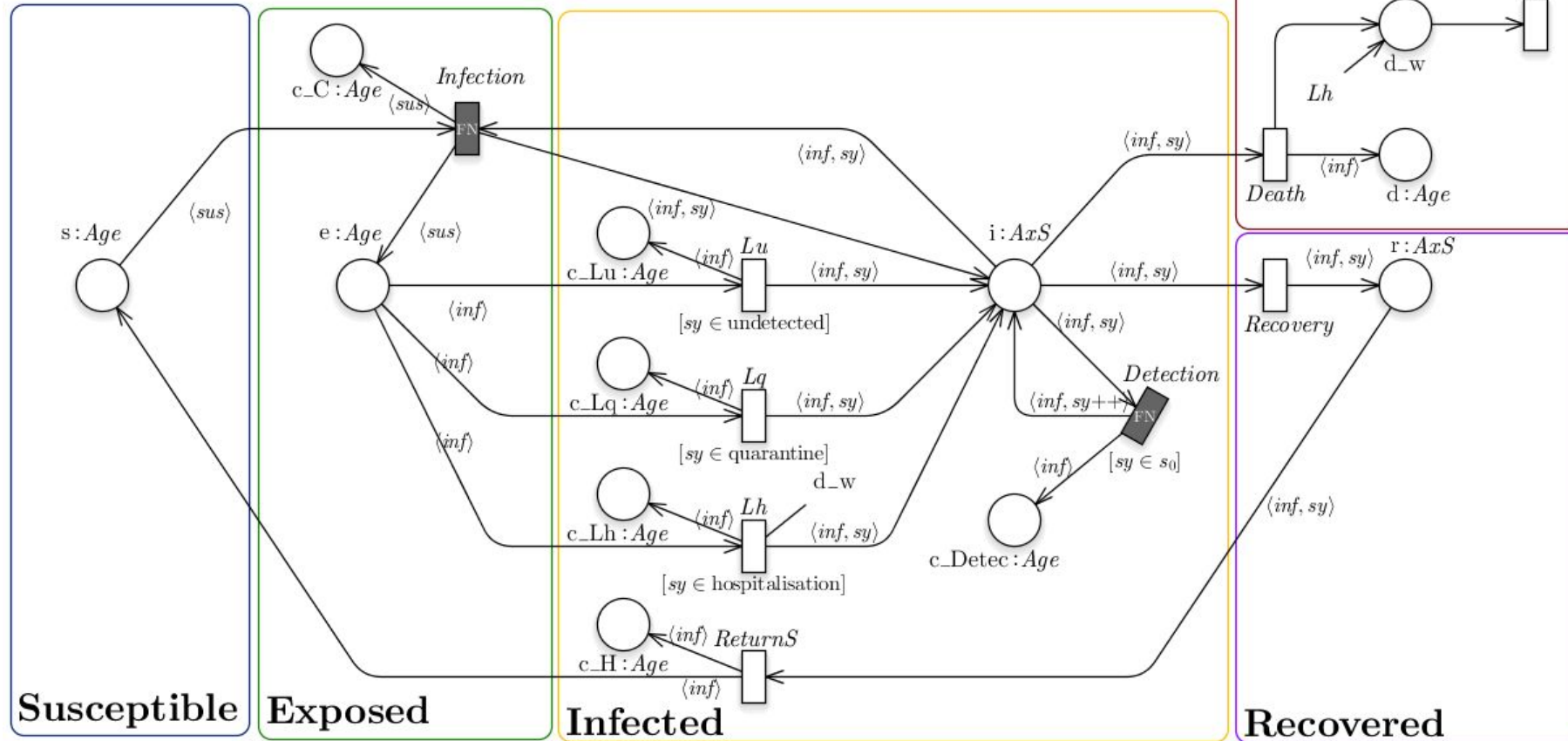
class *Symptom* = circular { s_0 } is undetected + { s_1 } is quarantine + { s_2 } is hospitalisation

class *Age* = { a_0 } is A_0 + { a_1 } is A_1 + { a_2 } is A_2

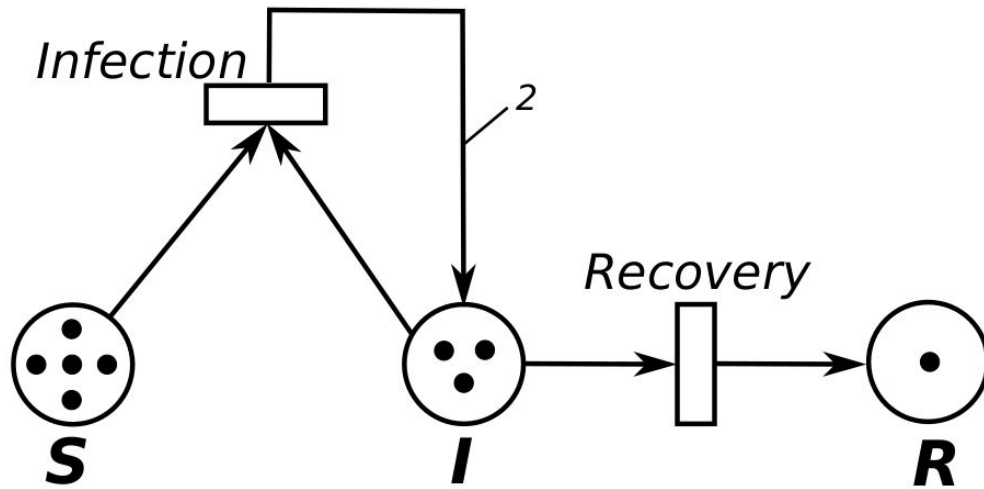
var *sus* : *Age*

var *sy* : *Symptom*

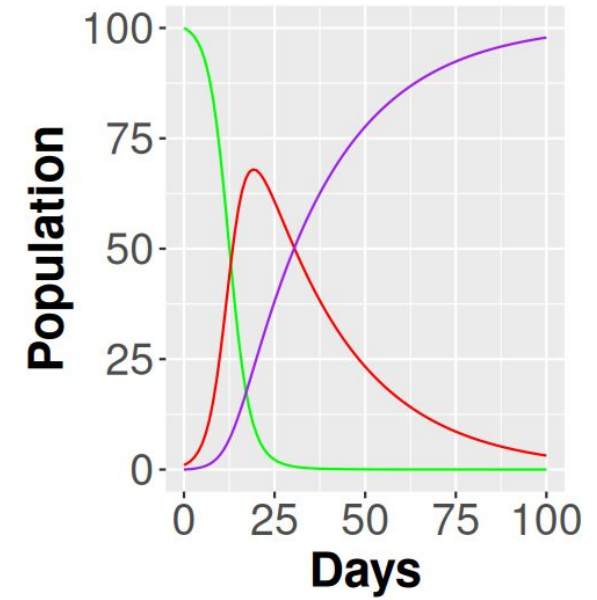
var *inf* : *Age*



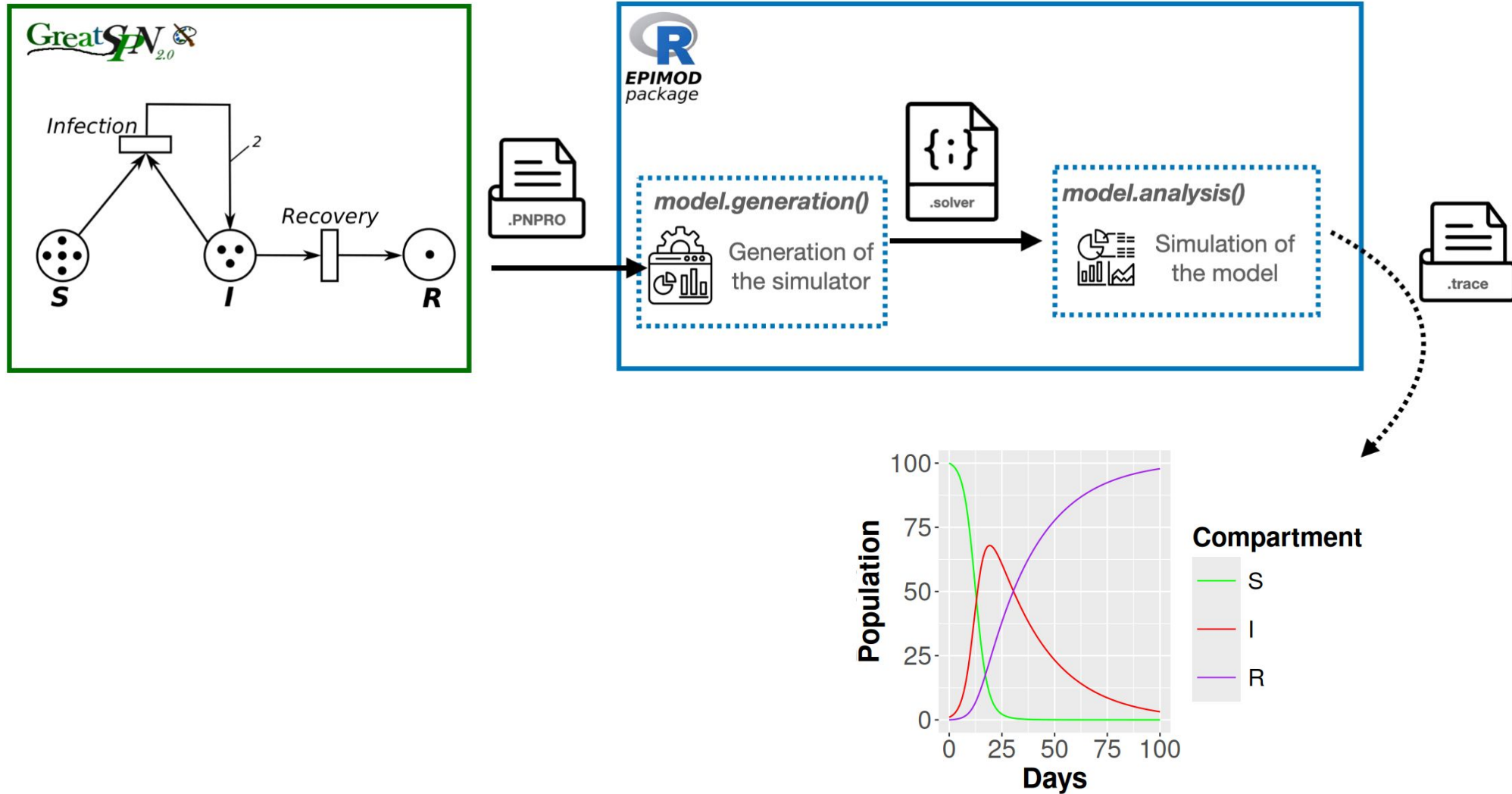
From graphics to dynamics



HOW?



From graphics to dynamics



Epimod

- General modeling framework for the analysis of epidemiological/biological systems;
- Tool easily accessible by any researcher even without advanced mathematical and computational skills.

Main features:

- **Graphical formalism:** you don't have to write equations, you draw them!
- **Modular:** you can shuffle the components to get what you need
- **R package:** no configuration is needed, you just install it
- **Virtualization with Docker:** the framework runs on on your laptop and rocks on your data-center

Environment and Analysis

R library



Model Generation
Sensitivity Analysis
Model Calibration
Model Analysis



Docker containers



Generation: deterministic/stochastic process
Sensitivity: PRCC and ranking
Calibration: optimization problem
Analysis: simulations and what-if analysis

Docker containerization



- Reproducibility is guaranteed by providing **Docker containers** packing all the required software and libraries frozen at a given version.
- Docker images are equivalent to Virtual Machines but ***Docker container provide an operating-system-level virtualization*** by abstracting the user space rather than the hardware, so that all the running docker images share the same kernel.
- Abstracting the user space guarantees better performance at the cost of minor portability (i.e., docker images built on Linux will not run on Windows systems.)

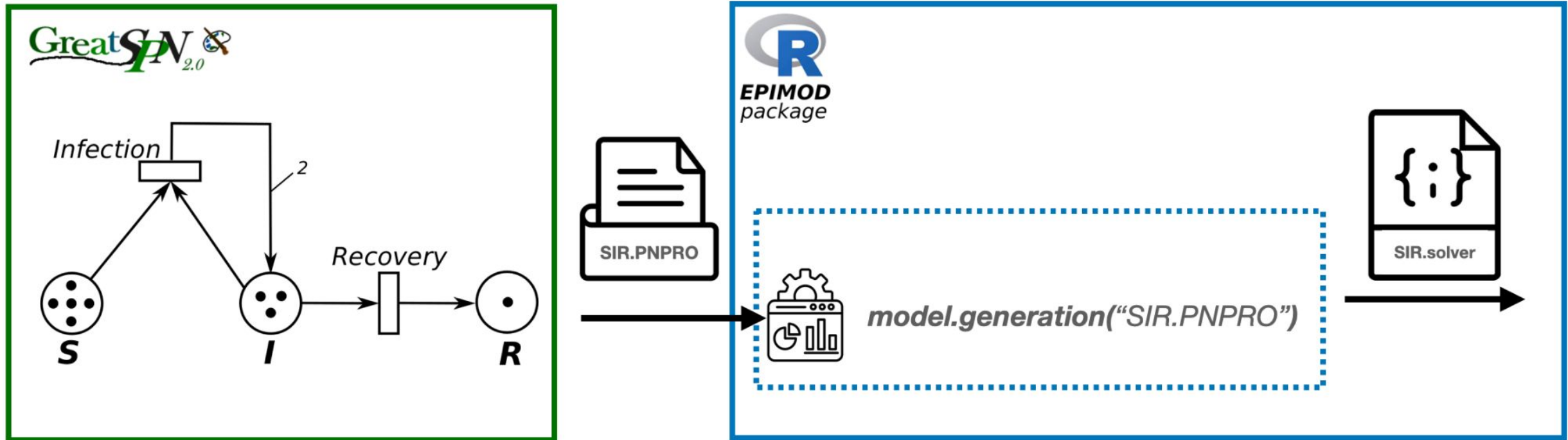
Ten Simple Rules for Reproducible Computational Research

1. For Every Result, Keep Track of How It Was Produced
2. **Avoid Manual Data Manipulation Steps**
3. Archive the Exact Versions of All External Programs Used
4. Version Control All Custom Scripts
5. Record All Intermediate Results, When Possible in Standardized Formats
6. For Analyses That Include Randomness, Note Underlying Random Seeds
7. Always Store Raw Data behind Plots
8. Generate Hierarchical Analysis Output, Allowing Layers of Increasing Detail to Be Inspected
9. Connect Textual Statements to Underlying Results
10. Provide Public Access to Scripts, Runs, and Results

Sandve et al. PLoS Comp Biol. 2013

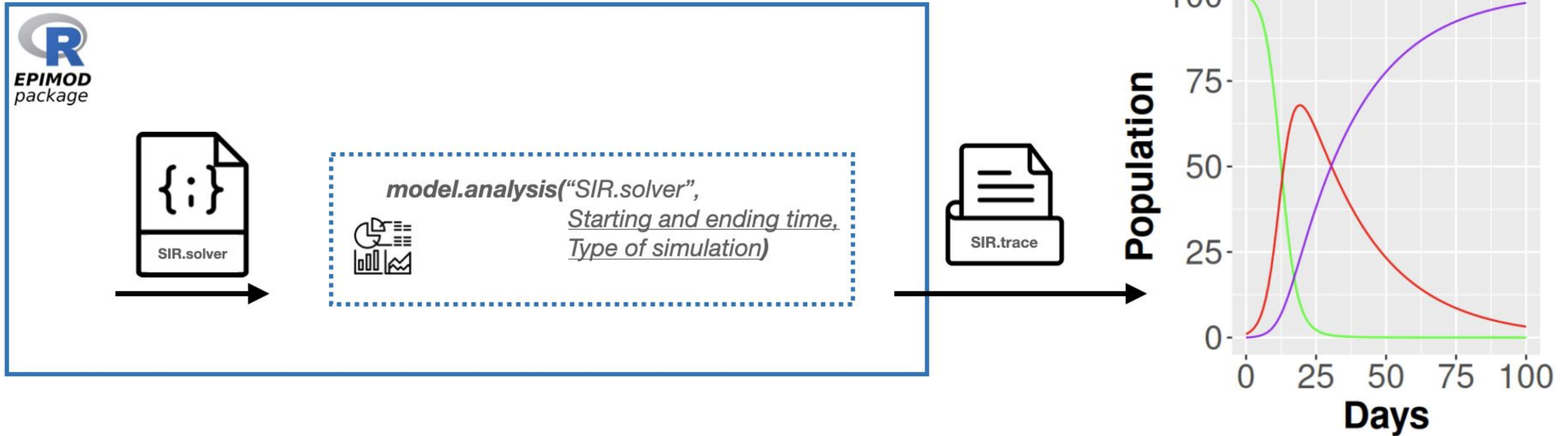
Model Generation

model_generation(): generate the implementation of the model (both deterministic and stochastic) from its graphical representation



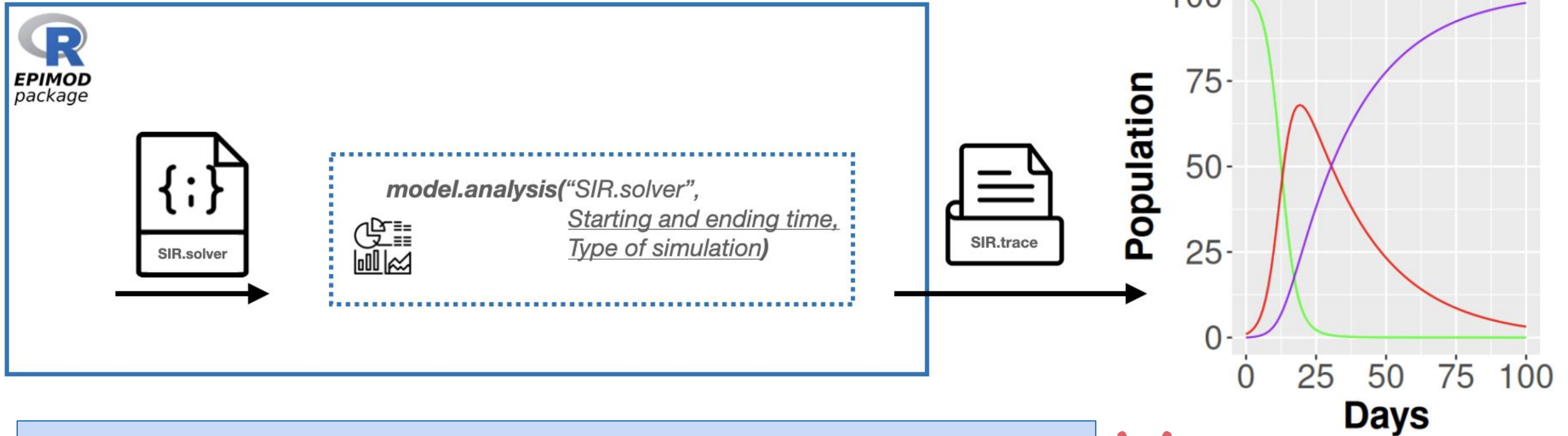
Model Analysis

model_analysis(): play with your brand new model



Model Analysis

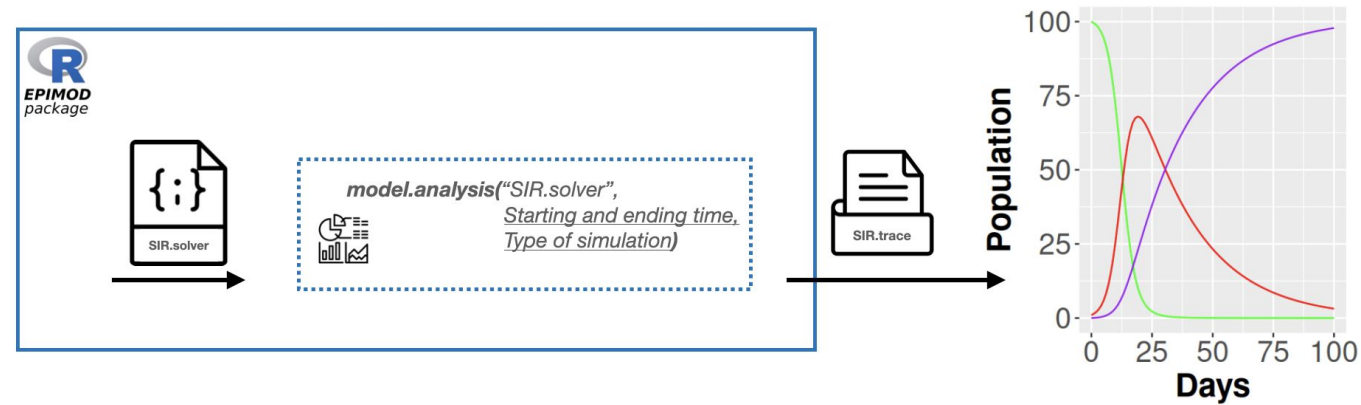
model_analysis(): play with your brand new model



The simulations depend on the parameters (initial state and transition rates) set from GreatSPN



Changing Model Parameters



The simulations depend on the parameters (initial state and transition rates) set from GreatSPN



Change parameters from GreatSPN

Change parameters from *model_analysis()*

Changing Model Parameters

The simulations depend on the parameters (initial state and transition rates) set from GreatSPN



Change parameters from *model_analysis()*



parameters_fname.csv

Describes what parameters vary without modifying the PNPRO file and how to define them

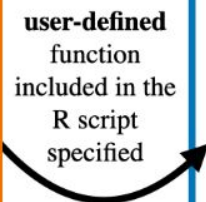
Tag & Meaning	Name	Value	Further Parameters
i= Complete initial marking	Usually <code>init</code>	Vector of numbers equal to the number of places or the name of a R function returning that vector	input parameters needed by the R function, which can be an R built-in function or a <i>user-defined function</i>
m= Initial marking of a specific place	Name of the place	An integer number or the name of a R function returning a integer number	
c= Constant rate for a transition	Name of the transition	An real number or the name of a R function returning a real number	
g= Rate for a general transition	Name from file in which will be saved the value	An real number or the name of a R function returning a real number	



functions_fname.R

Contains user-defined R functions used in parameters_fname and for analysis

Example	Purpose	Purpose	Notes
init_generation	Generates the initial marking values	parameters_fname (with tag i)	Initial conditions
target	Extracts the target values from model output (for PRCC calculation)	target_value	Selects one or more places to track over time.
mse	Computes the error/ distance between model output and reference data	distance_measure	For instance Mean Squared Error



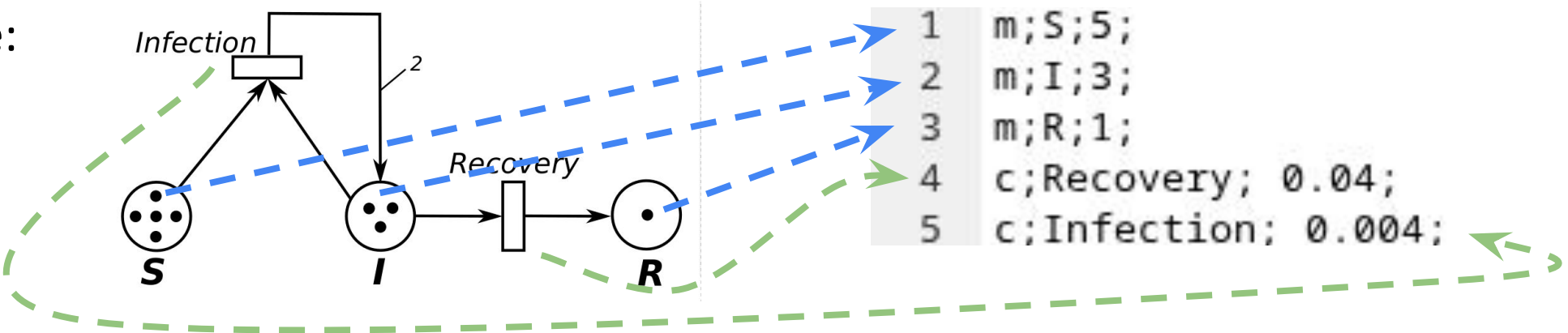
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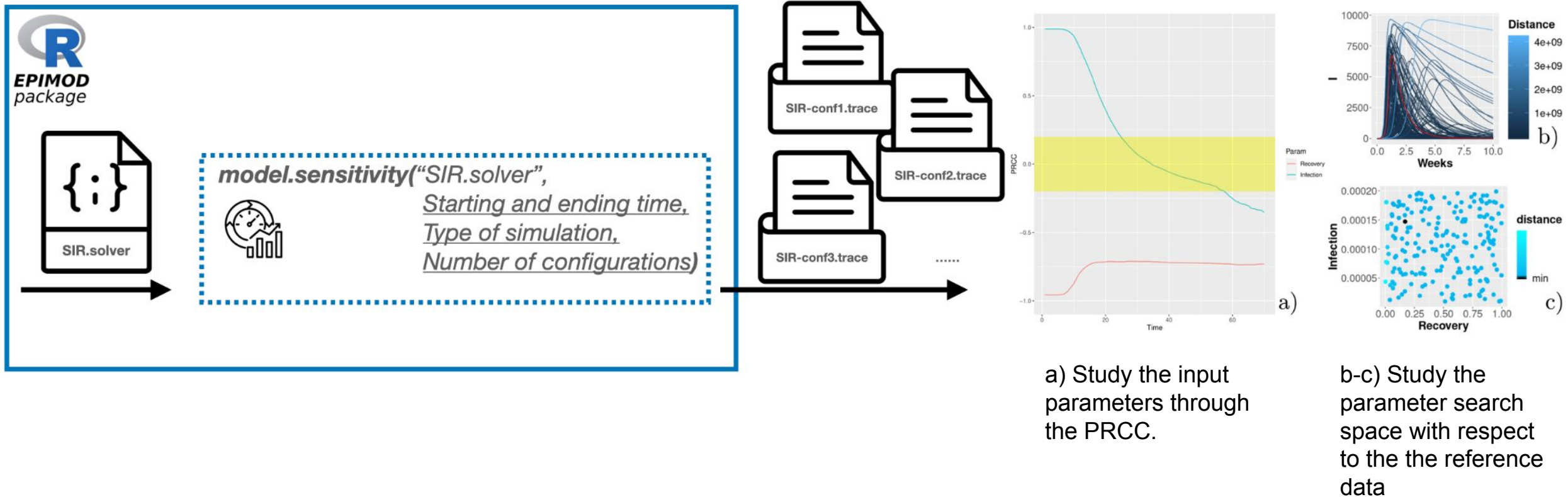
user-defined function included in the R script specified

Simple example:



Model Sensitivity

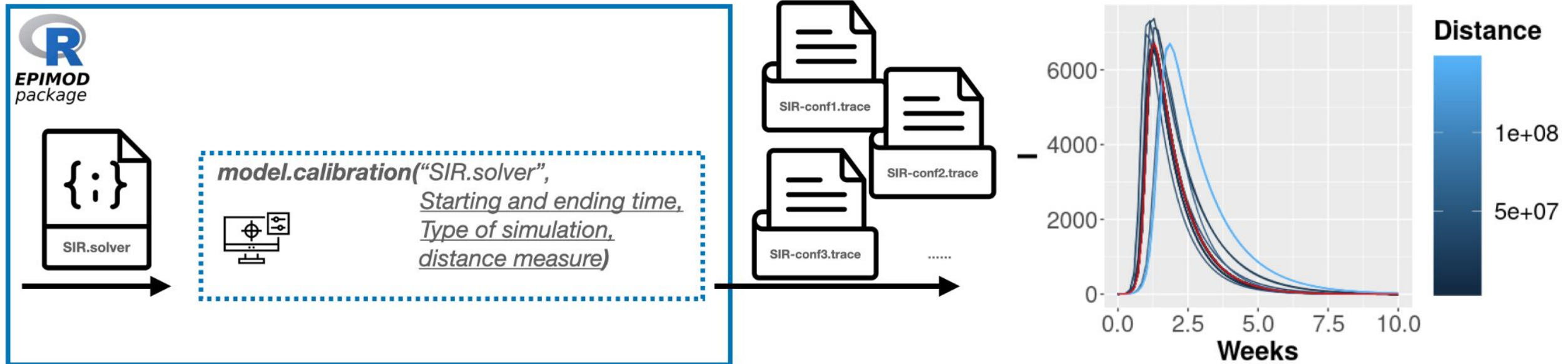
model_sensitivity(): identify the set of parameter that impact the most on the model outcome (dynamics) with the Partial Rank Correlation Coefficients



Model Calibration

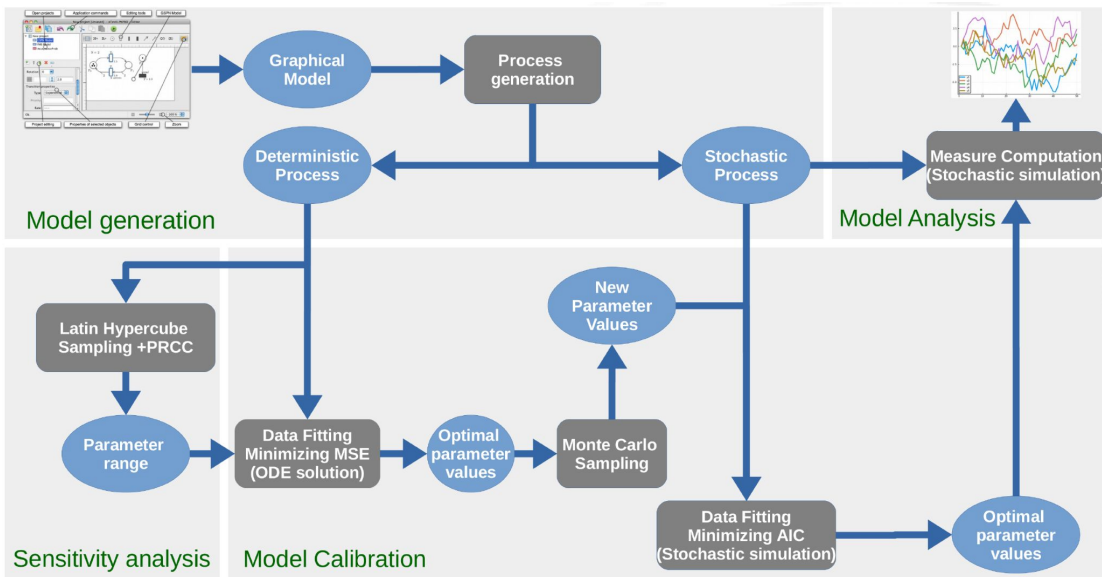
model_calibration(): estimate the parameters configuration which fit the best a given reference data.

Minimise the distance between the reference data and the model's output according to a given (user provided) distance metric.

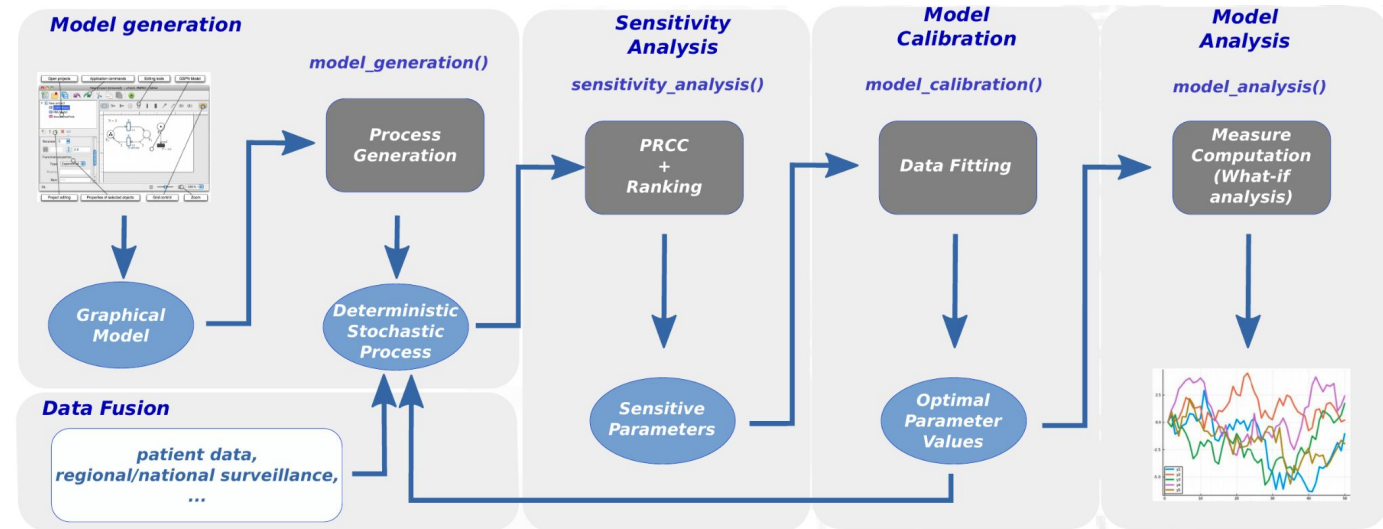


Epimod is modular

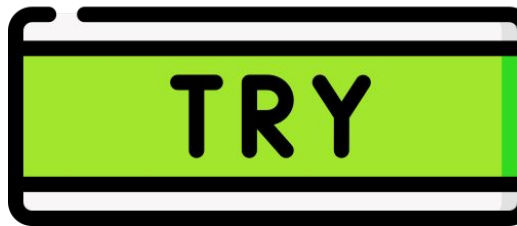
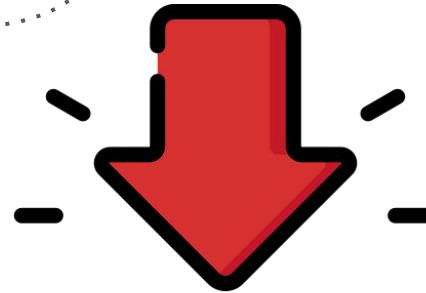
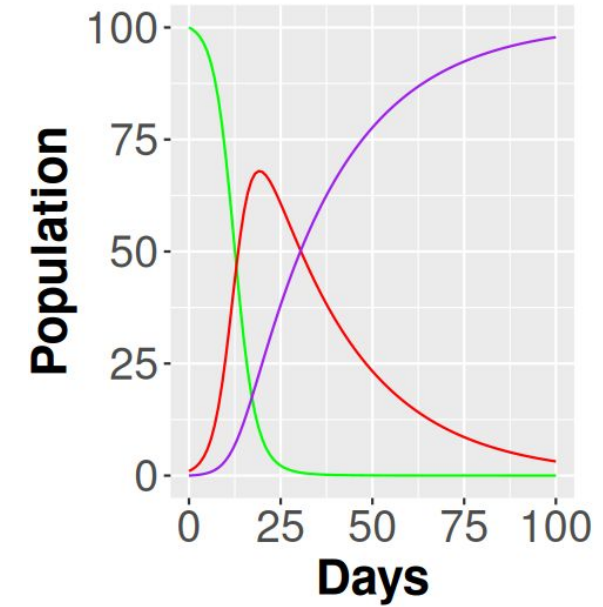
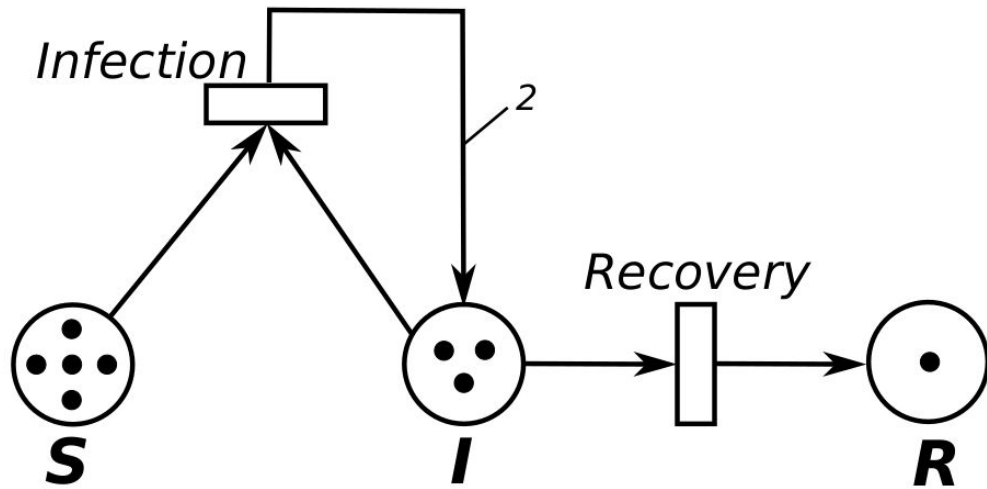
Analysis pipeline for Pertussis model



Analysis pipeline for COVID-19 model

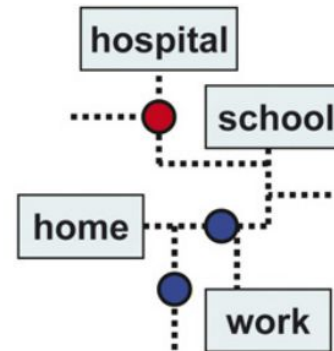


Case study: SIR model

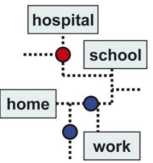




Micro-level models



Micro-level models



The second major class of modeling approaches: **Micro-Level Models**, often called **agent-based models (ABMs)**.

ABMs have two main components: agents and the environment. **Agents** represent individual people, each following their own rules, behaviors, and decision-making processes rather than being grouped. The **environment** provides the specific space or setting where these agents interact and carry out their decisions.

Some key features of these models include:

- Each agent can have **different** characteristics.
- Interactions depend on **who meets whom, when, and where**.
- Agents can move, change behavior, and react to their **environment**.
- From these individual interactions, **complex global patterns**, like waves of infection or clustering, can emerge over time.

The strength of these models lies in their **realism**. They help explore questions like:

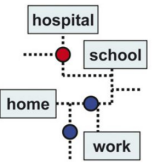
- *What if some people are super-spreaders?*
- *How do contact patterns affect the outbreak?*
- *Can we target specific groups or locations to slow the spread?*



We simulate interactions in detail, like a crowd of individuals, each making **their own decisions**.



While they're **more complex and slower to run**, they're **excellent for understanding detailed dynamics**.



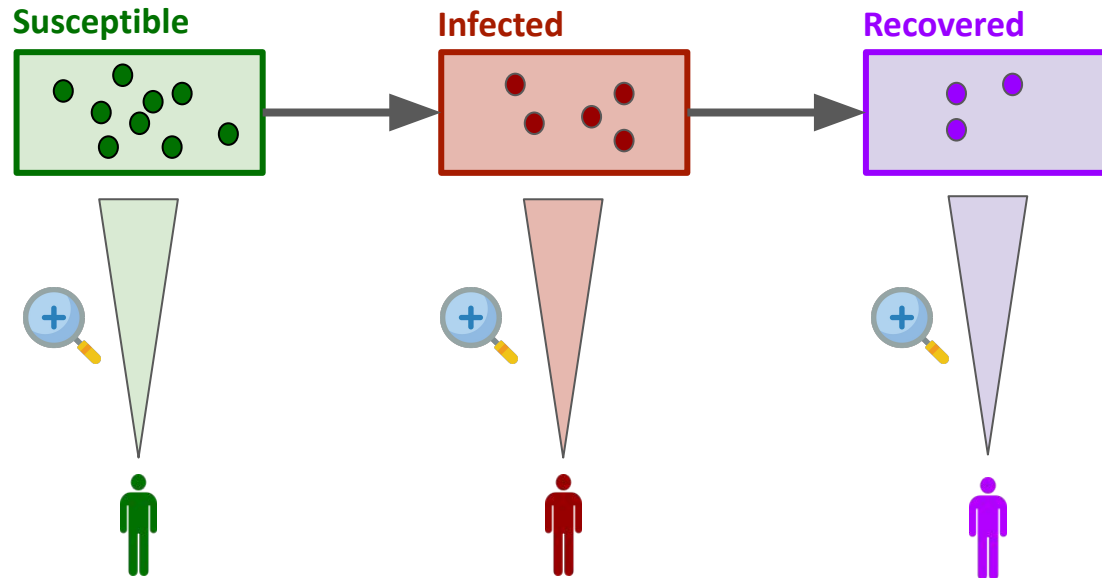
Agent-based models in infectious disease modeling



We simulate interactions in detail, like a crowd of individuals, each making **their own decisions**.



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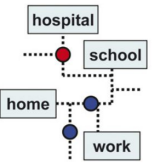


Agents can be **Susceptible (S)**, **Infected (I)**, or **Recovered (R)**, just like in the SIR model, but **interactions are local**, not global.

Unlike compartment models...

- **No assumption of perfect mixing.**
- Infection spreads through **local contact and shared spaces**, e.g., in neighborhoods, households, or workplaces.
- The outcomes are **stochastic** (random), and we often run simulations **many times** to explore possibilities.

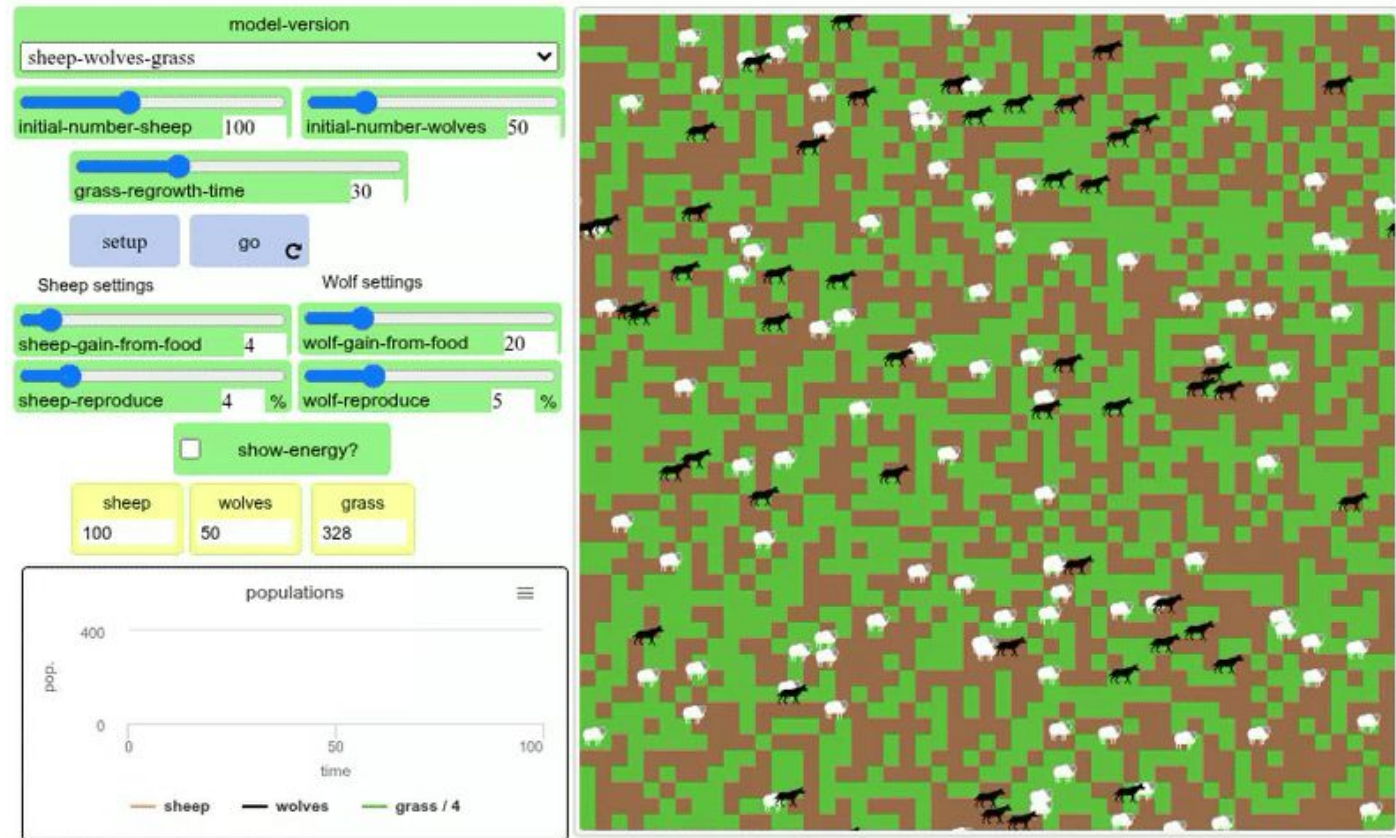
ABM environment



Wolf Sheep Predation

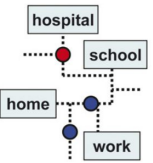
Spatial (grid or continuous space):

- Agents move across physical spaces (e.g., cities, maps).
- Supports location-based interactions.
- Examples: predator-prey models, traffic flow, school.

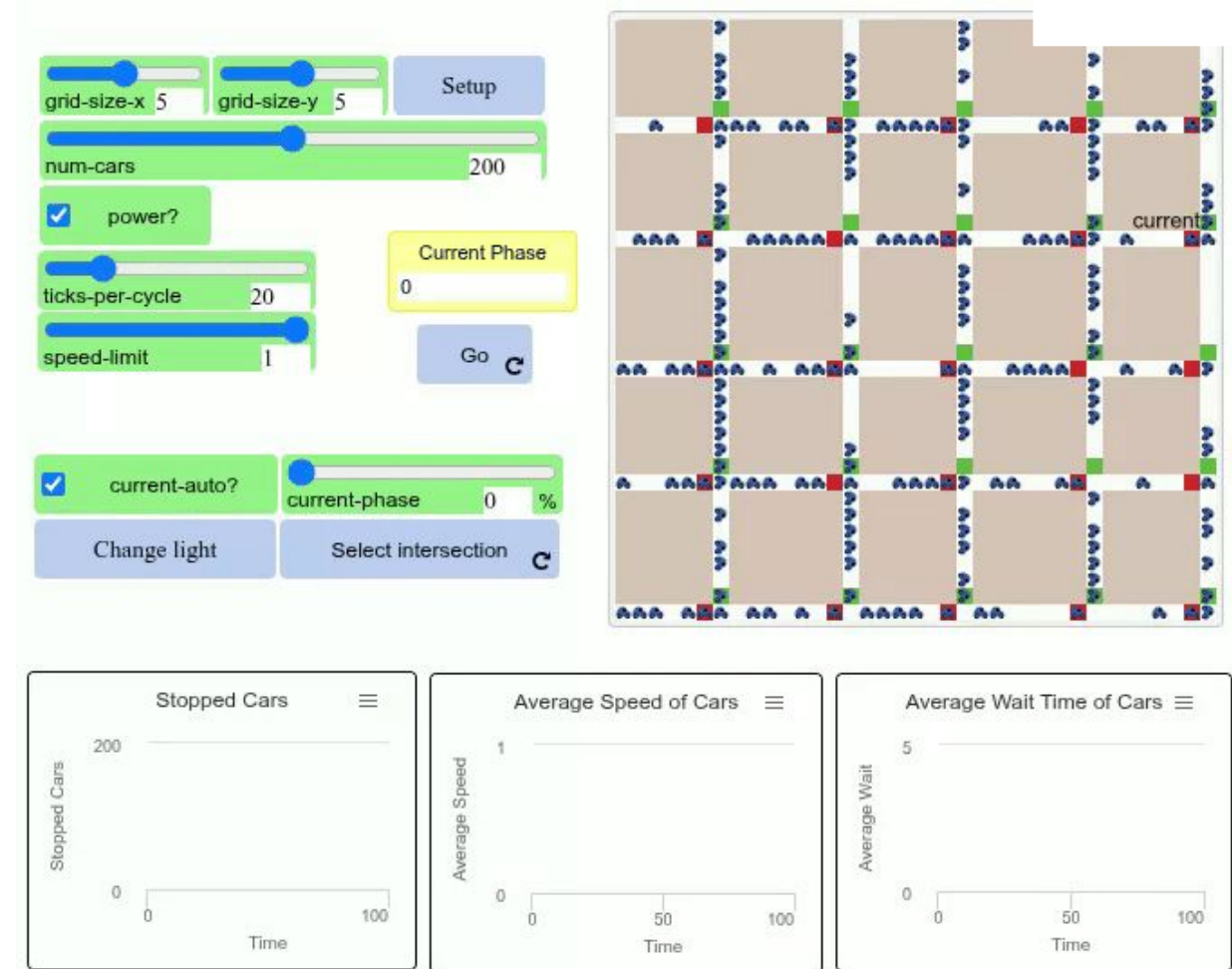


From <https://ccl.northwestern.edu/netlogo/models/>

ABM environment



Traffic Grid

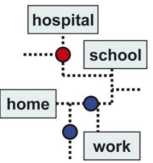


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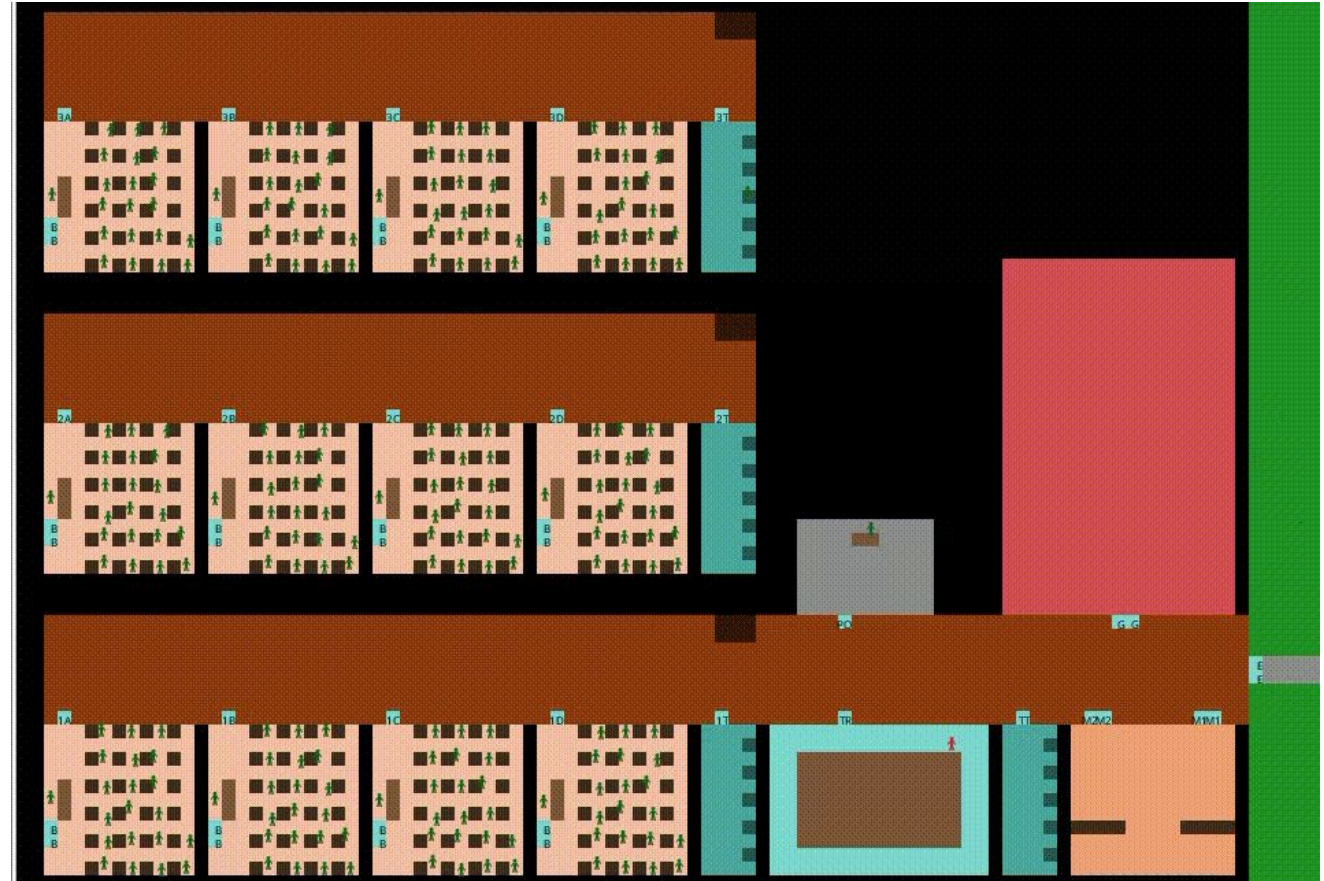
ABM environment



School model

Spatial (grid or continuous space):

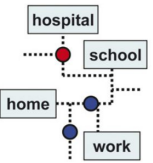
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Baccega, Daniele, et al. "An agent-based model to support infection control strategies at school." *JASSS* 25.3 (2022): 1-15.

Baccega, Daniele, et al. "Living along COVID-19: assessing contention policies through Agent-Based Models." *International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics*. Springer, Cham, 2025.

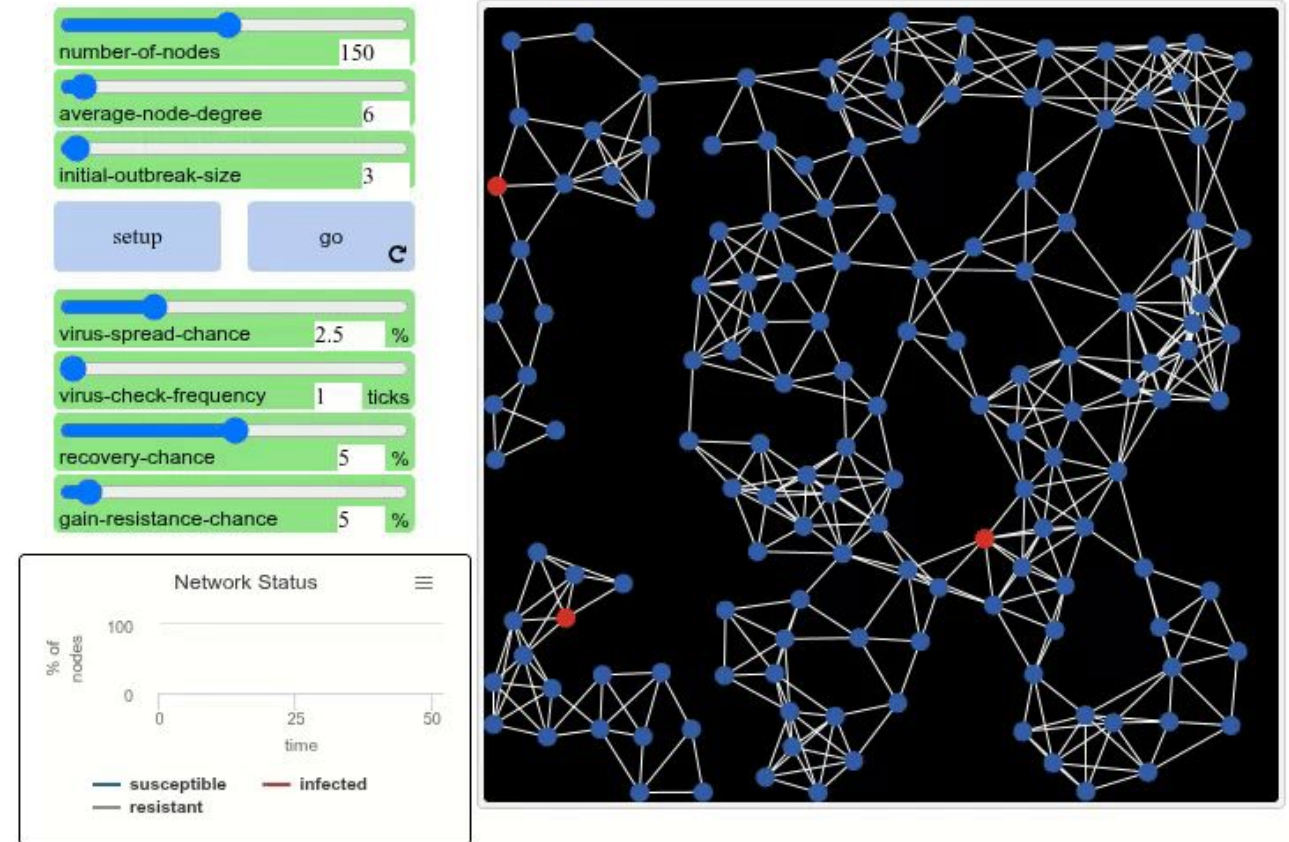
ABM environment



Network-based:

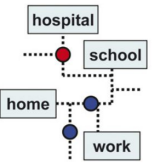
- Agents are nodes connected by relationships (edges).
- Useful for modeling social influence, disease spread, communication.
- Examples: social networks, power grids.

Virus on a Network



From <https://ccl.northwestern.edu/netlogo/models/>

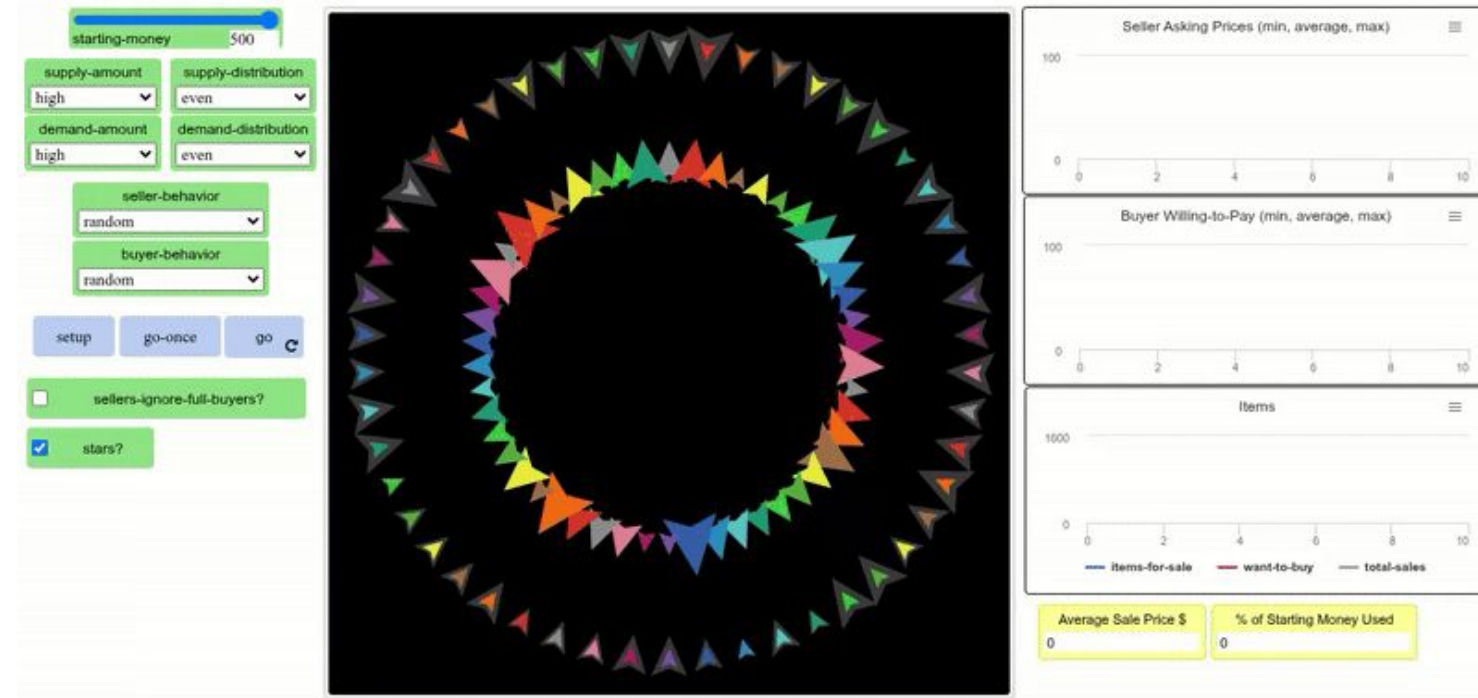
ABM environment



Bidding market

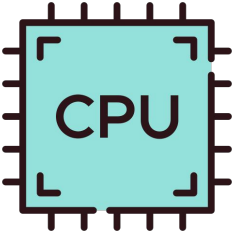
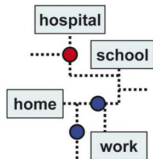
Abstract or conceptual:

- Represents non-physical dimensions (e.g., markets, belief systems).
- Agents interact based on logical conditions or data.
- Examples: bidding market, voting.



From <https://ccl.northwestern.edu/netlogo/models/>

ABM tools



[NetLogo](#)



[Repast Symphony](#)

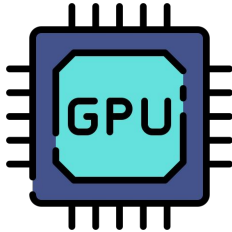
[Repast HPC](#)

[Mason](#)



[FLAME](#)

...



[FLAME GPU 2](#)

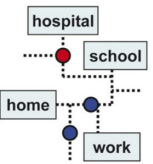


[BioDynaMo](#)

[GEMSim](#)

[SIMCoV-GPU](#)

...

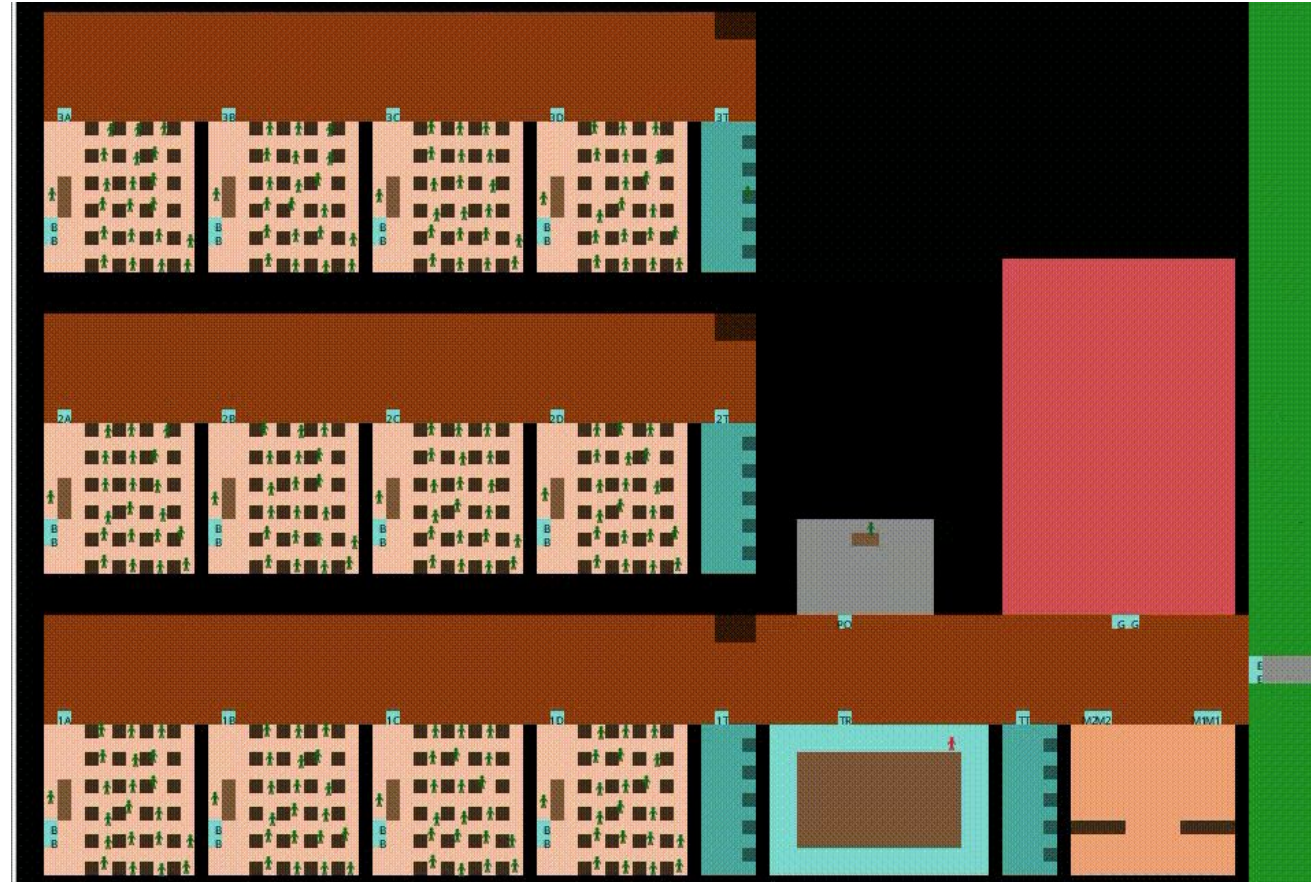


What is NetLogo?

NetLogo is a simulation environment for building and exploring agent-based models, designed to run on the CPU.

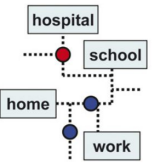
Why use it?

- Simulate real-world systems (e.g., ecosystems, traffic, markets).
- Visual + interactive = great for learning & experimentation.
- Easy to start, powerful to scale up.



Baccegga, Daniele, et al. "An agent-based model to support infection control strategies at school." JASSS 25.3 (2022): 1-15.
Baccegga, Daniele, et al. "Living along COVID-19: assessing contention policies through Agent-Based Models." International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics. Springer, Cham, 2025.

NetLogo

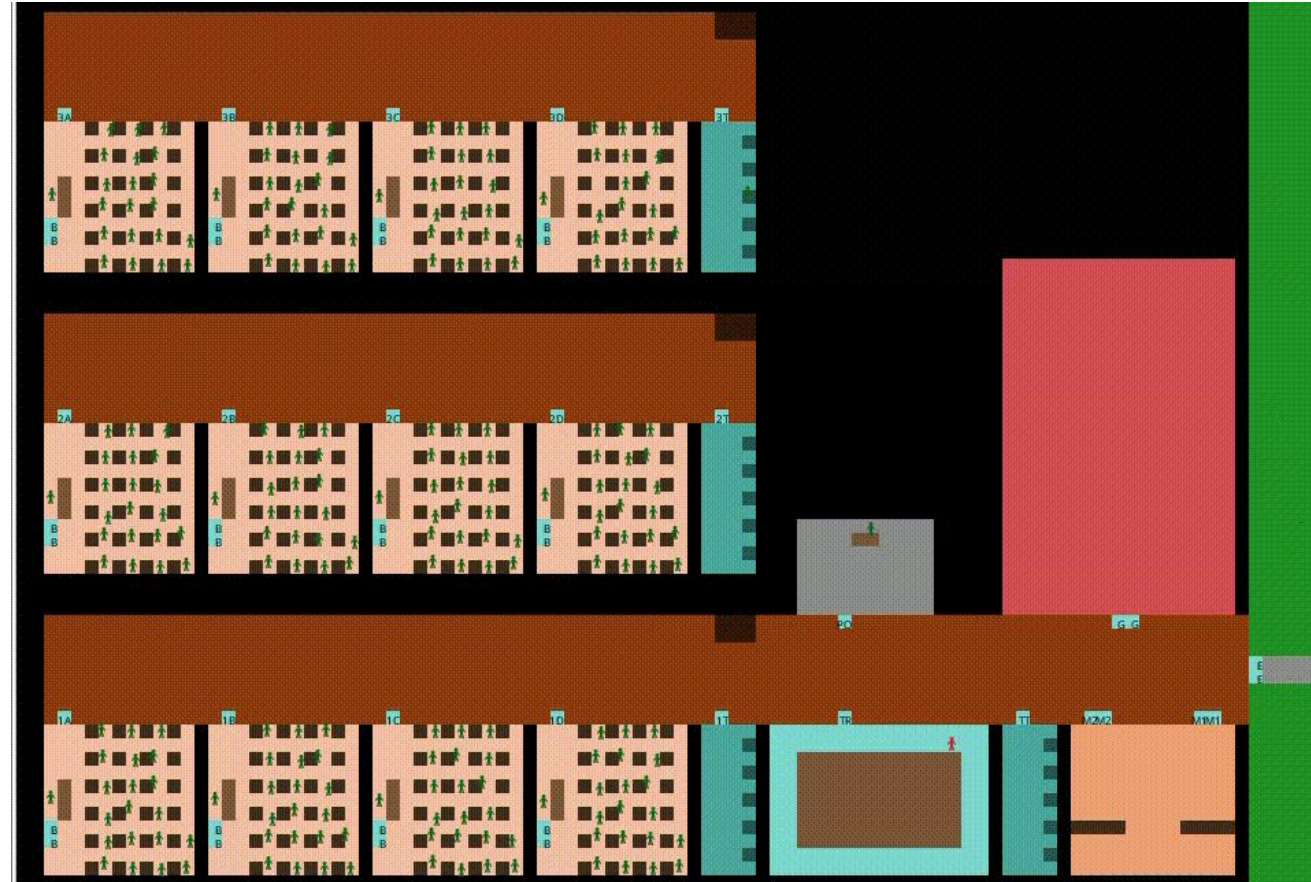


✓ Pros

- Easy to learn, beginner-friendly.
- Visual feedback for real-time learning.
- Huge built-in model library.
- Supports complex behavior with simple code.
- Cross-platform and free.

✗ Cons

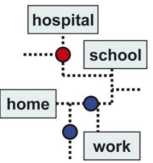
- Slower with very large agent counts.
- No parallelization.
- Not ideal for heavy numerical computations.
- Scripting language unique to NetLogo (not general-purpose).



Baccegga, Daniele, et al. "An agent-based model to support infection control strategies at school." *JASSS* 25.3 (2022): 1-15.

Baccegga, Daniele, et al. "Living along COVID-19: assessing contention policies through Agent-Based Models." *International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics*. Springer, Cham, 2025.

FLAME GPU 2

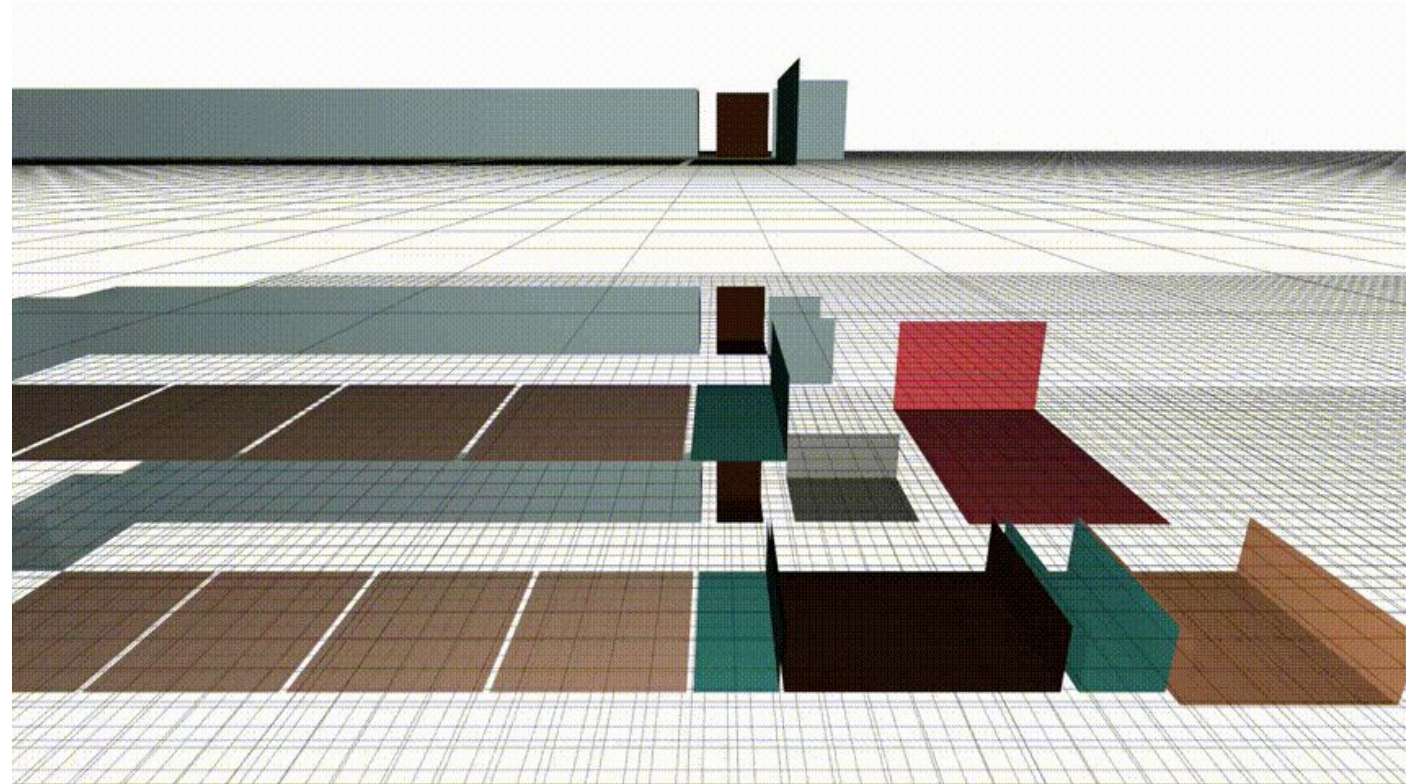


What is FLAME GPU 2?

FLAME GPU 2 is a high-performance agent-based modeling framework designed for GPU acceleration using NVIDIA CUDA.

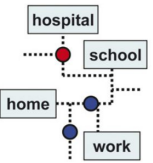
Why use it?

- Ideal for large-scale, computationally intensive models.
- Massively parallel: leverages the GPU to simulate millions of agents.
- Great for fields like biology, transportation, crowd simulation, and epidemiology.



Baccegá, D., et al. "Forge4Flame: An Intuitive Dashboard for Designing GPU Agent-Based Models to Simulate Infectious Disease Spread." Available at SSRN 5194584.

FLAME GPU 2

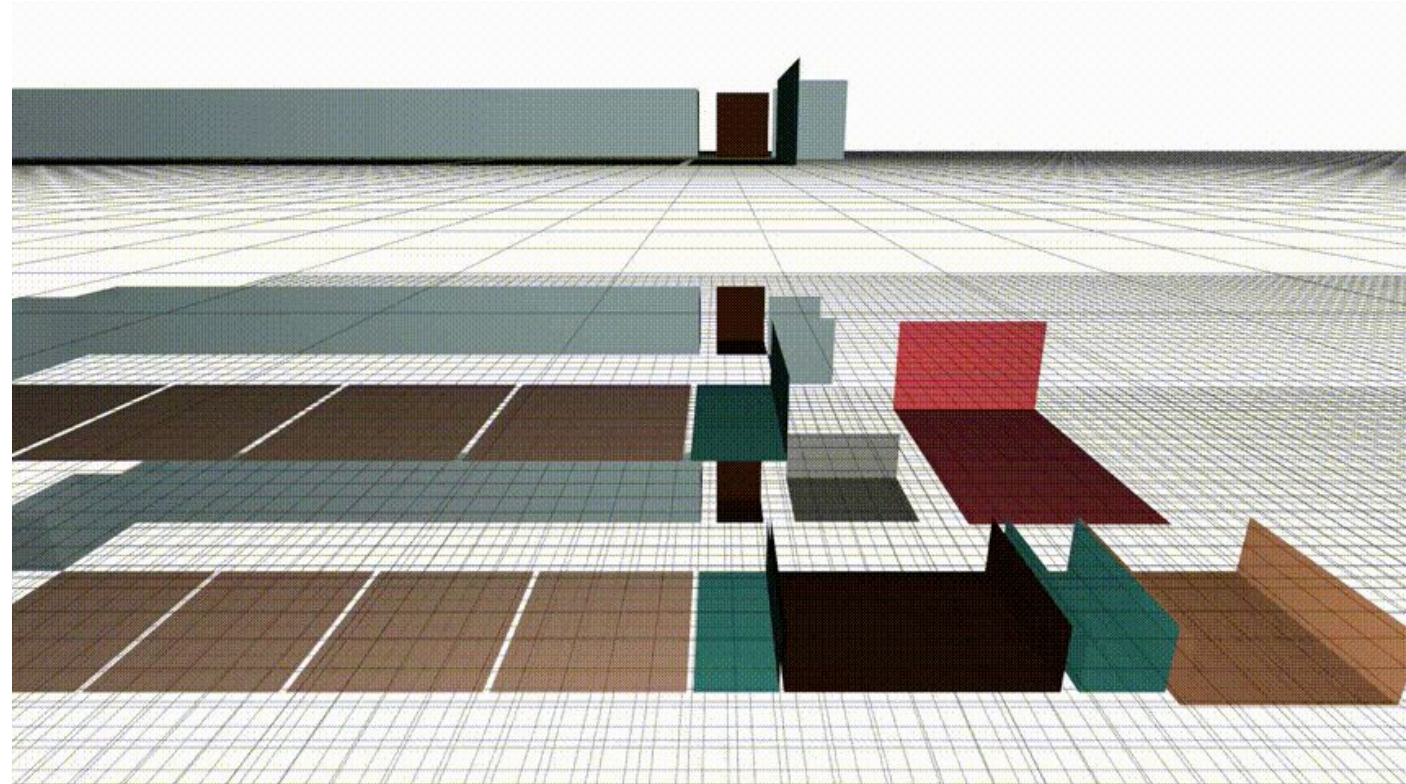


✓ Pros

- Extreme scalability: handles millions of agents.
- GPU acceleration = faster simulations.
- Custom C++ agent logic.
- Integration with Python.

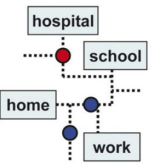
✗ Cons

- Steeper learning curve than GUI-based tools.
- Requires CUDA-capable GPU and some C++ knowledge.
- Less beginner-friendly for quick prototyping.
- Limited built-in visualization (compared to NetLogo).

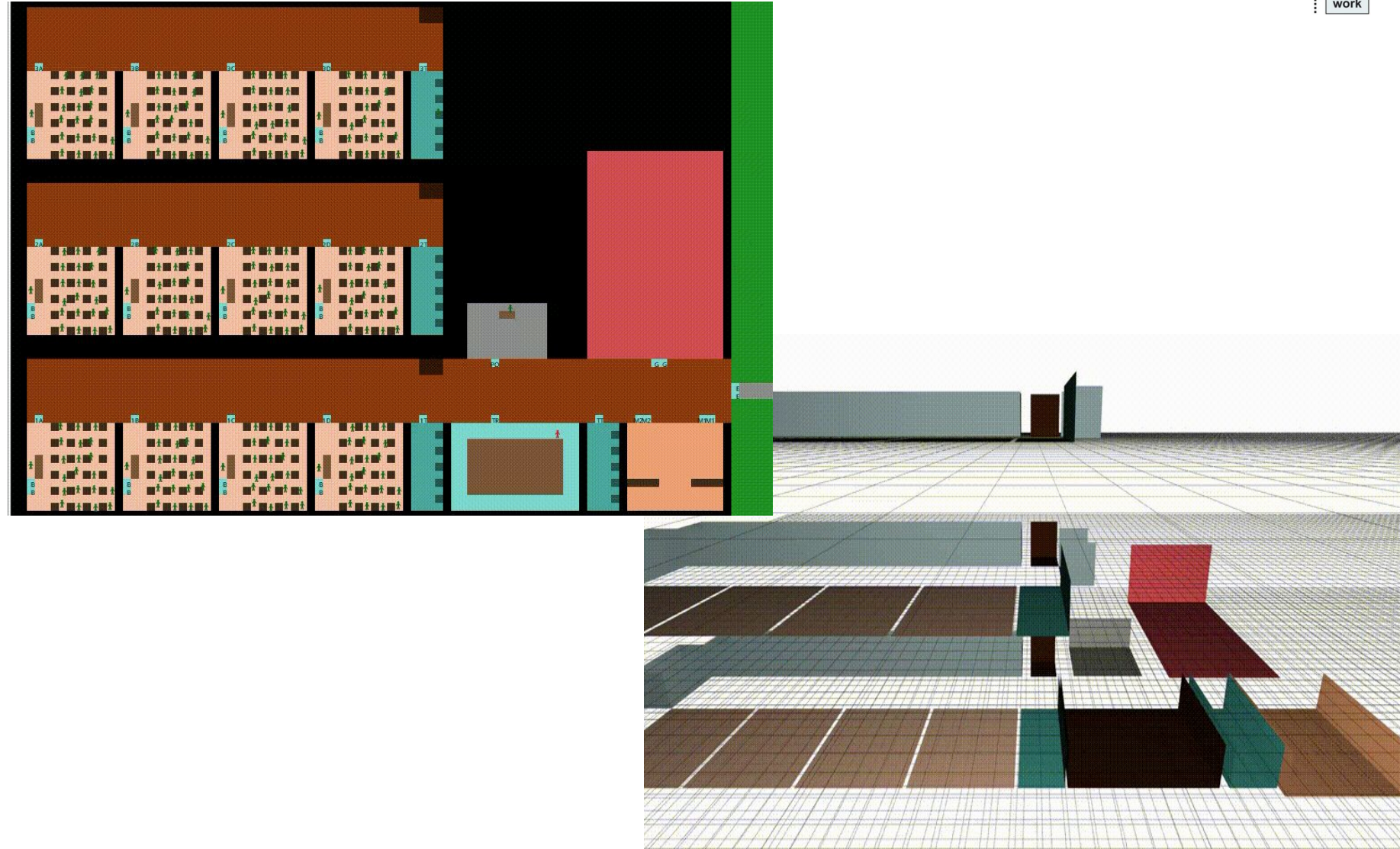


Baccegá, D., et al. "Forge4Flame: An Intuitive Dashboard for Designing GPU Agent-Based Models to Simulate Infectious Disease Spread." Available at SSRN 5194584.

Well-defined closed environment



- These scenarios feature detailed simulation of agents' movements within a confined and structured space.
- **Very complex** to define.
- Examples: school, hospital, supermarket, office.



Forge4Flame: An Intuitive Dashboard for Designing GPU Agent-Based Models to Simulate Infectious Disease Spread



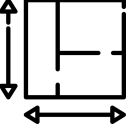
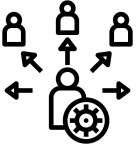
Baccegà, D., et al. "Forge4Flame: An Intuitive Dashboard for Designing GPU Agent-Based Models to Simulate Infectious Disease Spread." Available at SSRN 5194584.



Forge4Flame: introduction

User-friendly dashboard designed to simplify the definition of an **Agent-Based Model environment** without the need to write code.

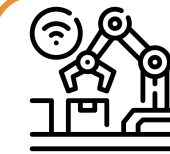
It enables to define:

1. model's environment 
2. the agents interacting within it 
3. the disease model 
4. other components relevant to an agent-based simulation of disease spread 
5. simple visualisations 



User-friendly

Approachable by scientists without specific computational knowledge



Automatic

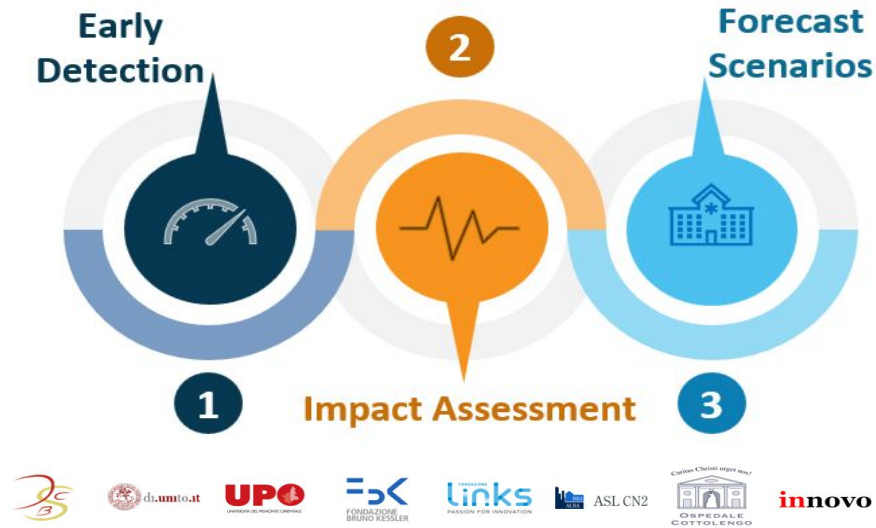
Automatic generation of 3D model environment for FLAME GPU 2



Reproducibility

Scientific results can be consistently replicated by independent researchers using the same methodology and data

How it started...



TrustAlert

Platform for the analysis of near real-time stream of healthcare data

Founded by Compagnia San Paolo

1M Euro Project
7 Collaborators

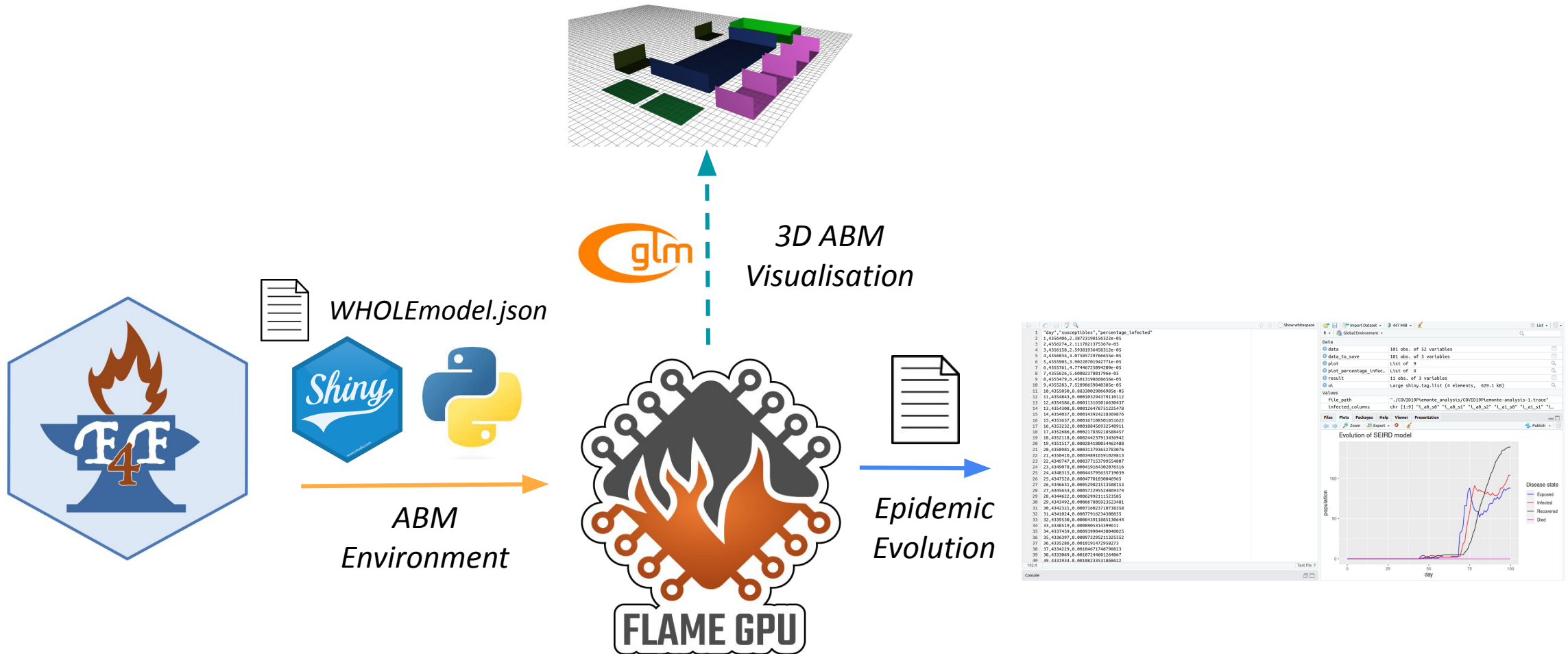
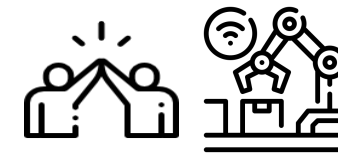
Our objectives in this project:

Evaluate the potential impacts of a health emergency in a real case scenario (Cottolengo Hospital in Turin) at different levels of detail, spanning from **different simulation paradigms** (e.g., ABM).

Expected results:

Being able to simulate the behavior of the "Hospital" system, for what happens outside, will allow Cottolengo Hospital to be able, in the medium term, not only to be effective but to improve the efficiency (cost / result ratio) and response capacity to ensure the delivery of health services.

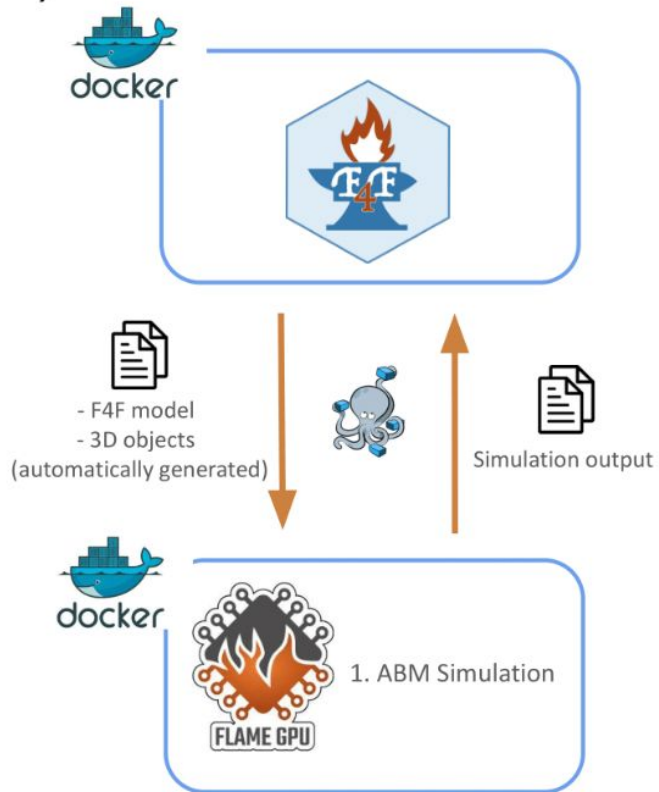
Pipeline



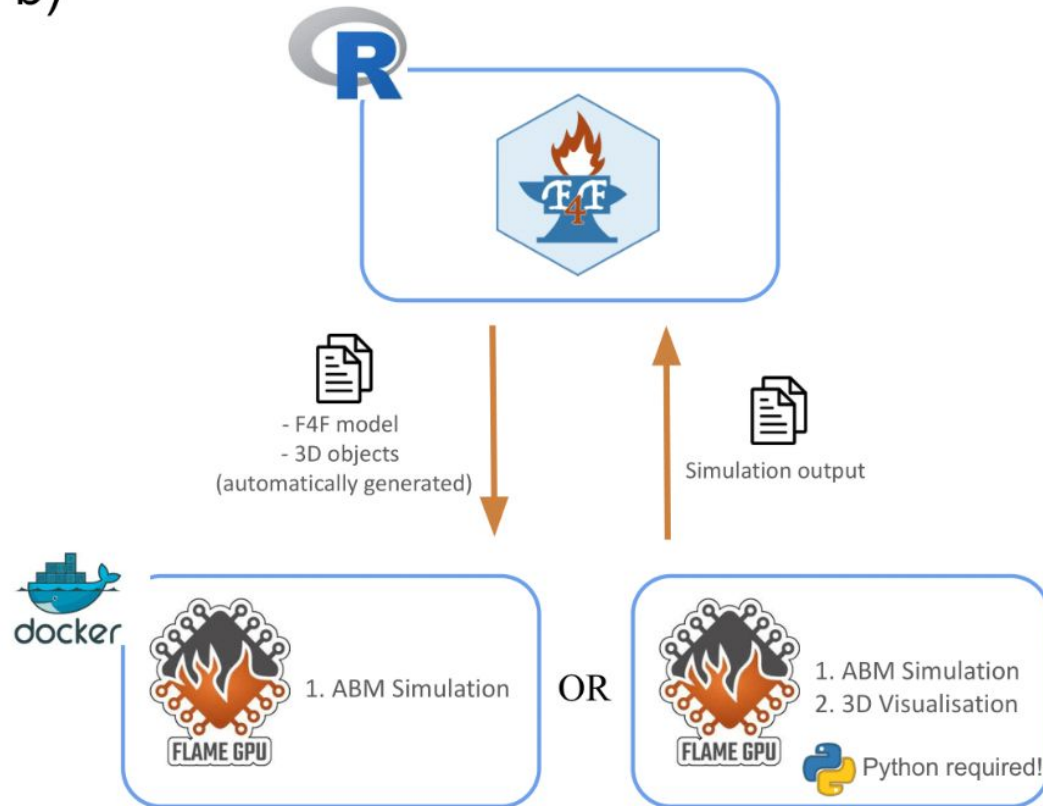
Docker pipeline



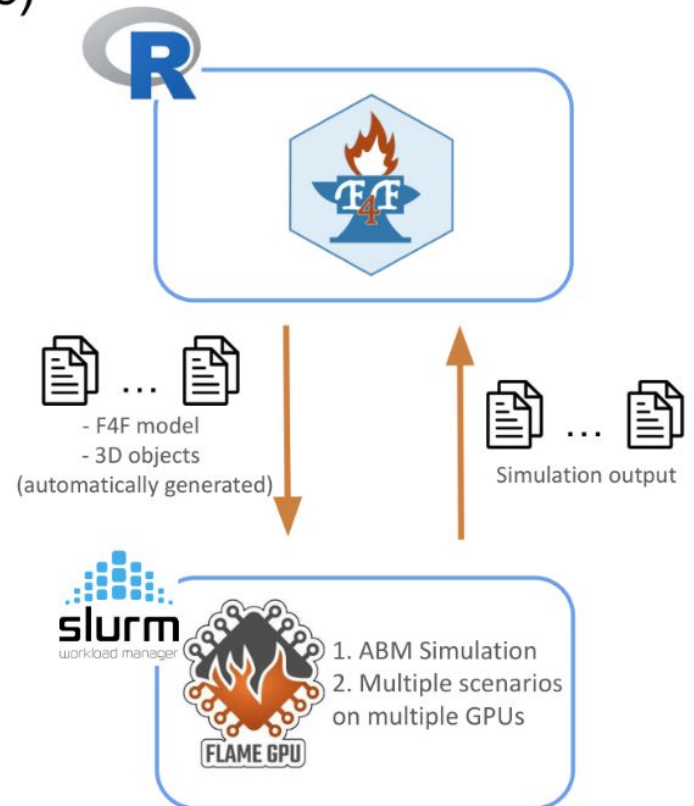
a)



b)



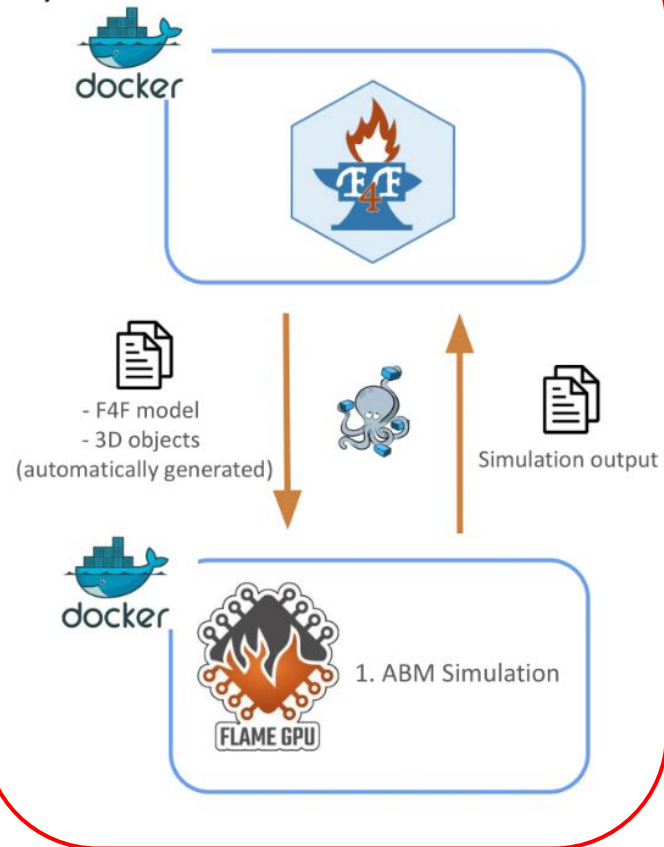
c)



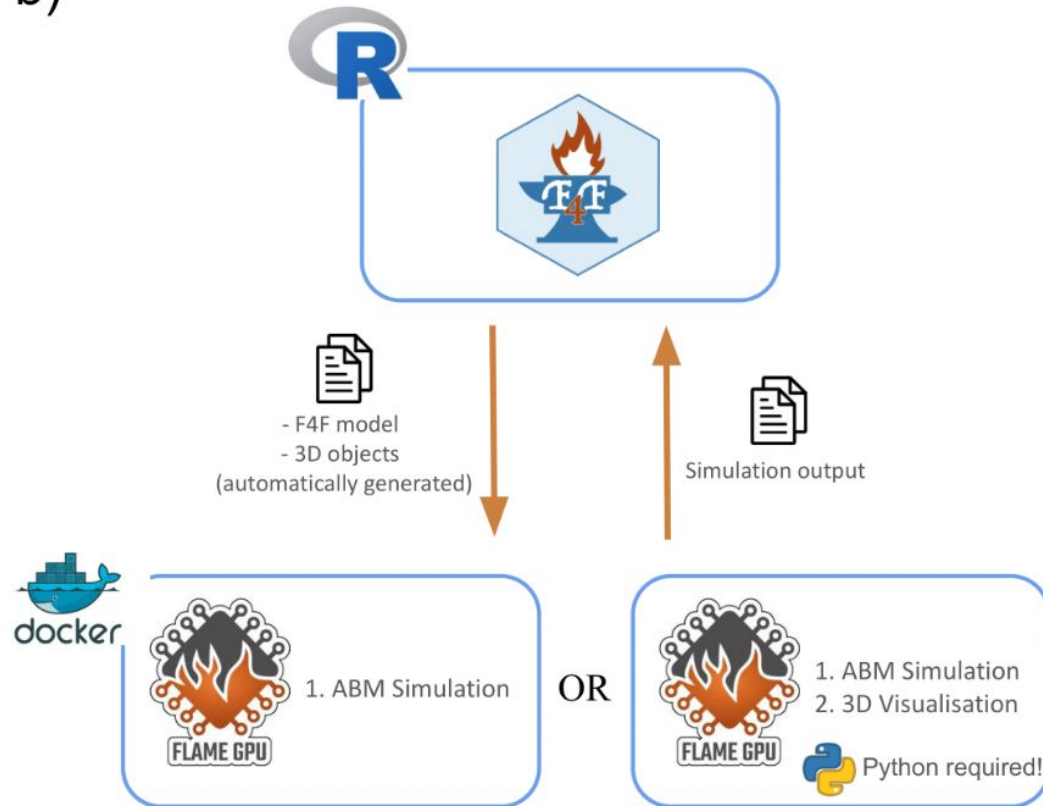
Docker pipeline



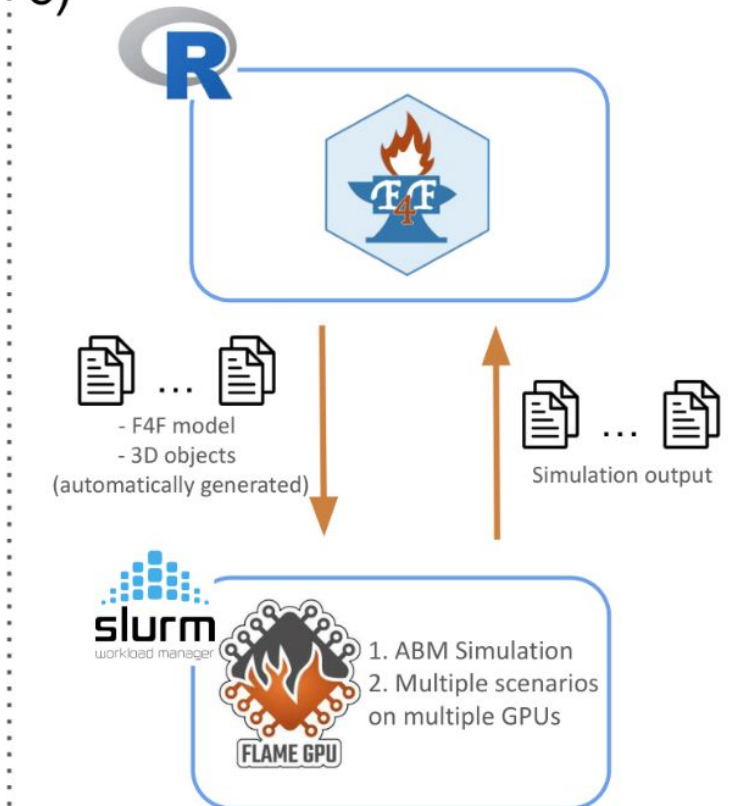
a)



b)



c)





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Forge4Flame (F4F)

F4F is a user-friendly dashboard (developed in R Shiny) designed to simplify the definition of an ABM environment for FLAME GPU 2 [5, 6] agent-based models, automatically generating the necessary code. It enables users to define the model's environment, the agents interacting within it, the disease model, and other components relevant to an ABM simulation. F4F is constituted by the following components (the images refer to the school model defined in [7, 8]):

- Canvas:** define the model's environment using a drag-and-drop interface for rooms.

Define floor

Define/Select floor

floor0

Delete floor

Drag the floors in the desired order

floor0

floor1

floor2

Add elements

Draw rooms

Select the room:

Length of selected room (length refers to the wall with the door):

Width of selected room:

Height of selected room:

Door position:

Remove rooms

Select the room to remove:

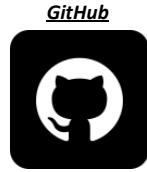
Remove room

Clear floor

Agents management

Epidemic Modelling Frameworks for Studying Infectious Disease Dynamics 76

How to install and run



Forge4Flame (F4F)



Forge4Flame (F4F) is a user-friendly R Shiny dashboard designed to streamline the creation of FLAME GPU 2 [1, 2] ABMs, automatically generating the necessary code. With its intuitive interface, users can easily define key components, including the model's environment, the agents interacting within it, the disease model, and other essential elements for agent-based simulations.

See [user guide](#) for more information.

How to download

Repository

Cloning the repository is optional, especially for users intending to use the F4F package, dockers, or Slurm—see the following sections for more details. To clone it, use the following Bash command:

```
git clone --recurse-submodules https://github.com/qBioTurin/Forge4Flame.git
```

The user can run F4F:

- Installing and running the R package.
- Downloading and running the provided Docker image.

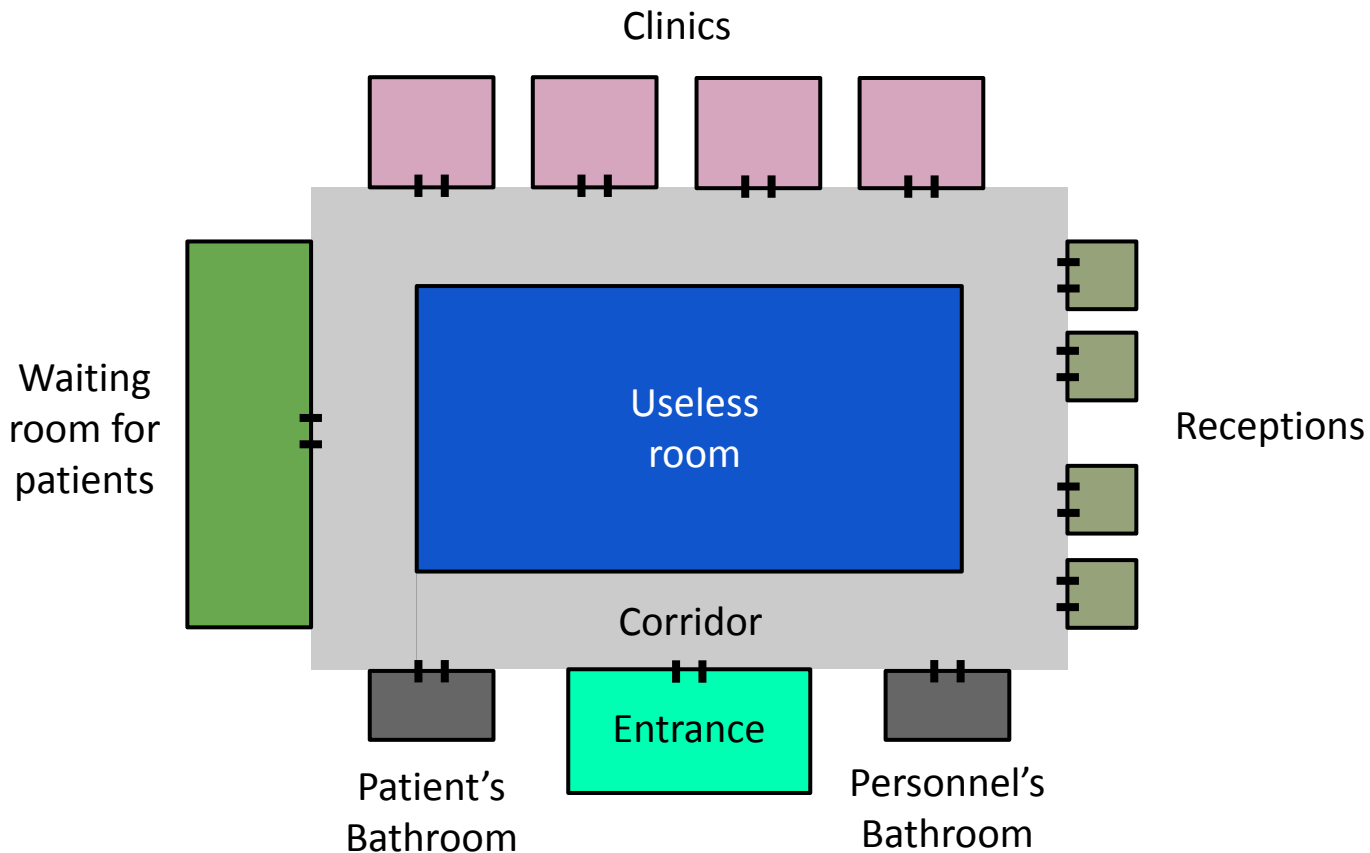
See the [GitHub page](#) for more details.



<http://github.com/qBioTurin/IEEEICH2025>

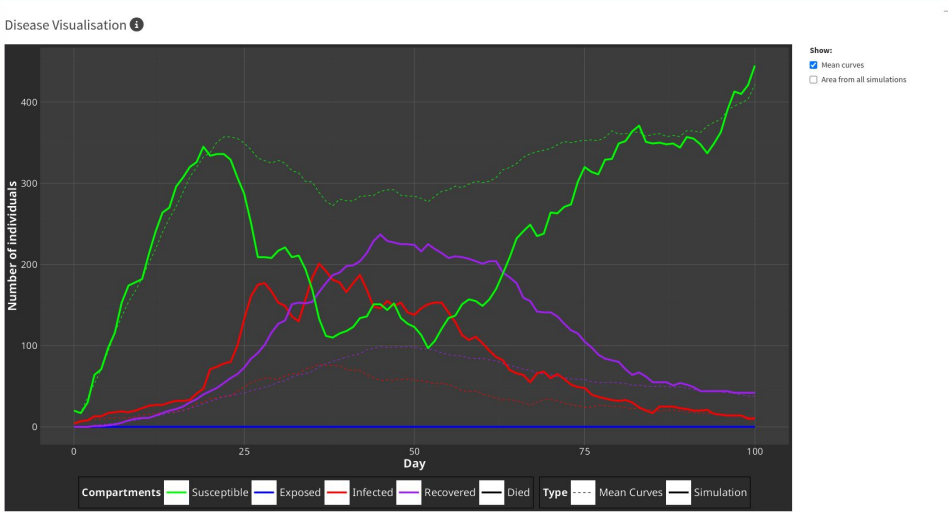
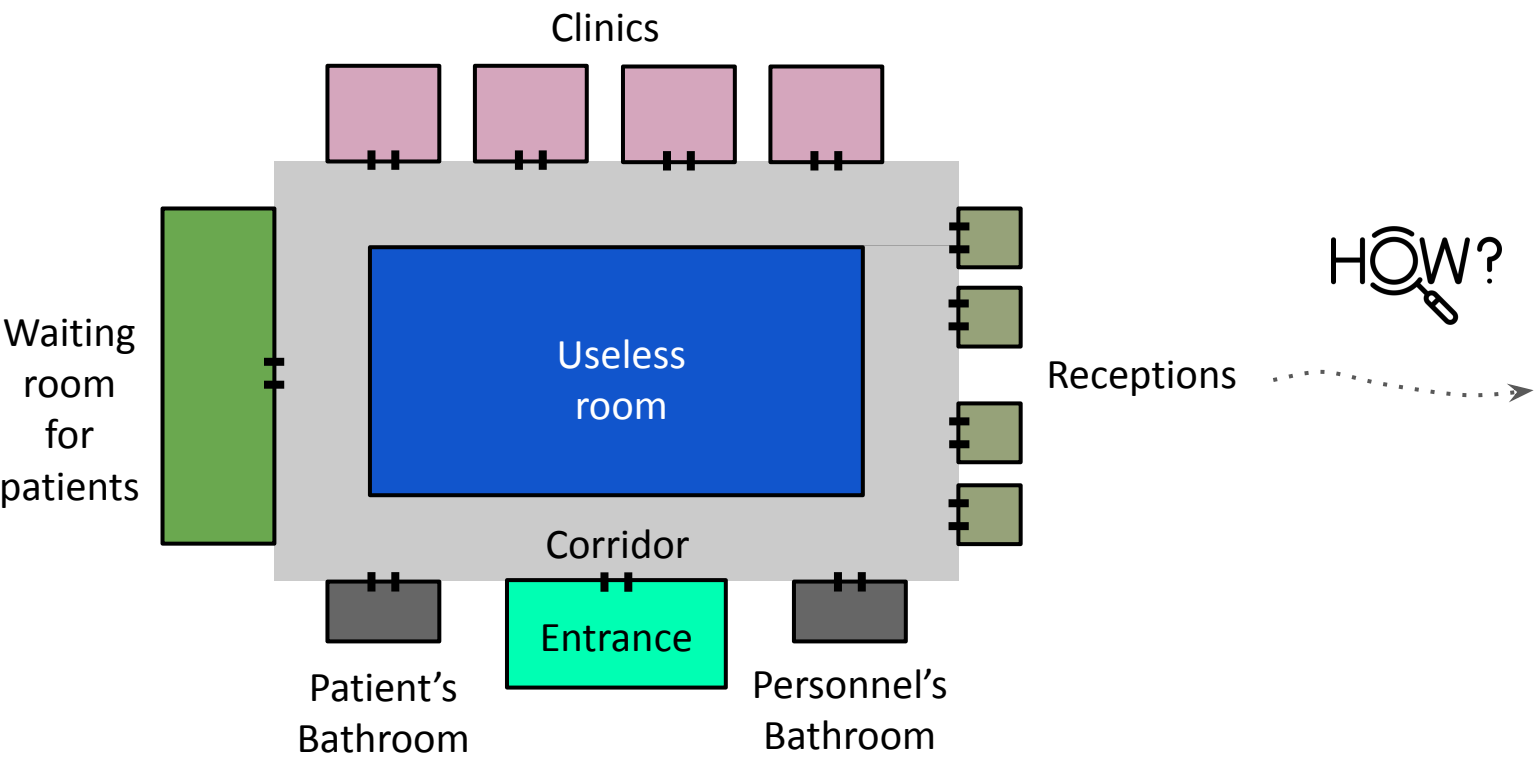
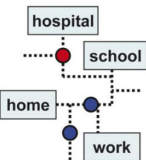


Case study: small ambulatory



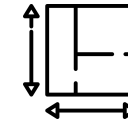
- 12 doctors and 12 receptionists are divided across three shifts (6 AM–2 PM, 2 PM–10 PM, and 10 PM–6 AM). Doctors remain in the clinic during their shifts, while receptionists stay at the reception.
- On average, 20 patients visit the clinic per day, each required to attend four weekly appointments. They check in at the reception (≈ 5 min) and proceed to the clinic (≈ 40 min), waiting their turn in the respective waiting rooms.
- Each individual may use the bathroom with a predefined probability: $1.4e^{-4}$ for both receptionists and doctors, and $2.8e^{-4}$ for patients.

Case study: small ambulatory





Canvas page with empty canvas



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Define floor

Define/Select floor

Define or select floor

Delete floor

Add elements

Draw rooms

Select the room:

Length of selected room (length refers to the wall with the door):

Width of selected room:

Height of selected room:

Door position:

right

Area:

None

Add room

Remove rooms

Select the room to remove:

Remove room

Clear floor

Agents management

Add graph point

Visualise graph

Remove last graph point

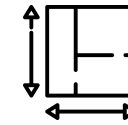
Colour fill by:

Room

The canvas represents a plane (or floor) where predefined rooms can be added and positioned using drag and drop. At least one floor has to be defined.

Before building the environment, rooms must be defined.

Rooms page



Build your ABM



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Define a new room

Name:

Length (meter):

Height (meter):

Type:

Width (meter):

Set colours legend

Select room:

Select type:

Select area:

#5FFB07

Select colour for Normal

#FF0000

Select colour for None

#000000

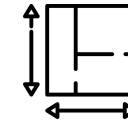
It is possible to define and change the color of rooms, types and areas displayed in the *Canvas*.

A room is defined by:

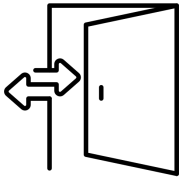
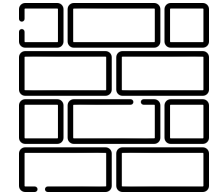
- 3 dimensions (*Length, Width, Height*),
- name (*room's id*), and
- type (*to group rooms into clusters*)



Rooms page: predefined rooms' type

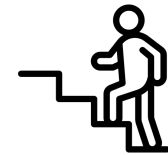


Filling-rooms: room type utilized by agents to simulate walls or useless rooms.



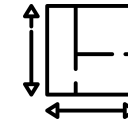
Spawn-rooms: room type utilized by agents to enter (and exit) the environment.

Stairs: room type utilized by agents to move among floors.



Waiting-rooms: room type utilized by agents to wait to enter in another room.

Canvas page



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Define floor

Define/Select floor

Define or select floor

Delete floor

Add elements

Draw rooms

Select the room:

Length of selected room (length refers to the wall with the door):

Width of selected room:

Height of selected room:

Door position:

Area:

Add room

Remove rooms

Select the room to remove:

Remove room

Clear floor

Agents management

Add graph point

Remove last graph point

Visualise graph

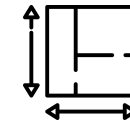
Colour fill by:

Room

Each room may be assigned to a particular area (e.g., a specific area of a hospital). This is useful restrict the areas in which agents can go.



Canvas page: floor0



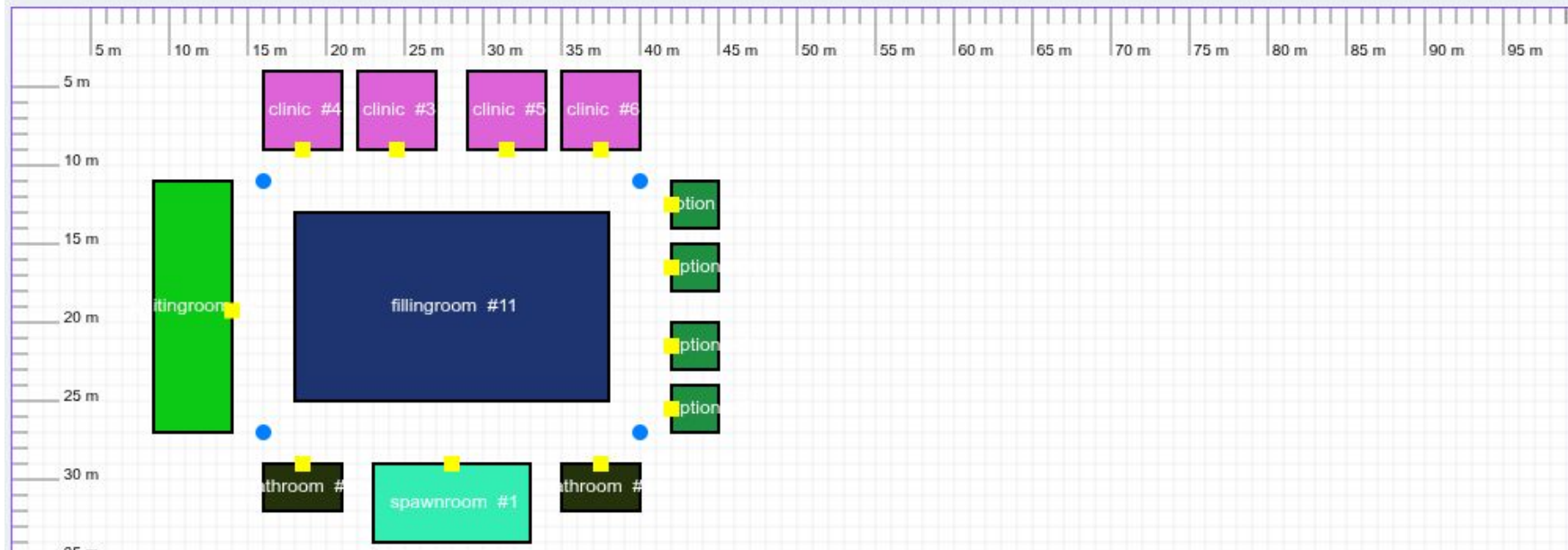
Define floor

Define/Select floor

floor0

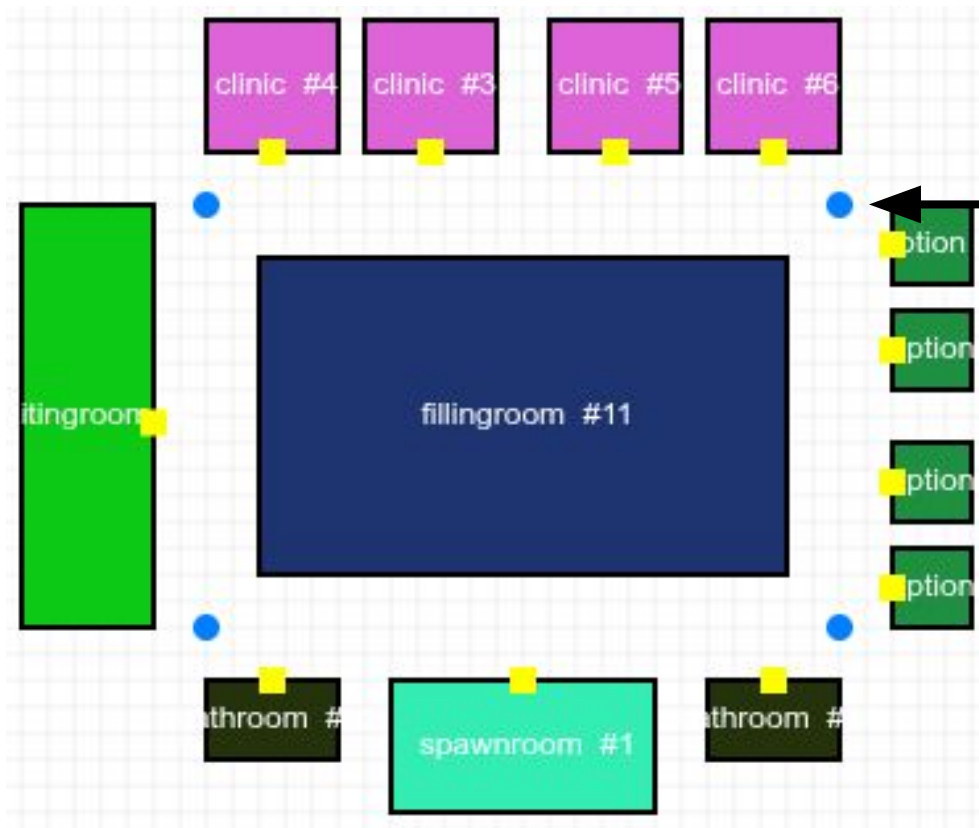
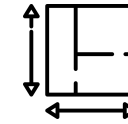
Delete floor

Add elements





Canvas page: graph points

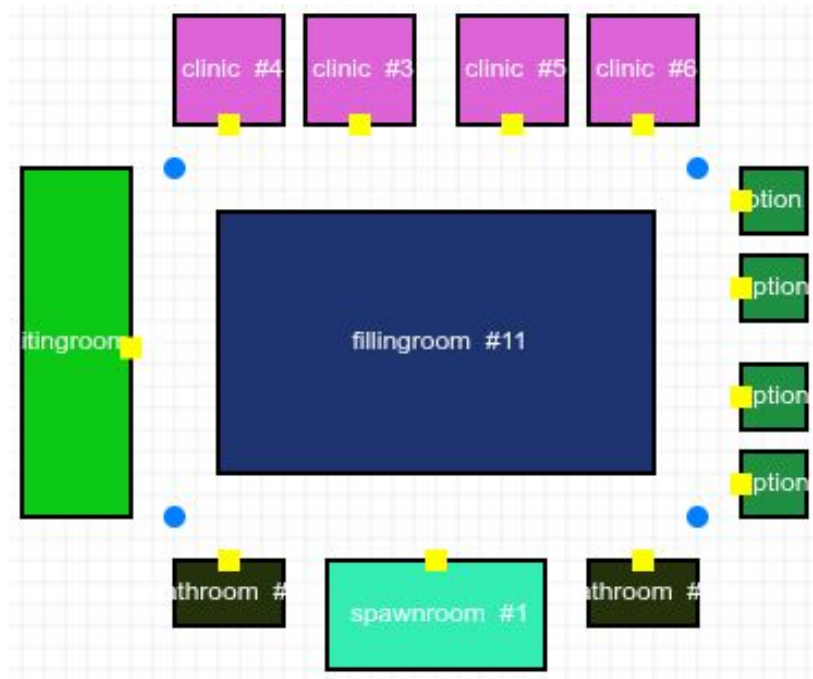
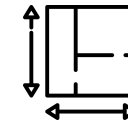


It is possible to add **graph points** to insert a new node into the agents' movement graph.

By default, each door and room center corresponds to a node in this graph.



Canvas page: graph points



It is possible to generate and visualise all the possible paths that the agents might follow.



Agents



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Agent definition

Agent name:

Remove agent

Agent class:

Doctor

Copy information from:

Copy

Number of agents:

The number must be a positive integer

Determined flow

Type:

Activity:

Deterministic

Stochastic

Fixed deterministic value:

Value

Add room

Remove last room

Add flow

Remove flow

Random flow

Type:

Activity:

Weight:

Deterministic

Stochastic

Fixed deterministic value:

Value

Entry flow

Select type of entrance:

Daily Rate

Time window

Save time

1 slot

Entry time:

hh:mm

Associate with a determined flow:

Add slot

Remove slot

Select Days of the Week

☒ Monday

☒ Tuesday

☒ Wednesday

☒ Thursday

☒ Friday

☐ Saturday

☐ Sunday

Agent definition**Determined flow****Random flow****Entry/Exit flow**



Agent page: agent definition

Agent definition 

Agent name:	doctor_firstshift	Remove agent	Copy information from:		Copy
Agent class:	Doctor		Number of agents:	4	

An agent is defined by:

- a name,
- a class (category that groups different types of agents), and
- the number of agents in the simulation.



Agent page: agent determined flow

Determined flow 

Type:

Activity:

Deterministic

Stochastic

Add room

Fixed deterministic value:

Value

Remove last room

Add flow

Remove flow

1 flow

Drag the rooms in the desired order

Spawnroom-None - Exponential 5 min - Very Light

Clinic-None - Deterministic 480 min - Light

(2) Spawnroom-None - Exponential 5 min - Very Light

The user can select the intensity of the activity that the agent performs in the selected room. This is important to determine the quantity of quanta exhaled by an infected agent.

Options: very light, light, quite hard, hard.

The flow must always begin and end with the spawn-room.

Agent page: agent random flow



The user can add random events that interrupt the agent's determined flow.

Random flow ⓘ

Type:

Activity:

Weight:

Deterministic

Stochastic

Add room

Fixed deterministic value:

Value

Click on an event to remove it (except the 'Do nothing' event)

Show

5

 entries

Search:

Room	Distribution	Activity	Time	Weight
Do nothing	Deterministic	Light	0	0.99986
Bathroom-Personnel	Exponential	Light	3	0.00014

Showing 1 to 2 of 2 entries

Previous

1

Next

By default the event *Do nothing* is present with probability 1.

The probability that an event might occur will be checked every step.



Agent page: agent time window

Entry flow

Select type of entrance:

☐ Daily Rate ☒ Time window

Save time

1 slot

Entry time:

06:00

Associate with a determined flow:

1 flow

Select Days of the Week

- ☒ Monday
- ☒ Tuesday
- ☒ Wednesday
- ☒ Thursday
- ☒ Friday
- ☒ Saturday
- ☒ Sunday

Add slot

Remove slot

The user can define the entry time and the specific days of the week when the agent will enter the environment.

Copy information from:

Number of agents:

4

The number of agents is specified in the agent definition panel.



Agent page: agent daily rate

Entry flow

Select type of entrance:

☒ Daily Rate ☐ Time window

Save time

The user can define an entrance rate for the agent using either a deterministic value or a stochastic distribution, along with specifying an initial and a final generation time and the specific days of the week when the agent will enter the environment.

1 slot

Entrance rate:

Deterministic

Stochastic

Distribution:

Truncated Positive Normal

Mean:

20

Sd:

10

Initial generation time:

00:00

Final generation time:

23:59

Select Days of the Week

- ☒ Monday
- ☒ Tuesday
- ☒ Wednesday
- ☒ Thursday
- ☒ Friday
- ☒ Saturday
- ☒ Sunday

Add slot

Remove slot

Resources page



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Set resources

Select global resources for each room and agent type:

Select second choice for each agent:

Select type and area:

For each room, the list of all agents that might reach it is showed. The user can set the rooms' resources available for each agent.

Resources: the number of agents that can occupy a room simultaneously.

Resources page



Set resources

Select global resources for each room and agent type:

0

Select second choice for each agent:

Same room

Set

Select type and area:

Bathroom-Personnel

Room	Maximum	receptionist_firstshift	reception_secondshift	reception_thirdshift	doctor_firstshift	doctor_secondshift	doctor_thirdshift
bathroom	4	4	4	4	4	4	4

Select second choice room in Random Flow for receptionist_firstshift:

Skip room

Select second choice room in Random Flow for reception_secondshift:

Skip room

Select second choice room in Random Flow for reception_thirdshift:

Skip room

Select second choice room in Random Flow for doctor_firstshift:

Skip room

Select second choice room in Random Flow for doctor_secondshift:

Skip room

Select second choice room in Random Flow for doctor_thirdshift:

Skip room

Infection page



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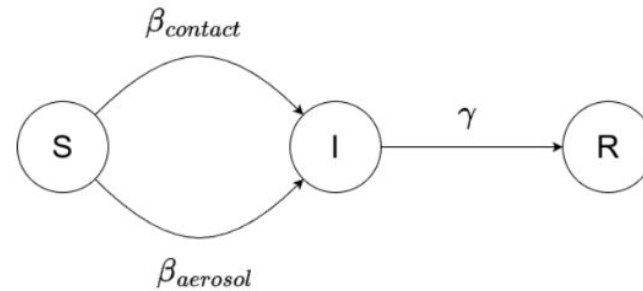
Disease model

Model:

SIR

Disease model: SIR. Beta (contact): 0.024, Beta (aerosol): 410, Gamma: 0.1 (Deterministic)

SIR: Susceptible - Infected - Recovered.



The evolution of the infection is modeled through a compartmental model.

Description:

The SIR model is a straightforward compartmental model used to study the dynamics of infectious diseases. It divides the population into three key groups. The first group, susceptible (S), includes individuals who are vulnerable to the disease. When these individuals come into contact with infected people or share a confined space with them, they can contract the disease and transition to the infected group. The second group, infected (I), consists of individuals who are carrying the disease and can transmit it to those who are susceptible. The third group, recovered (R), includes individuals who have overcome the infection and gained immunity. Overall, the SIR model offers a simplified framework to analyze the spread of infectious diseases within a population, focusing on the interactions between the susceptible, infected, and recovered groups.

1 $\beta_{\text{contact}} =$

0.024

1 $\beta_{\text{aerosol}} =$

410

1 $\gamma =$

Deterministic

Stochastic

Fixed deterministic value:

0.1

Save

Two contagion processes: contact [1] and aerosol [2].

[1] Hoertel, N., et al. *A stochastic agent-based model of the SARS-CoV-2 epidemic in France.*

[2] Gkantonas, S., et al. *airborne.cam: A risk calculator of SARS-CoV-2 aerosol transmission under well-mixed ventilation*



Infection page: SEIRDS

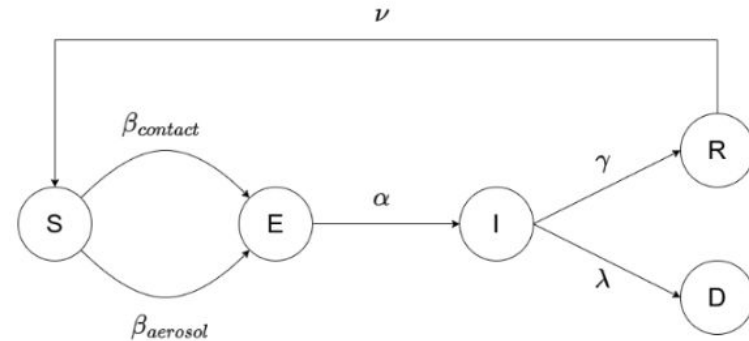


Disease model

Model:

SEIRDS

β_{contact} and β_{aerosol}
are not the
infection rates!



SEIRDS: Susceptible - Exposed
- Infected - Recovered - Dead -
Susceptible.

Description:

The SEIRDS model is an extension of the SEIRD model that introduces the possibility of waning immunity, adding a dynamic aspect to the compartmental framework. It includes five compartments. The first compartment, susceptible (S), represents individuals who are vulnerable to the disease. These individuals can contract the disease and transition to the exposed compartment upon contact with infected individuals or sharing a closed space with them. The second compartment, exposed (E), consists of individuals who have been infected but are in an incubation period, during which they cannot yet transmit the disease. The third compartment, infected (I), includes individuals who are actively infected and capable of spreading the disease to susceptible individuals. The fourth compartment, recovered (R), represents individuals who have overcome the infection and gained immunity. The fifth compartment, dead (D), accounts for individuals who have died as a result of the disease. Unlike the SEIRD model, the SEIRDS model assumes that immunity is temporary. After a certain period, individuals in the recovered compartment lose their immunity and return to the susceptible compartment, re-entering the cycle. This feature makes the SEIRDS model particularly relevant for analyzing diseases where immunity wanes over time.

What-If page



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Countermeasures

Saved Countermeasures 0

Measure	Type	Parameters	From	To
All	All	All	All	All
No data available in table				

Measure	Type	Parameters	From	To
All	All	All	All	All
No data available in table				

Ventilation

Masks

Vaccination

Swabs

Quarantine

External screening

Virus parameters

Saved Virus Parameters 0

Virus

Virus variant factor: 1

Virus severity: 0.22

Save

Initial infected: 0

Initial infected:

What-If scenarios are useful to understand the potential impact of hypothetical situations.

Options:

- Environmental ventilation
- Masks usage
- Virus parameters
- Vaccination
- Swabs (internal screening)
- Quarantine
- External screening
- Initial infected
- Outside contagion



What-If page: ventilation

Global

Ventilation

Ventilation:

- ☒ Global
☐ Different for each room

Ventilation (in ACH): ⓘ

0.3 (poorly ventilated) ▼

From (day):

1

To (day):

60

Ventilation is a process that replace contaminated with clean air at n Air Changes per Hour (ACH).

Different for each room

Ventilation

Ventilation:

- ☐ Global
☒ Different for each room

Room:

Waitingroom-None ▼

Ventilation (in ACH): ⓘ

5 (well ventilated) ▼

From (day):

1

To (day):

60

Options: no ventilation, poorly ventilated, domestic, offices/school, well ventilated, typical maximum, hospital setting.



What-If page: masks

Global

Masks reduce the virus inhalation and exhalation rate.

Masks

Mask:

- ☒ Global
☐ Different for each agent

Mask type:

No mask

% mask: ⓘ

1

From (day):

1

To (day):

60

Different for each agent

Options: no mask, surgical, FFP2.

Masks

Mask:

- ☐ Global
☒ Different for each agent

Agent:

doctor_firstshift

Mask type:

FFP2 mask

% mask: ⓘ

1

From (day):

1

To (day):

60



What-If page: vaccination

Global

Vaccination

Vaccination:

- ☒ Global
☐ Different for each agent

Fraction of vaccinated agents:

Vaccine efficacy:

Vaccine coverage (day):

Fixed deterministic value:

At (day):

Vaccination immunizes people by putting them in the Recovered state.

Different for each agent

Vaccination

Vaccination:

- ☐ Global
☒ Different for each agent

Agent:

Fraction of vaccinated agents:

Vaccine efficacy:

Vaccine coverage (day):

Fixed deterministic value:

At (day):

Vaccine efficacy models the fraction of the vaccinated population that is effectively immune to infection.



What-If page: swab

Global

Swabs

Swab:

☐ Global

☒ Different for each agent

Sensitivity:

1

Specificity:

1

Agent:

doctor_firstshift

Swab:

☐ No swab

☒ Swab

A swab every how many days?

Deterministic

Stochastic

Fixed deterministic value:

3

From (day):

1

To (day):

60

Internal screening campaigns are useful to detect infected people and to monitor the spread of the infection in the environment.

Different for each agent

Swabs

Swab:

☒ Global

☐ Different for each agent

Sensitivity:

1

Specificity:

1

A swab every how many days?

Deterministic

Stochastic

Fixed deterministic value:

7

From (day):

1

To (day):

60

These screening campaigns are conducted **within** the environment!



What-If page: quarantine

Global

Quarantine

Quarantine:

- ☒ Global
☐ Different for each agent

Quarantine days:

Fixed deterministic value:

Swab:

- ☐ No swab
☒ Swab

Sensitivity:

Specificity:

A swab every how many days?

Fixed deterministic value:

From (day):

To (day):

Quarantine room for severe cases: ⓘ

Isolating infected people is useful to slow down the spread of the virus.

Different for each agent

Quarantine

Quarantine:

- ☐ Global
☒ Different for each agent

Agent:

Quarantine:

- ☐ No quarantine
☒ Quarantine

Quarantine days:

Fixed deterministic value:

Swab:

- ☐ No swab
☒ Swab

Sensitivity:

Specificity:

A swab every how many days?

Fixed deterministic value:

From (day):

To (day):

Quarantine room for severe cases: ⓘ



What-If page: external screening



Global

External screening

External screening:

- ☐ Global
☒ Different for each agent

Agent:

doctor_firstshift

Screening campaigns:

0.01

Symptoms:

0.05

From (day):

1

To (day):

60

External screening campaigns are useful to account for interactions **outside** the environment.

Different for each agent

External screening

External screening:

- ☒ Global
☐ Different for each agent

Screening campaigns:

0.0003

Symptoms:

0.0075

From (day):

1

To (day):

60

Two probabilities:

- test an individual because he/she follows activities that involve screening campaigns (like practise sports)
- test an infected individual due to symptoms.



What-If page: virus



Virus variant factor models the virus or a particular variant of the virus under study [1].

Virus

Virus variant factor: 

Virus severity: 

Virus severity models the fraction of infected with severe symptoms [2].

[1] J.L. Jimene, et al. *COVID-19 Aerosol Transmission Estimator*.

[2] Tolksdorf K, et al. *Influenza-associated pneumonia as reference to assess seriousness of coronavirus disease (COVID-19)*.



What-If page: initial infected

Random

Initial infected

Initial infected: ⓘ

☒ Random
☐ Global
☐ Different for each agent

Initial infected:

1

Global

Initial infected

Initial infected: ⓘ

☐ Random
☒ Global
☐ Different for each agent

Initial infected:

1

Different for each agent

Initial infected

Initial infected: ⓘ

☐ Random
☐ Global
☒ Different for each agent

Agent:

doctor_firstshift ▼

Initial infected:

1

The number of initial infected within the environment can pilot the evolution of the infection.



What-If page: outside contagion

Outside contagion ⓘ

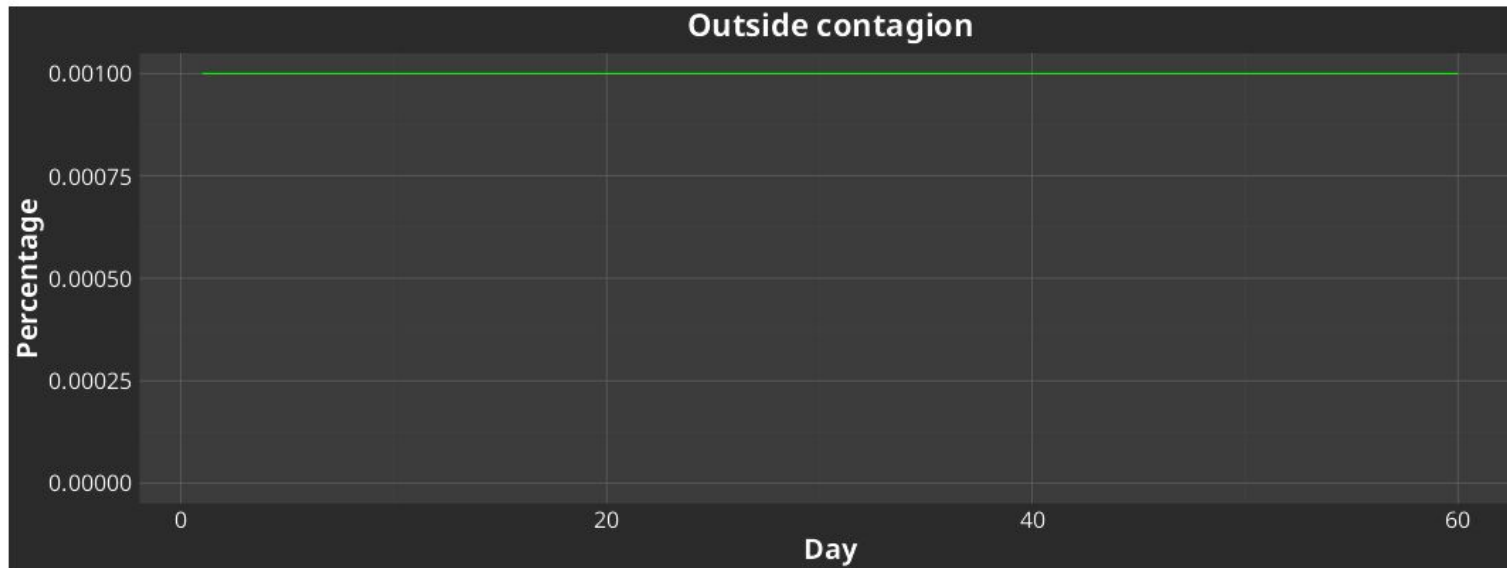
Browse... outside_contagion.csv

Load

Upload complete

Population: ⓘ

1



Outside contagion accounts for the fact that any individual will likely catch the infection outside the environment.



Configuration page



Build your ABM

Home
Canvas
Rooms
Agents
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Infection
What-If
Configuration
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Post Processing

Configuration parameters

Simulation days:

Initial day:
☒ Monday
☐ Tuesday
☐ Wednesday
☐ Thursday
☐ Friday
☐ Saturday
☐ Sunday

Initial time:

Step:

Definition of some basic configurations

Run page



Build your ABM

Home

Canvas

Rooms

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Post Processing

Run models

Seed:

1749561324

Number of simulations:

10

Number of parallel simulations:

10

Select run type:

Docker

Local with 3D visualisation

Local

Docker

Simulation Log

Run a defined model.

Stop

Setting page



Build your ABM

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Set floor dimension

Height (meter)
Floor height dimension (default 80m)

Width (meter)
Floor width dimension (default 100m)

[Set dimension](#)

Load a saved model

[Browse...](#) [Select an RDs file.](#) [Load](#)

Save the model

[Check model](#) [Save the model](#) [Link the model to FLAME GPU 2](#)

References

[1] HOERTEL, N., Blachier, M., Blanco, C., Olsson, M., Massetti, M., Rico, M. S., Limosin, F., & Leleu, H. (2020). A stochastic agent-based model of the SARS-CoV-2 epidemic in France. Nature Medicine, 26(9), 1417–1421. doi:<https://doi.org/10.1038/s41591-020-1001-6>

[2] GKANTONAS, S., Zabotti, D., Mesquita, L. C., Mastorakos, E., & de Oliveira, P. M. (2021). airborne.cam: A risk calculator of SARS-CoV-2 aerosol transmission under well-mixed ventilation conditions. Available at: <https://airborne.cam>

[3] J.L. Jimenez and Z. Peng, COVID-19 Aerosol Transmission Estimator. <https://tinyurl.com/covid-estimator>

[4] Tolksdorf K, Buda S, Schuler E, Wieler LH, Haas W. Influenza-associated pneumonia as reference to assess seriousness of coronavirus disease (COVID-19). Euro Surveill. 2020 Mar;25(11):2000258. doi: <https://doi.org/10.2807/1560-7917.ES.2020.25.11.2000258>. Epub 2020 Mar 16. PMID: 32186278; PMCID: PMC7096775

FLAME GPU 2 references:

[5] Richmond, P., Chisholm, R., Heywood, P., Leach, M., Chimeh, M. K. FLAME GPU. Version 2.0.0-rc. Dec. 2022. <https://doi.org/10.5281/zenodo.7434228>

[6] Richmond, P., Chisholm, R., Heywood, P., Chimeh, M. K., Leach, M. FLAME GPU 2: A framework for flexible and performant agent based simulation on GPUs. In: Software: Practice and Experience (2023). <https://doi.org/10.1002/spe.3207>

ABM school references:

[7] Baccaga, Daniele, Pernice, Simone, Terna, Pietro, Castagno, Paolo, Moirano, Giovenale, Richiardi, Lorenzo, Sereno, Matteo, Rabellino, Sergio, Maule, Milena Maria and Beccuti, Marco (2022) 'An Agent-Based Model to Support Infection Control Strategies at School' Journal of Artificial Societies and Social Simulation 25 (3) 2 <http://jasss.soc.surrey.ac.uk/25/3/2.html>. doi: <https://doi.org/10.18564/jasss.4830>

[8] Daniele Baccaga, Simone Pernice, Paolo Castagno, Matteo Sereno, and Marco Beccuti. Evaluating the Impact of Mask and Quarantine Policies on the Spread of COVID-19 in Schools using computational modeling. In the 18th Conference on Computational Intelligence Methods for Bioinformatics & Biostatistics (CIBB 2023)

Definition of some settings, load/save models, references

Post Processing page



Build your ABM

Home

Canvas

Rooms

Agents

Resources

Infection

What-If

Configuration

Run

Settings

Post Processing

Uploading simulation ⓘ

Select Folder

Load

Query on Disease Status ⓘ

Resulting Simulations ⓘ

Disease Visualisation ⓘ

2D Simulation Visualisation ⓘ

Contact Matrix



Post Processing page: query



Uploading simulation ⓘ

Select Folder /home/dbaccega/Desktop/results_SIR/Clinic_NoCountermeasures

Query on Disease Status ⓘ

Select range for Day

Select range for Susceptible

Select range for Exposed

Select range for Infected

Select range for Recovered

Select range for Died

Simulations can be filtered using queries on disease status.

Resulting Simulations ⓘ

Show 5 entries Search:

FolderNames

seed1749561324

seed1749561325

seed1749561326

seed1749561327

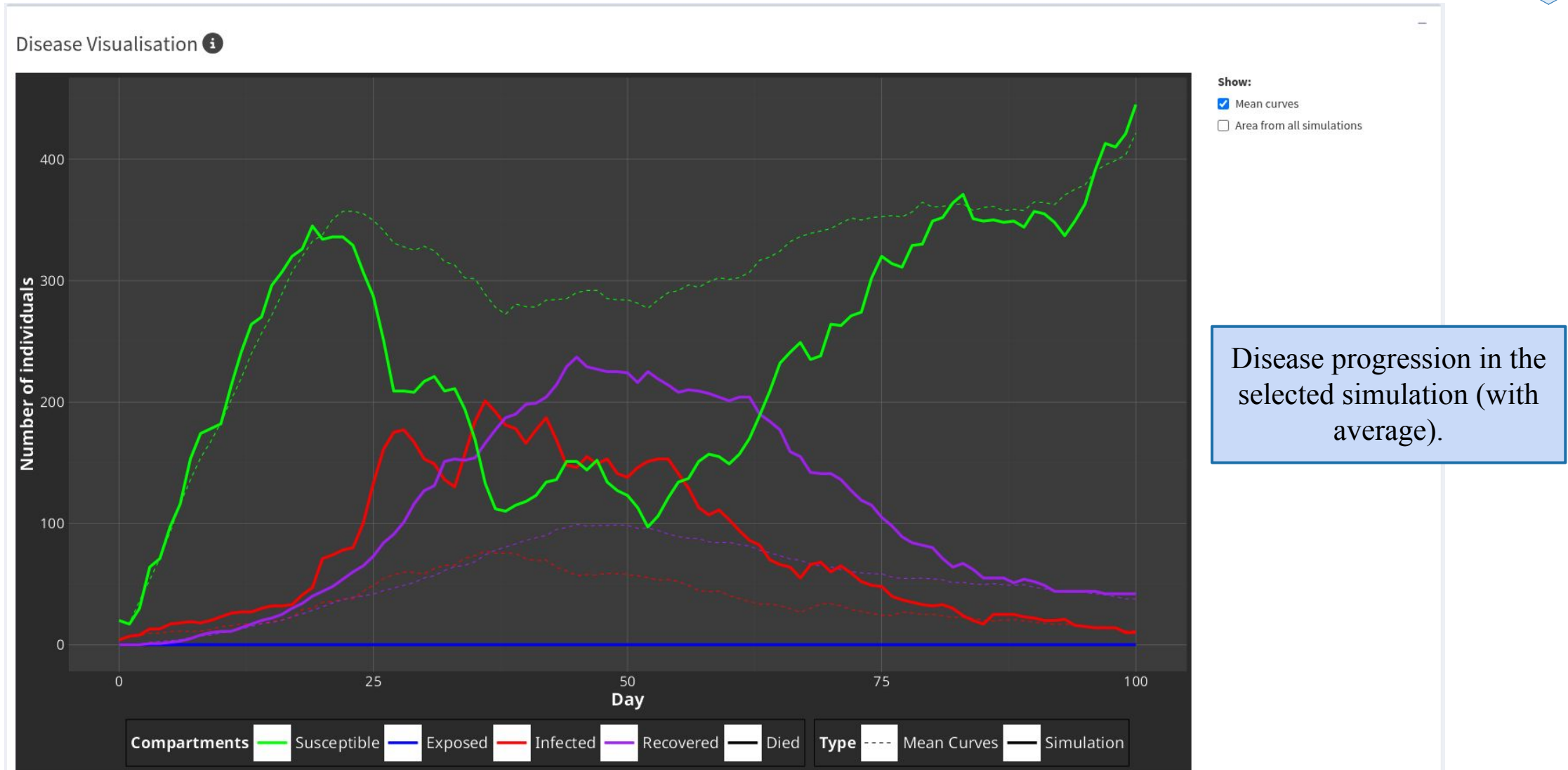
seed1749561328

Showing 1 to 5 of 10 entries Previous 1 2 Next

The user can inspect a specific simulation.

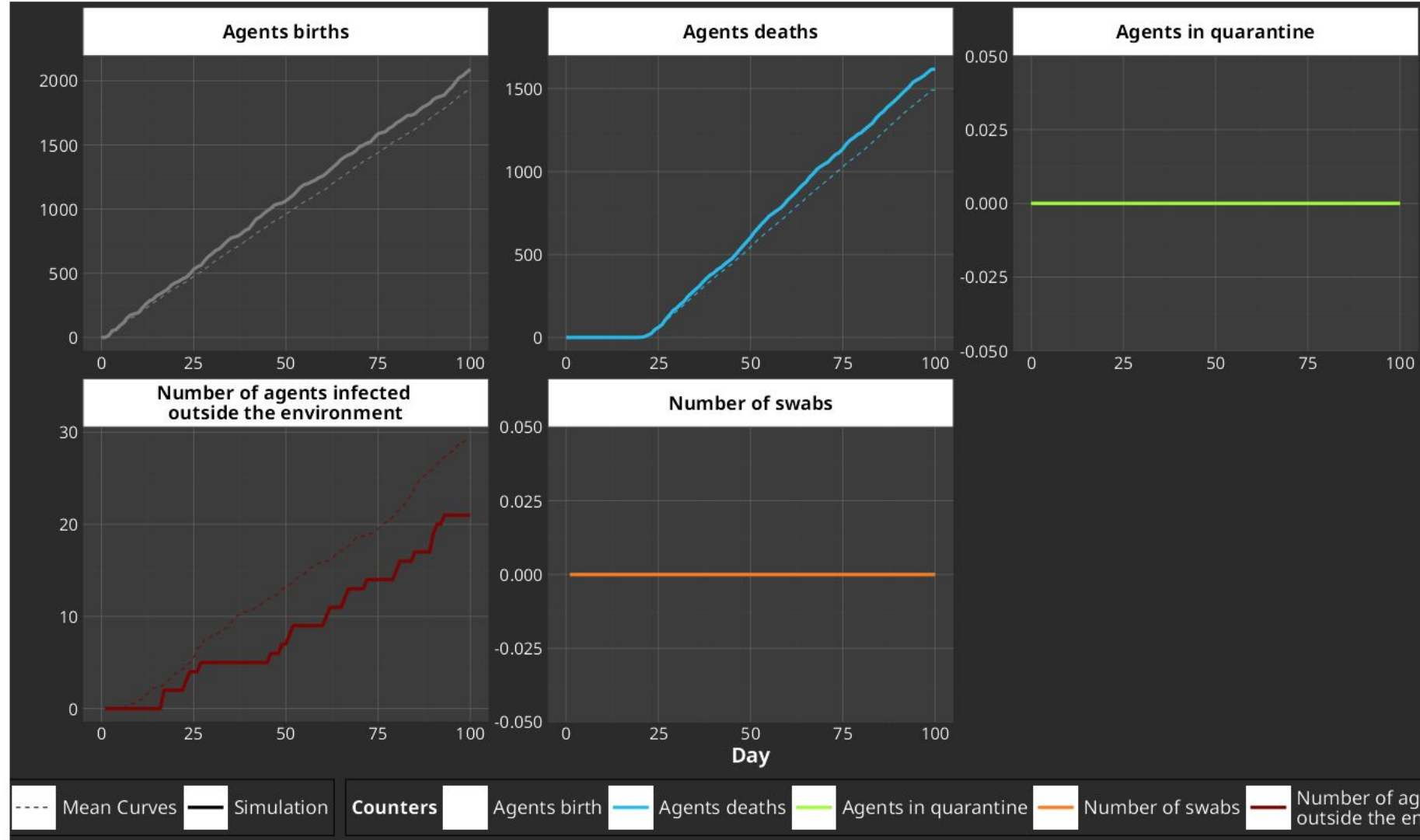


Post Processing page: disease evolution





Post Processing page: other plots



Show:

- ☒ Mean curves
- ☐ Area from all simulations

Some other useful information in the selected simulation (with average).



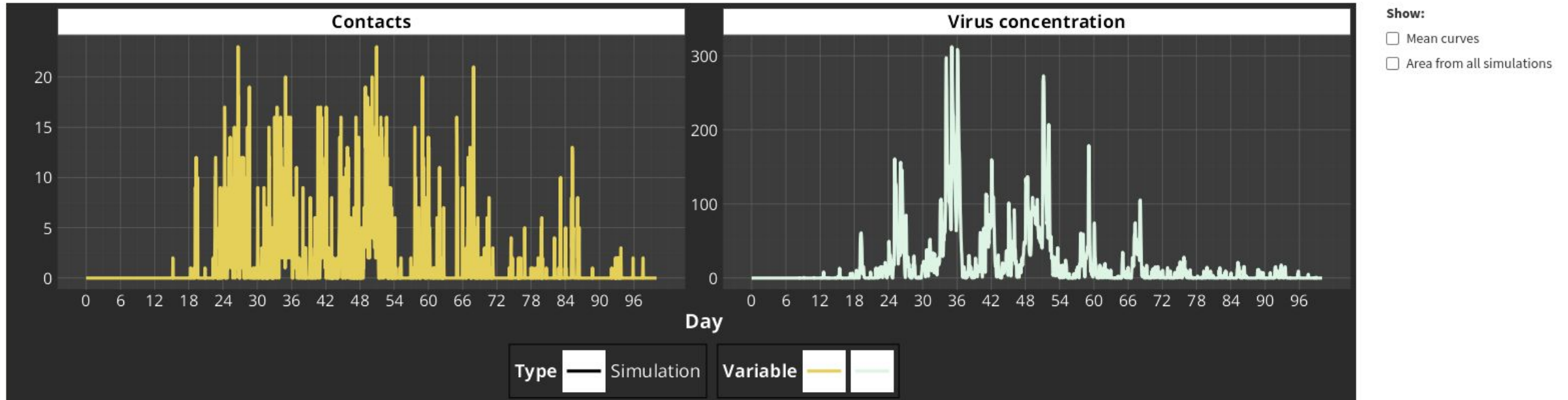
Post Processing page: contacts and virus concentration



Number of contacts and virus concentration inside the waiting room.

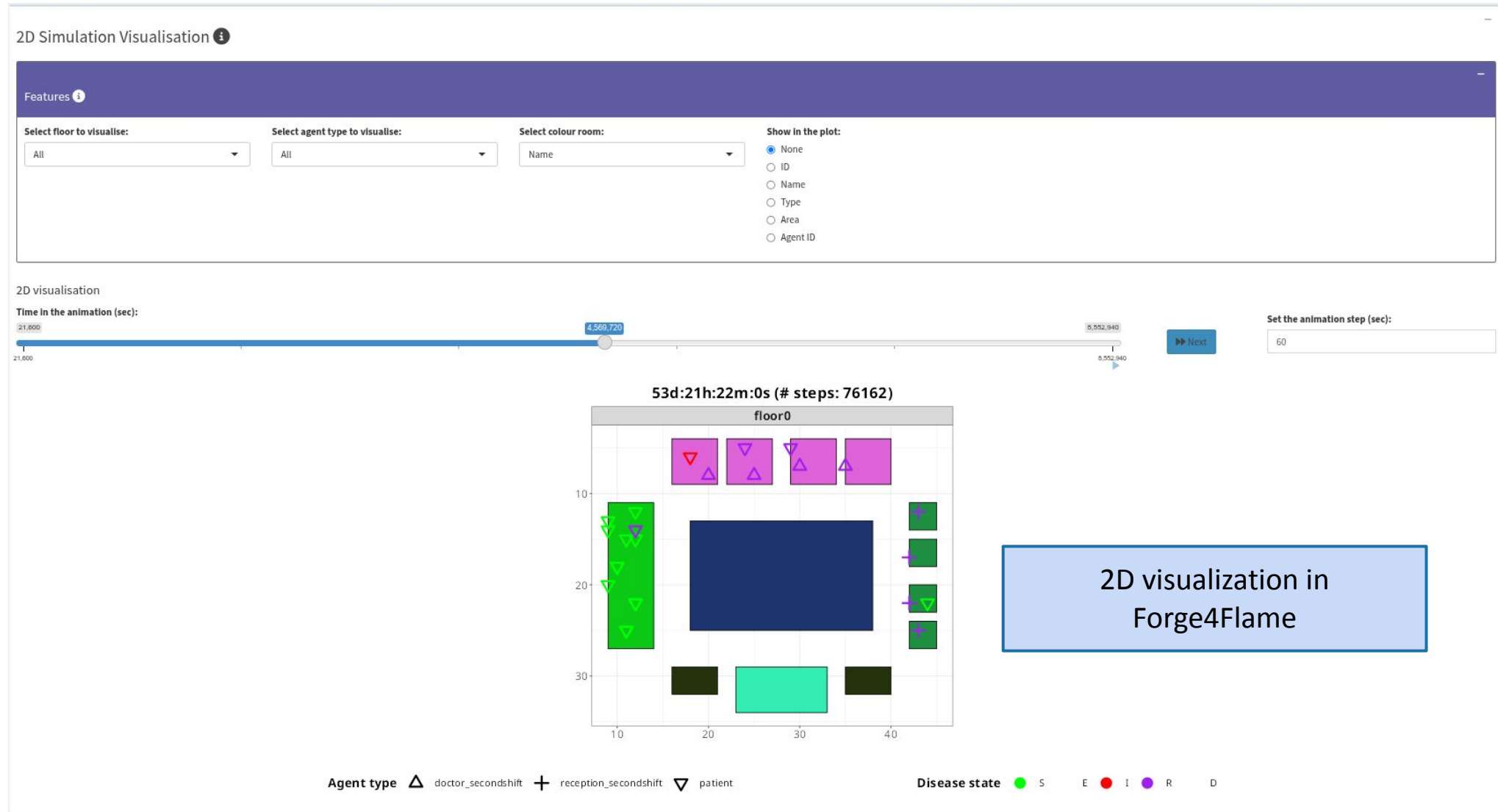
Choice of the room: 

floor0 ; None ; waitingroom ; ID 3



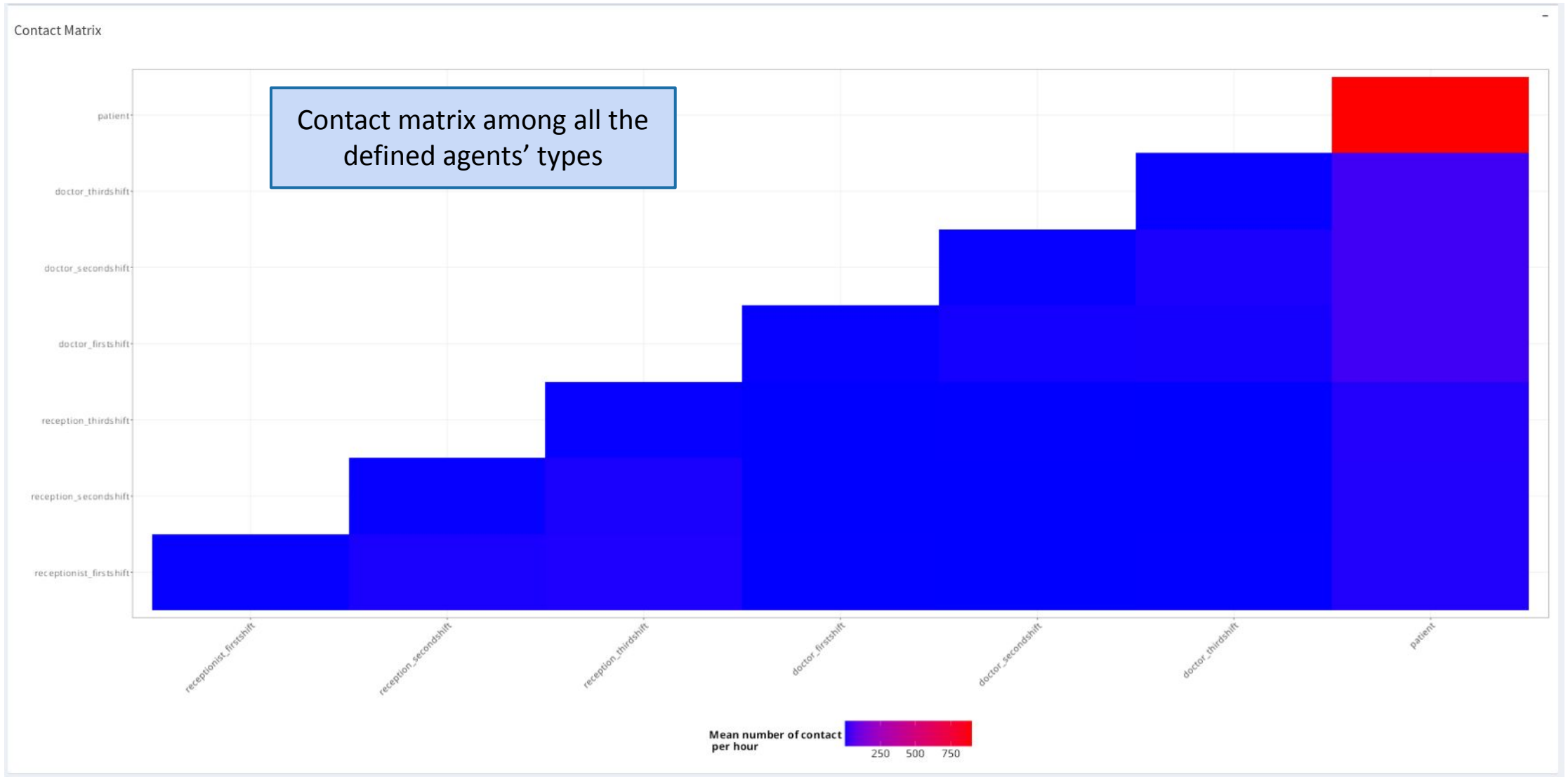


Post Processing page: 2D visualization



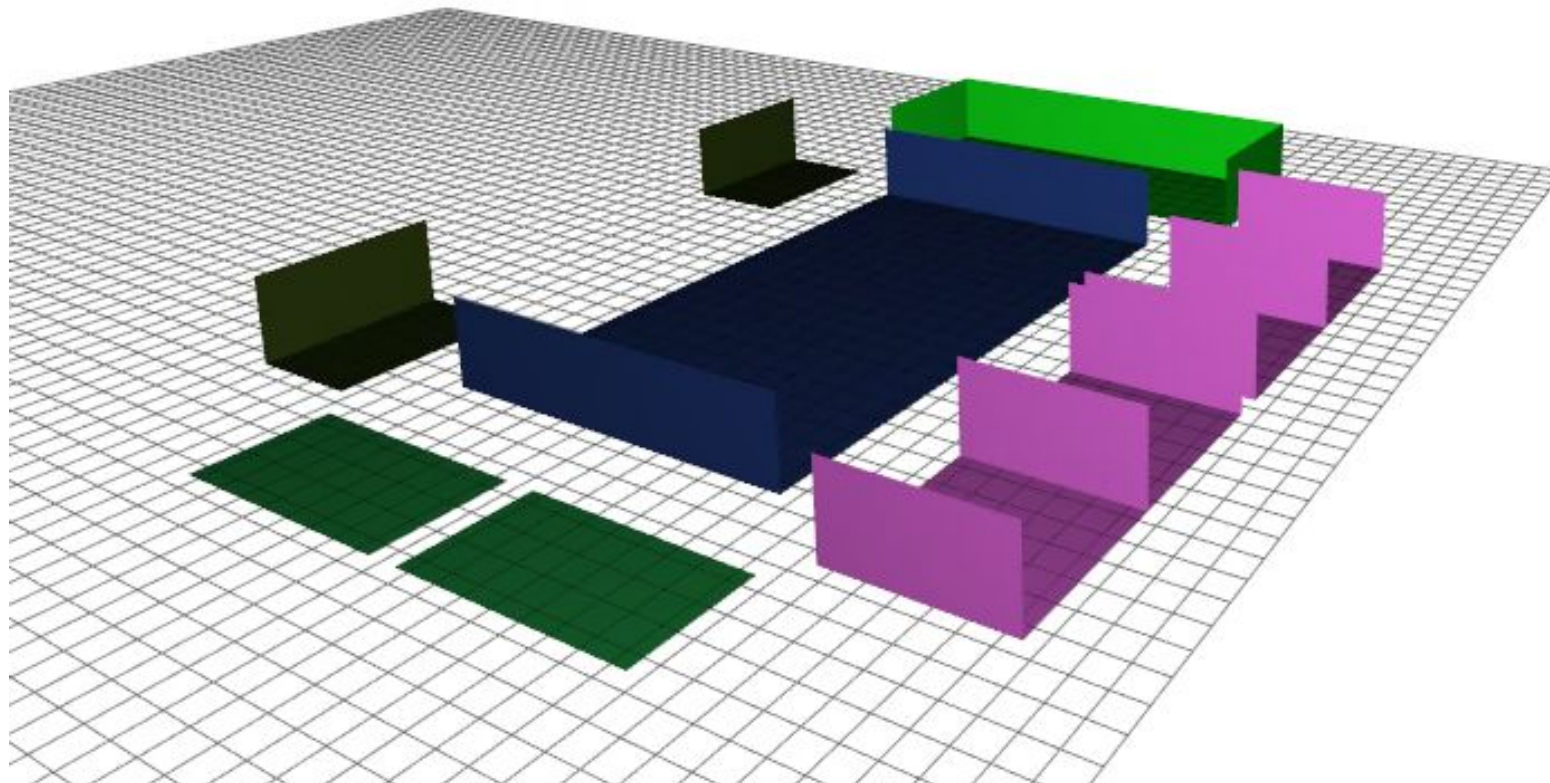


Post Processing page: contact matrix



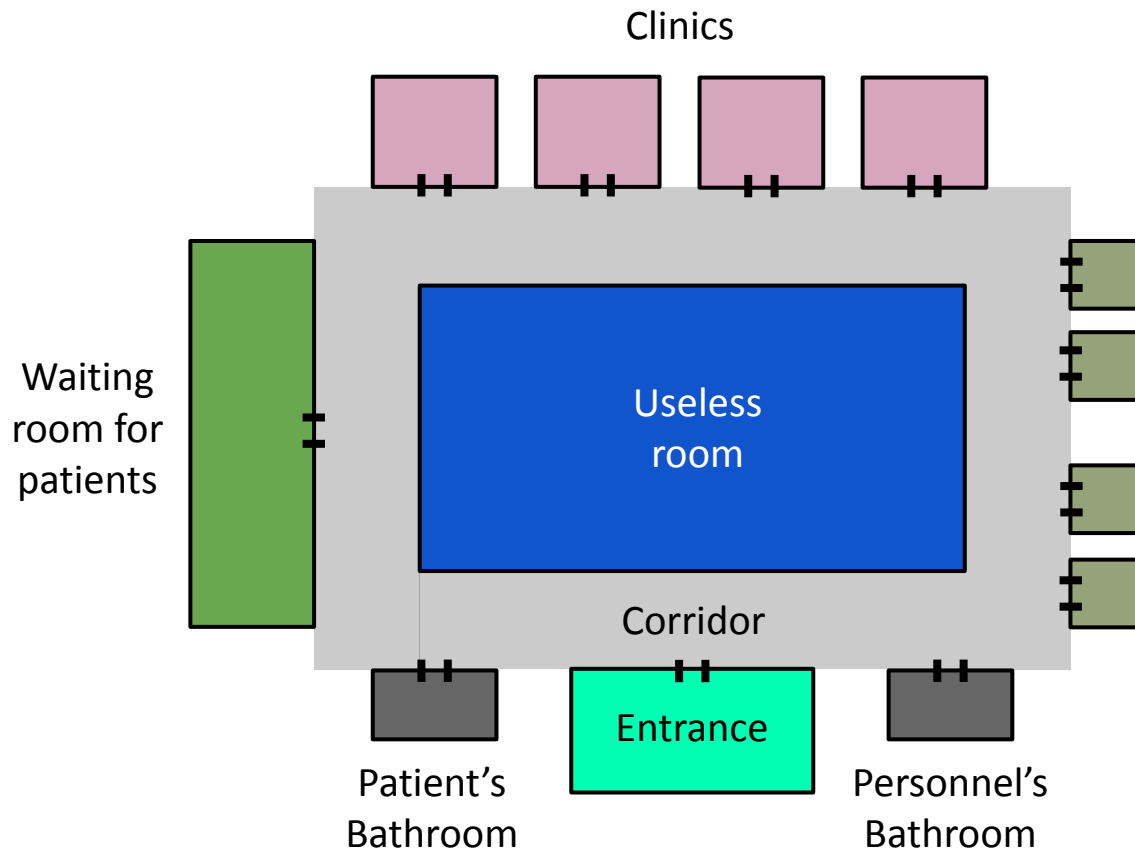


3D visualization in FLAME GPU 2

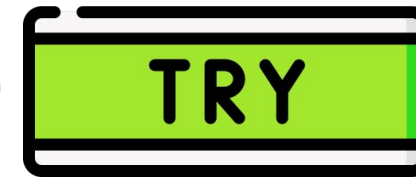
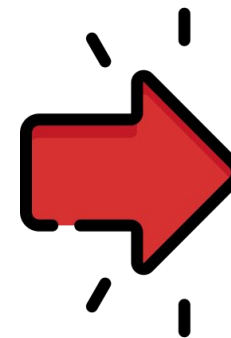


3D visualisation
provided by FLAME
GPU 2

Case study: small ambulatory



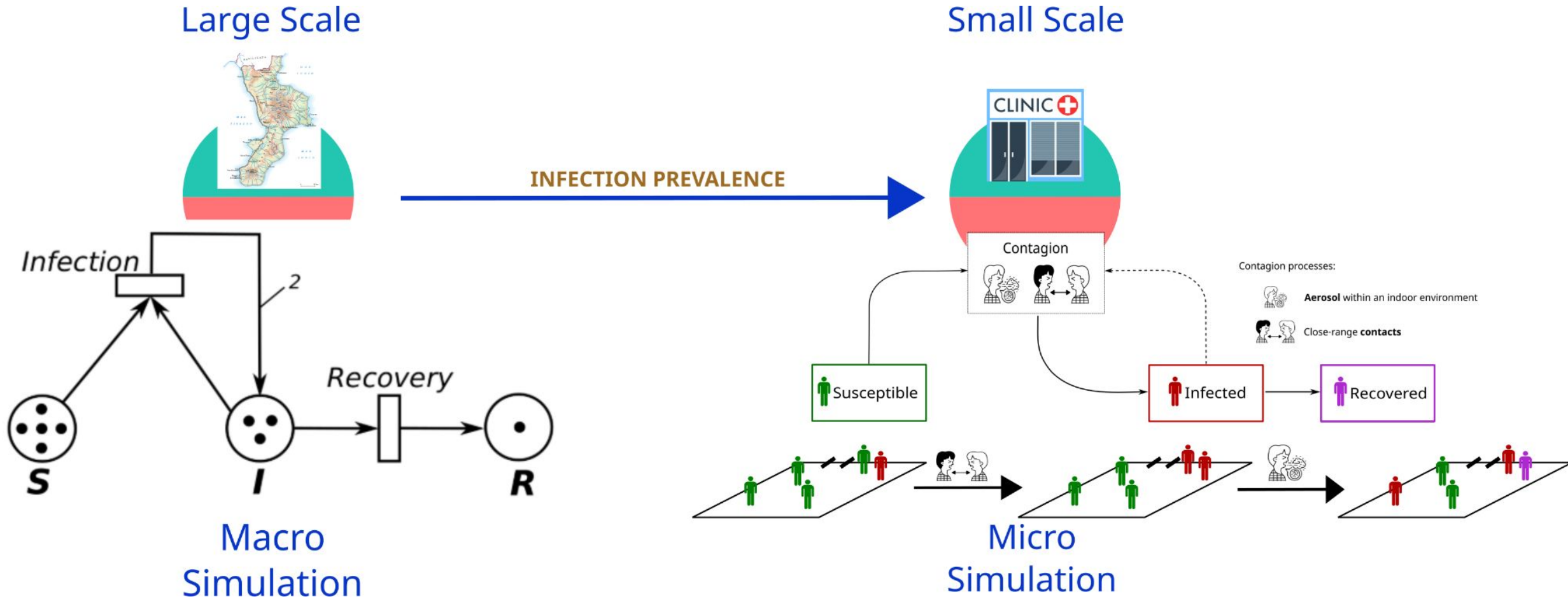
Receptions



<http://github.com/qBioTurin/IEEEICHI2025>



From macro to micro-level models

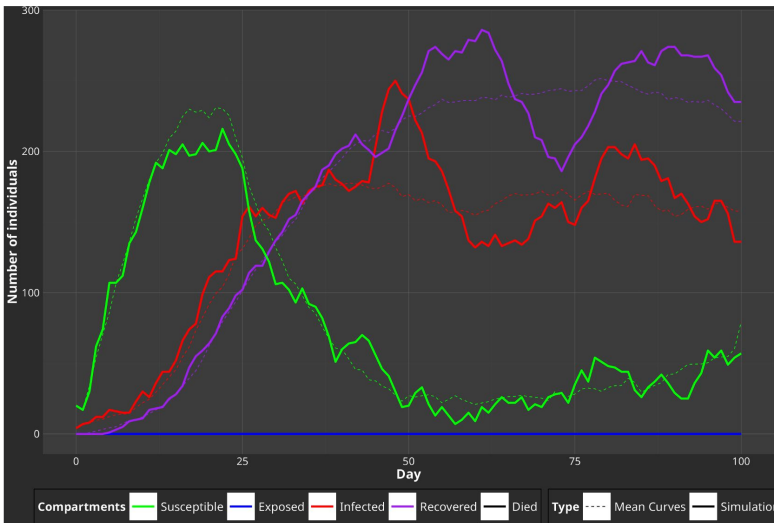
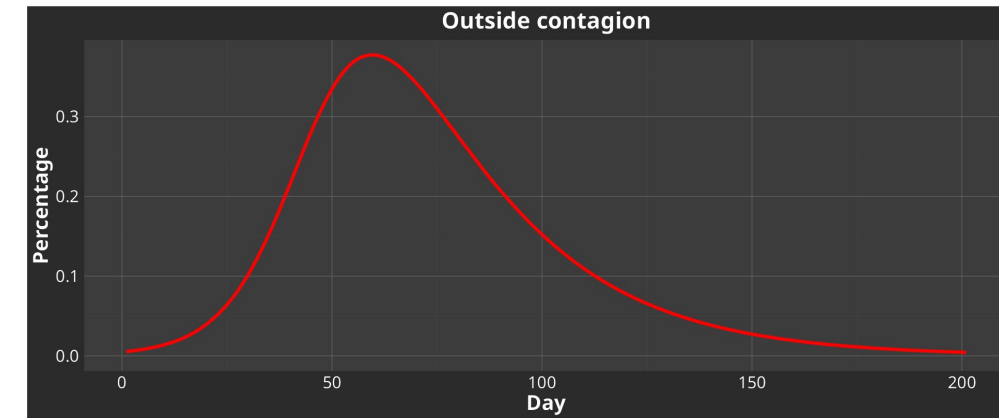
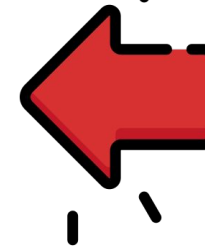
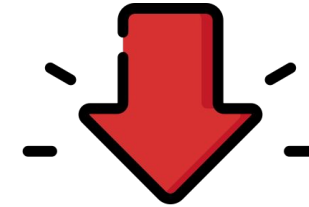
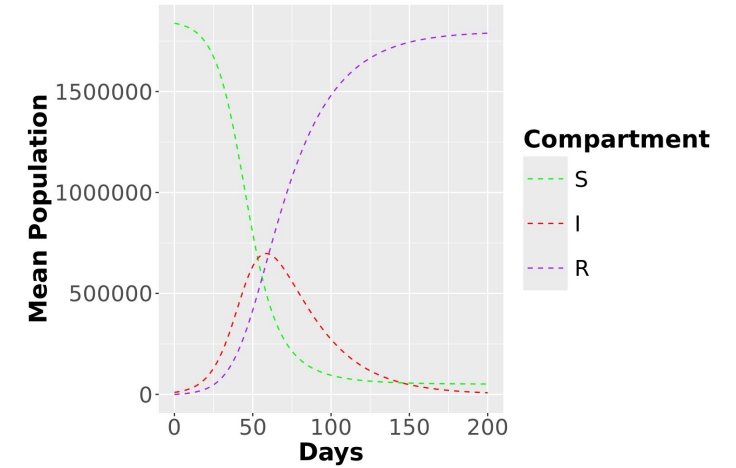
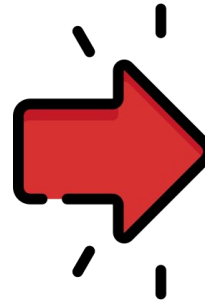
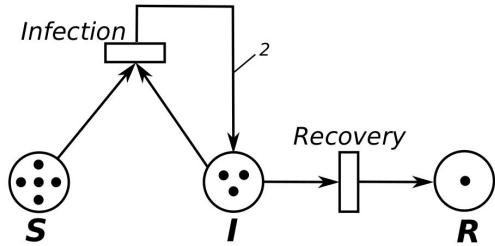


From macro to micro-level models



```

1 m;S;1838568;
2 m;I;10000;
3 m;R;0;
4 c;Recovery; 0.04;
5 c;Infection; 0.00000008;
    
```



Conclusion

Estimating Intervention Impact

Both macro and micro models help assess policies like lockdowns and vaccinations, each with their own strengths and limitations.

Individual Behavior Matters

Micro-level models offer detailed insights into how personal choices shape epidemic dynamics.

Simplicity vs. Detail

A key modeling challenge—striking the right balance depends on your goal: explanation, forecasting, or policy evaluation.

Choosing the Right Model

- **Macro models:** Best for broad trends, quick insights, low data requirements.
- **Micro models:** Ideal for detailed scenarios, heterogeneous behavior, and exploring emergent effects.



We hope you've gained a clearer understanding of when and how to use different computational models to study and respond to epidemics.

 **We'll be around during the entire conference!**

 **Found bugs or have suggestions? Leave feedback on our GitHub.**

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Thank you for the attention!



<https://qbio.di.unito.it/>

